

Classical Hydrodynamics

UE HAP707P

Master degree in Fundamental Physics

2025-2026

Frédéric Geniet

frederic.geniet@umontpellier.fr

University of Montpellier

Outline

1	Introduction - Ideal Fluid - The Euler Equation	5
1.1	The hydrodynamics description	5
1.1.1	Micro and macroscopic scales	5
1.1.2	Hydrodynamic quantities	5
1.1.3	Euler and Lagrange description of the motion	6
1.1.4	Particle or Material Derivative	7
1.1.5	Current lines, particular trajectories	7
1.2	Equation of Mass Conservation	8
1.2.1	differential approach	8
1.2.2	Integral approach	9
1.2.3	Lagrangian approach	10
1.3	Euler's Equation	10
1.4	Boundary conditions	12
1.5	Non-compressible fluid	12
1.6	Hydrostatic and Archimedes theorem	13
1.7	Bernoulli Theorem	14
1.7.1	Venturi	15
1.7.2	Clepsydra	15
1.7.3	Pitot tube	16
1.8	Potential flow	17
1.8.1	Potential flow past a sphere	18
1.9	Vortices and eddies	20
1.9.1	Rankine Vortex	20
1.9.2	Kelvin theorem	21
1.9.3	Helmholtz theorem	22
2	Navier Stokes Equation	24
2.1	Introduction	24
2.1.1	Newtonian Fluid	24
2.1.2	The role of viscosity : Boundary Layers	25
2.2	The Navier Stokes Equation	26
2.2.1	Viscous stress tensor	26
2.2.2	Navier Stokes (NS) Equation	26
2.2.3	NS Boundary conditions.	26
2.3	Reynolds Number	27
2.3.1	Adimensionalization of N.S.	28
2.4	Navier-Stokes Classical flows	29

2.4.1	Planar Couette flow	29
2.4.2	Non Stationary Couette flow	30
2.4.3	Inclined plane flow	31
2.5	Forced flow and Poiseuille Formula	32
2.5.1	Planar Poiseuille flow	32
2.5.2	Cylindrical Poiseuille flow	33
2.6	Rotating flows	33
2.6.1	Orthoradial solutions	33
2.6.2	Taylor-Couette flow	34
2.6.3	Non-stationary orthoradial flow - Vortex decay	34
2.7	2-dimensional flow and vorticity equation	35
2.8	Creeping flows	36
3	Boundary Layer	37
3.1	Introduction	37
3.1.1	Context	37
3.1.2	Size of the Boundary Layer	37
3.1.3	Boundary Layer Separation	38
3.2	The planar Boundary Layer	39
3.2.1	Orders of magnitude analysis	39
3.2.2	Planar Boundary Layer velocity profile	40
3.2.3	Boundary Layer at a stagnation point	42
3.3	Secondary flow	43
3.4	Scaled adimensional equations of the BL	44
3.5	The fate of BL - Turbulence	45
4	Water Waves	48
4.1	Introduction	48
4.1.1	General assumptions	48
4.1.2	General wave equations	49
4.1.3	Boundary conditions	50
4.1.4	Sakharov's Equations	51
4.2	Deep water small amplitude gravity waves	51
4.2.1	Linearization	51
4.2.2	Solution	52
4.2.3	Dispersion relation	52
4.3	Finite Depth	53
4.4	Propagation and Dispersion	54
4.4.1	Phase velocity	54
4.4.2	Group velocity	55
4.5	Waves between two fluids	58
4.6	Capillary Waves	59
4.7	Conclusion	60
5	Stability of flows	62
5.1	Introduction	62
5.1.1	Short historical outline	62
5.1.2	Linear stability analysis	63

5.2	Rayleigh-Taylor instability	64
5.3	Kelvin-Helmholtz instability	65
5.3.1	General equations	65
5.3.2	Unperturbed solution	66
5.3.3	Perturbed solution	66
5.3.4	Stability condition	67
5.4	Stability of the planar Couette and Poiseuille flows	68
5.5	Other instability mechanism	71
5.6	Taylor-Couette instability	71
6	Turbulence	75
6.1	Introduction - What is turbulence ?	75
6.1.1	Irregular flow	76
6.1.2	Intermittency	77
6.1.3	Rotational flow and mixing	78
6.1.4	Structures (eddies) at all scales	79
6.1.5	A more precise statement of the question	79
6.2	Richardson-Kolmogorov cascade	80
6.3	Reynolds stress tensor	82
6.4	The log velocity profile	83
6.4.1	Kinetic model of viscosity	84
6.4.2	Turbulent viscosity model	84
6.5	Conclusion	86
A	Some Useful Formula	88
A.1	operators	88
A.1.1	Vector identities :	88
A.1.2	Cartesian Coordinates:	88
A.1.3	Cylindrical Coordinates :	89
A.1.4	Spherical Coordinates :	89
A.2	Hydrodynamics Equations	89
A.2.1	Vectorial form:	89
A.2.2	Cartesian Coordinates:	90
A.2.3	Cylindrical Coordinates:	91
A.2.4	Spherical Coordinates:	91
B	Tensor Calculus	93
B.1	Basic correspondence	93
B.2	Tensor product	93
B.3	Scalar product, vector product	94
B.4	Totally antisymmetric tensor	95
B.5	Differential calculus	95
B.6	Covariance	96
C	Proof of Reynolds Theorem	97

D	Hydrodynamics in a rotating frame	99
D.1	The Coriolis force	99
D.2	The vorticity equation in a rotating frame	99
D.3	The Ekman layer	100
E	The Diffusion Equation	102
E.1	The equation	102
E.2	Fundamental solution	102
E.3	General solution in unbounded medium	103
E.4	Solutions 'à la Fourier'	103
E.5	Numerical solution in 1D by the 'line method'	104
E.6	Boundary conditions	105
E.7	Stability of the line method	106
F	Stokes Flow past a sphere	107
F.1	Axisymmetric incompressible flow - Stokes current function	107
F.2	Stokes flow	108
F.3	Stokes formula	109
G	The kinematic boundary condition at a free surface	110
G.1	Lagrangian derivation of the BC	110
G.2	Eulerian derivation of the BC	110
H	Remark on the continuity of pressure at a wave interface.	112
I	The Miles mechanism	114

Chapter 1

Introduction - Ideal Fluid - The Euler Equation

1.1 The hydrodynamics description

1.1.1 Micro and macroscopic scales

The hydrodynamics description of a fluid is valid at a meso scale : we will consider "fluid particles" large enough to assume a local equilibrium and neglect the fluctuations of mean quantities. On the other side, the particle is small enough to admit a local description. As a consequence, one can define a local temperature $T(\vec{r}, t)$, a local pressure $p(\vec{r}, t)$, and finally a local density $\rho(\vec{r}, t)$ of the fluid particle. Let us give some order of magnitude :

- in water the mean free path of a molecule is $l \sim 1\text{\AA}$. This is the typical distance responsible of thermalization. A non thermal molecule will be brought to equilibrium with the rest of the fluid in a distance of the order of l .
- Let us consider a fluid particle of size $L \sim 0.1\text{ }\mu\text{m}$. Its volume is $V \sim 10^{-21}\text{ m}^3$. For water, $M \sim 18\text{ g/Mol}$ and $\rho \sim 1000\text{ kg/m}^3$, and this particle contains $N \sim 5 \cdot 10^{-17}\text{ Mol}$ or $n \sim 3 \cdot 10^7$ molecules. As we know from statistical physics, the relative size of the fluctuations (number, density, pressure...) is $\sim 1/\sqrt{n} \sim 2 \cdot 10^{-4}$. We see from this example that it makes sense to speak of a fluid particle with definite values for the thermodynamic quantities.

Exercise 1.1. *Compute at which scale the hydrodynamics description ceases to be valid. You may also do the computation for air.*

1.1.2 Hydrodynamic quantities

As a consequence, the local state of the fluid is defined by its mean values at scale L :

$$p(\vec{r}, t), T(\vec{r}, t), \rho(\vec{r}, t), \vec{v}(\vec{r}, t) \dots$$

respectively the mean local pressure, temperature, density and velocity of the fluid. This description is completed by some empirical parameters describing the fluid behavior : thermal conductivity, viscosity, surface tension.... In principle these parameters are also defined locally, and should be derived from a microscopic description, using quantum mechanics and

statistical physics. This goal is in fact very hard to reach and we will use a phenomenological approach, guided by symmetry principles and conservation laws to provide relations between the macroscopic quantities defined above (see examples later in these lectures).

The goal of hydrodynamics is to relate these functions, and ultimately determine them if possible.

1.1.3 Euler and Lagrange description of the motion

Understanding these two descriptions of the motion of a fluid and their relationship is fundamental. Many theorems are easily demonstrated by going from one point of view to the other.

Lagrangian description

In mechanics of systems of material points, one uses a Lagrangian description : particles are labeled by an index $i = 1 \dots N$ and one follows the trajectories of these particles as a function of time : $\vec{r}_i(t)$. The velocities and accelerations are given in the usual way :

$$\vec{v}_i(t) = \dot{\vec{r}}_i(t) \quad \vec{a}_i(t) = \dot{\vec{v}}_i(t) \quad (1.1)$$

using the usual notation $\dot{x}(t) \equiv \frac{dx(t)}{dt}$.

One can extend this to a continuous description using a continuous label¹ σ , namely $\vec{r}(\sigma, t)$ is the position of particle σ at time t . One then has

$$\vec{v}(\sigma, t) = \frac{\partial \vec{r}(\sigma, t)}{\partial t} \quad \vec{a}(\sigma, t) = \frac{\partial \vec{v}(\sigma, t)}{\partial t} \quad (1.2)$$

It is often useful to take the position $\vec{r}_0$ of the particle at a given initial time t_0 as a label, namely $\vec{r}(\vec{r}_0, t)$. One then has of course $\vec{r}(\vec{r}_0, t_0) = \vec{r}_0$.

Eulerian description

Although the Lagrangian description is crucial in understanding the laws of hydrodynamics, it is rarely used from a practical point of view, since the equations it produces are rather intricate. In the Euler description one does not follow a fluid particle, but rather one looks at the content of a fixed volume in space. Such a volume is called a 'control volume', a notion that will play an important role in the following. If this volume is small enough (linear size L defined above) one may speak of $\vec{v}(\vec{r}, t)$, the velocity of the particle sitting at the position $\vec{r}$ at time t .

N.B. Be careful that these expressions look like those in the Lagrange description, but they do not mean the same thing at all. In particular the fluid particle sitting at $\vec{r}$ at time t will change with time.

Exercise 1.2. In the Euler description one never encounters $\vec{r}(\vec{r}, t)$. Why ?

Exercise 1.3. One considers a 1D flow given by the Lagrangian description $v_L(x_0, t) = at + bx_0$. Find the Eulerian expression of its velocity.

For any physical quantity T the relation between the two descriptions is given by (look at it carefully !)

$$T_L(\sigma, t) = T_E(\vec{r}_L(\sigma, t), t) \quad (1.3)$$

¹in general a vector

1.1.4 Particle or Material Derivative

From the preceding paragraph, the variation rate of the quantity T following the fluid particle is given by

$$\frac{DT}{Dt} \equiv \frac{dT_L(\sigma, t)}{dt} = \frac{dT_E(\vec{r}_L(\sigma, t), t)}{dt} = \frac{\partial T_E(\vec{r}, t)}{\partial t} + \vec{v}_E(\vec{r}, t) \cdot \vec{\nabla} T_E(\vec{r}, t) \quad (1.4)$$

with the correspondence $\vec{r} = \vec{r}_L(\sigma, t)$ and $\vec{\nabla} \equiv (\partial_x, \partial_y, \partial_z)$ is the 'Nabla' operator. The material derivative symbol D/Dt is just a bookkeeping device for the ordinary time rate in the Euler description.

From now on we will use the Euler description except when stated and we will omit the E subscript.

Exercise 1.4. Check the previous result by making explicit the Euler and Lagrange correspondence at every step.

Exercise 1.5. Consider the following Euler flow $\vec{v}(\vec{r}, t) = (\alpha x, -\alpha y)$ and a concentration profile given by $c(\vec{r}, t) = \beta x^2 y e^{-\alpha t}$, where α and β are positive constants. Compute Dc/Dt and interpret this result.

1.1.5 Current lines, particular trajectories

Current lines are curves tangential to the velocity flow at any given time. In principle they evolve with time. If we denote $\vec{r}(s) = (x(s), y(s), z(s))$ a parametric representation of such a line, we have $\frac{d\vec{r}}{ds} = k(s) \vec{v}(\vec{r}(s), t)$ (the tangent to the current line is proportional to the velocity). We can redefine the parameter s so that $k(s) = 1$ and define current lines as solutions of

$$\frac{d\vec{r}}{ds} = \vec{v}(\vec{r}(s), t) \quad (1.5)$$

From this it is clear that t plays here the role of a parameter, and does not mingle with the ODE.

Exercise 1.6. Find and plot the current lines of the 2D flow $\vec{v}(\vec{r}, t) = (\alpha x, -\alpha y)$.

Exercise 1.7. Find and plot the current lines of the 2D flow $\vec{v}(\vec{r}, t) = (\alpha(x - x_0(t)), -\alpha(y - y_0(t)))$, where $(x_0(t), y_0(t))$ is a given trajectory, and α is a positive constant.

By contradistinction, the particular trajectories follow immediately from the Lagrange description : these are the curves $\vec{r}(\sigma, t)$ when t varies. In the Euler description they are obtained by solving the ODE

$$\frac{d\vec{r}}{dt} = \vec{v}(\vec{r}(t), t) \quad (1.6)$$

In this case we see that t is no more a parameter but the variable in the ODE. Some remarks are in order :

- $Df/Dt = 0$ tells that f is constant along a particle trajectory.
- $(\vec{v} \cdot \vec{\nabla}) f = 0$ tells that f is constant along a current line.
- for a stationary flow, $\partial_t f = 0$ and the current lines and particular trajectories are identical.

Exercise 1.8. find and plot the current lines and trajectories of the flow $\vec{v} = (u_0, kt)$, where u_0 and k are positive constants.

Exercise 1.9. find and plot the current lines of the flow $\vec{v} = (-\Omega y, \Omega x, 0)$. Compute the corresponding acceleration field $\vec{a} = D\vec{v}/Dt$. Express these results in the polar coordinate basis $(\vec{e}_r, \vec{e}_\theta, \vec{e}_z)$. Interpret your results.

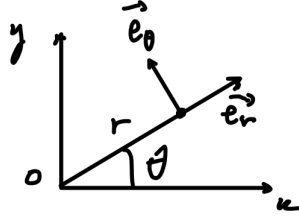


Figure 1.1: Polar coordinates.

1.2 Equation of Mass Conservation

This is often misnamed as 'continuity equation'. We will establish it using different methods that will become useful later. This equation is in fact a balance of mass for a control volume, and the different methods correspond to different choices of control volume.

1.2.1 differential approach

We use a small control volume as shown in figure (1.2). The mass contained in this volume is $dm = \rho(x, y, z, t) dx dy dz$, and its rate of variation with time is $\partial_t \rho d^3V$. This variation is due to

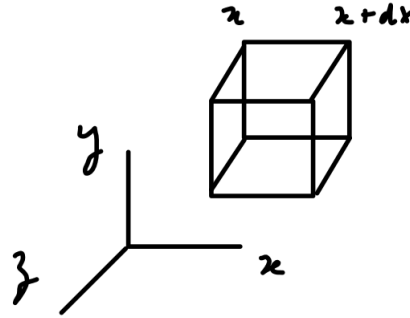


Figure 1.2: Infinitesimal control volume.

the incoming and out-coming fluxes of mass. In a time dt ,

- the mass flux entering through the x face is $\rho(x, y, z)v_x(x, y, z) dy dz dt$
- the mass flux coming out through the $x + dx$ face is $\rho(x + dx, y, z)v_x(x + dx, y, z) dy dz dt$

giving a mass variation rate through x :

$$dm_x/dt = (\rho(x, y, z)v_x(x, y, z) - \rho(x + dx, y, z)v_x(x + dx, y, z)) dy dz = -\frac{\partial}{\partial x} \{ \rho(\vec{r})v_x(\vec{r}) \} d^3V$$

If we replicate this computation for directions y and z and equate this to the mass variation in d^3r we obtain the mass equation conservation

$$\frac{\partial}{\partial t}\rho + \vec{\nabla} \cdot (\rho \vec{v}) = 0 \quad (1.7)$$

Exercise 1.10. Is mass conserved in the flow $\vec{v}(\vec{r}, t) = (\alpha x, -\alpha y)$, with a concentration profile given by $\rho(\vec{r}, t) = \beta x^2 y e^{-\alpha t}$?

1.2.2 Integral approach

This is actually the same approach with a macroscopic, fixed control volume V as in figure (1.3) : The mass included in volume V is $M(t) = \iiint_V \rho(\vec{r}, t) d^3V$, and since the volume is fixed,

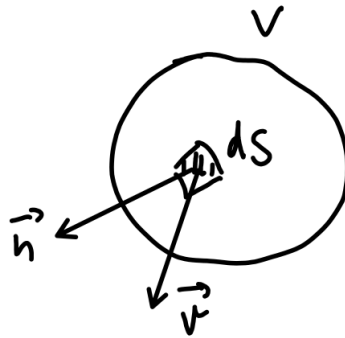


Figure 1.3: Macroscopic control volume.

its variation is

$$\frac{dM}{dt} = \iiint_V \partial_t \rho(\vec{r}, t) d^3V$$

This mass variation is related to the flux coming through the surrounding surface S : for an element of surface d^2S the incoming flux is $-\rho \vec{v} \cdot \vec{n} d^2S$ (where the normal $\vec{n}$ is pointing outwards from V). Summing on the whole surface we obtain the mass balance

$$\frac{dM}{dt} = - \oint_S \rho \vec{v} \cdot \vec{n} d^2S$$

Using then the Stokes-Ostrogradski theorem we have the mass balance on V

$$\iiint_V \left[\partial_t \rho + \vec{\nabla} \cdot (\rho \vec{v}) \right] d^3V = 0 \quad (1.8)$$

and since this is valid for any volume V we obtain Equation (1.7).

Exercise 1.11. Show that the continuity equation can be written as $\frac{D\rho}{Dt} + \rho \vec{\nabla} \cdot \vec{v} = 0$. Note that $\vec{\nabla} \cdot \vec{v} = 0$ does not imply that ρ is constant everywhere, since it can vary from a particular trajectory to another (check with the previous exercise). Conversely a constant value of ρ implies that $\vec{\nabla} \cdot \vec{v} = 0$.

1.2.3 Lagrangian approach

This last method will be useful to understand better the integral derivation of Euler's equation. Here the control volume $V(t)$ is following the fluid. In that case mass conservation is simply expressed as

$$\frac{dM}{dt} = \frac{d}{dt} \iiint_{V(t)} \rho(\vec{r}, t) d^3V = 0$$

To derive the local mass conservation we use the *Reynolds transport theorem*

$$\frac{d}{dt} \iiint_{V(t)} f d^3V = \iiint_{V(t)} \partial_t f d^3V + \oint_{S(t)} (\vec{v} \cdot \vec{n}) f d^2S \quad (1.9)$$

valid for any function $f(\vec{r}, t)$. Note the surface integral due to the displacement of the control volume.

Exercise 1.12. Show directly this theorem in 1D. See a general proof in appendix C.

Using this theorem and Ostrogradski formula in the equation above immediately yields (1.7).

1.3 Euler's Equation

This equation is in fact Newton's second law of motion applied to an ideal fluid without any viscous forces, as superfluid Helium. The question of where and when this is a good approximation for a real fluid is of central importance to these lectures and will be addressed in chapters 2 and 3. Here we will simply neglect these forces and derive the corresponding Euler equation. We consider a system consisting of the material particles enclosed within a volume

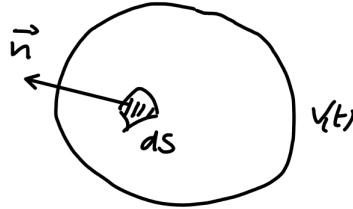


Figure 1.4: Control volume moving with the fluid.

$V_L(t)$ moving with the fluid and denote $\vec{P}$ its total momentum, see figure(1.4). The external forces acting on this system are its weight $M\vec{g}$ and the pressure exerted by the rest of the fluid $\vec{R}_{pr}$. The fundamental relation then writes

$$\frac{d\vec{P}}{dt} = M\vec{g} + \vec{R}_{pr}$$

and using an integral representation of these quantities

$$\frac{d}{dt} \iiint_{V(t)} \rho \vec{v} d^3V = \iiint_{V(t)} \rho \vec{g} d^3V - \oint_{S(t)} p \vec{n} d^2S \quad (1.10)$$

the minus sign due to $\vec{n}$ pointing outwards V . We will use a tensor notation to avoid confusion when using the Reynolds theorem, see appendix B for a quick introduction. We note

- v_i for $i = 1 \dots 3$ the three component of the velocity.
- ∂_i for $i = 1 \dots 3$ the three component of the Nabla operator.

We use Einstein convention : every repeated index is summed (and dummy). As examples

- $v_i v_i = v_x^2 + v_y^2 + v_z^2 = ||\vec{v}||^2$
- $\partial_i v_i = \vec{\nabla} \cdot \vec{v}$
- $v_i \partial_i = (\vec{v} \cdot \vec{\nabla})$

We then have, using Newton's law and Reynolds theorem

$$\iiint_V \partial_t(\rho v_i) d^3V + \oint_S v_j n_j \rho v_i d^2S = \iiint_V \rho g_i d^3V - \oint_S p n_i d^2S$$

Using now the Green-Ostrogradski theorem written in tensor form for any function f as

$$\oint_S f n_j d^2S = \iiint_V \partial_j f d^3V$$

we obtain

$$\iiint_V [\partial_t(\rho v_i) + \partial_j(\rho v_i v_j)] d^3V = \iiint_V [\rho g_i - \partial_i p] d^3V .$$

This equation is a momentum balance that gives the local form

$$\frac{\partial \rho v_i}{\partial t} + \frac{\partial \Pi_{ij}}{\partial x_j} = \rho g_i \quad (1.11)$$

where we have introduced the momentum transfer tensor $\Pi_{ij} = \rho v_i v_j + p \delta_{ij}$. This expression expresses the momentum (non)conservation. To collect the pressure term with current flow of momentum seems natural when one realizes that pressure is indeed a manifestation of microscopic momentum transfer (think about $p = mn u^2/3$ in the kinetic theory of gases). Using Leibniz rule to expand the differentials $\partial_t \rho v_i = \rho \partial_t v_i + v_i \partial_t \rho$ and $\partial_j \rho v_i v_j = \rho v_j \partial_j v_i + v_i \partial_j \rho v_j$ and using mass conservation we finally obtain $\rho [\partial_t v_i + v_j \partial_j v_i] = -\partial_i p + \rho g_i$, or

$$\rho [\partial_t \vec{v} + (\vec{v} \cdot \vec{\nabla}) \vec{v}] = -\vec{\nabla} p + \rho \vec{g} \quad (1.12)$$

This is Euler's fluid equation. It is valid even for compressible fluids when ρ is not constant. At this stage it is useful to give a look at the traditional derivation of this equation which can be written

$$\rho \frac{D\vec{v}}{Dt} = -\vec{\nabla} p + \rho \vec{g}$$

and not

$$\frac{D\rho\vec{v}}{Dt} = -\vec{\nabla} p + \rho \vec{g}$$

Exercise 1.13. Explain why this last result is incorrect.

Exercise 1.14. By integrating the conservation equation (1.11) over a fixed volume, show that Π_{ij} can be interpreted as the i component of the flux of momentum across a surface normal to $\vec{e}_j$. Stated otherwise $\Pi_{ij} dS_j$ is the flux of impulsions i across dS_j .

1.4 Boundary conditions

The Euler equation (1.12) is a (set of) partial differential equations (PDE) of first order. In order to solve it, one has to give an initial condition $\vec{v}(\vec{r}, t_0) = \vec{v}_0(\vec{r})$, and also some boundary conditions (BC) at the border of the fluid domain. This question of the good choice of BC is crucial since an incorrect one can lead to many different solutions, or inversely no solution at all. For Euler's equation in a vessel, without a free surface² experience has shown that the *normal component* of the velocity has to be continuous at the boundaries. For a rigid recipient, this writes

$$\vec{v} \cdot \vec{n}|_{boundary} = 0 \quad (1.13)$$

Some remarks :

- Note that there is no BC for pressure on walls, and it is false to specify one. If you come across a software asking for pressure BC on walls, you can trash it !
- It might seem more natural to impose a condition of null velocity $\vec{v}|_{boundary} = 0$ at the boundary. However in the case of Euler's equation this produces a solution $\vec{v} = 0$ everywhere. We will discuss this point further in chapter 3 with the concept of *Boundary Layer* introduced by Prandtl in 1905.
- See also the paragraph E.6 "Boundary conditions" in the appendix E on the diffusion equation to get a better insight about the role of BC.

1.5 Non-compressible fluid

In the remaining chapters of these lectures, we will assume that the fluid is incompressible and that $\rho = Cte$. From this condition we immediately derive the incompressibility equation :

$$\vec{\nabla} \cdot \vec{v} = 0 \quad (1.14)$$

This, together with the Euler equation, form the fundamental equations of an ideal fluid. Indeed, if we neglect heat exchanges in the fluid, temperature T does not enter the description of the fluid, and we have only 4 unknown quantities $\vec{v}(\vec{r}, t)$ and $p(\vec{r}, t)$. On the other side we have 4 EDP. To resume the equations describing an ideal incompressible fluid are

$$\frac{\partial}{\partial t} \vec{v} + (\vec{v} \cdot \vec{\nabla}) \vec{v} = -\frac{1}{\rho} \vec{\nabla} p + \vec{g} \quad \text{and} \quad \vec{\nabla} \cdot \vec{v} = 0 \quad (1.15)$$

We will see that these equations are generic enough to describe a variety of situations, and complex enough to keep mathematicians, physicists and engineers busy for 200 years.

Let us now ask when the incompressibility condition is valid :

- for a liquid as water, alcohol or glycerin, this seems to be a very good description, but it excludes all the convection phenomena as in Rayleigh-Benard convection.
- For a gas such as air this hypothesis seems absurd. As a matter of fact, it is totally valid when the fluid velocity satisfies $v \ll c$, where c is the sound velocity in the gas³.

²to be considered in chapter 4.

³The case $v \gtrsim c$ is called 'Gas dynamics' and corresponds to supersonic flows.

- Compression is indeed transmitted through sound waves. The condition $v \ll c$ writes $l/t \ll \lambda \nu$ where l and t are typical length and time order of magnitude of the flow, and λ and ν are the wavelength and frequency of the compression wave. Since $t\nu \sim 1$ the above condition writes $l \ll \lambda$ hence pressure is uniform over a typical length l of the flow⁴.

Exercise 1.15. Find in air the wavelength corresponding to a typical time evolution $t \sim 0.01\text{s}$. Deduce the maximal size for which the incompressibility hypothesis is valid.

Exercise 1.16. Going deeper about it : consider gas of mass m enclosed in a cylinder, see figure (1.5)

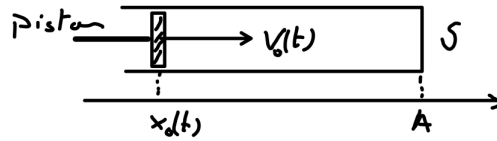


Figure 1.5: An incompressible fluid !!.

- Find the density $\rho(t)$ assuming it is uniform.
- Guess the expression of the velocity flow assuming it is 1-dim. The gas velocity should match that of the piston at $x_0(t)$ and A .
- Check that mass is conserved in this flow, namely that equation (1.7) is indeed satisfied.
- Obtain the pressure field p by using Euler's equation .
- Whenever the piston velocity V_0 is constant, check that p is uniform, making the initial assumption plausible. What happens if $V_0 \ll c$? The conclusion of this exercise is that 'incompressible' in fluid dynamics do not mean the same as in thermodynamics !

1.6 Hydrostatic and Archimedes theorem

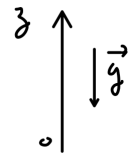
Hydrostatic refers of course to the case $\vec{v} = 0$. One then has $\partial_t \rho = 0$ by mass conservation, but ρ can vary from place to place as e.g. in the atmosphere. The fundamental hydrostatic equation (or Pascal Law in France) writes

$$\vec{\nabla} p = \rho \vec{g} \quad (1.16)$$

Some remarks :

- Pascal Law says that pressure increases in the direction of $\vec{g}$ as it should.

- If density is constant we get $p(z) = p_0 - \rho g z$, with the z direction pointing up.



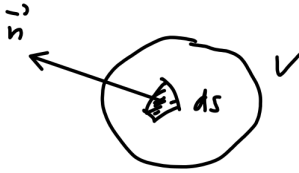
⁴This is similar to the quasi stationary approximation in electrokinetic, where one assume that intensity is constant over the whole circuit.

- For an isotherm perfect gas atmosphere we have $p/\rho = RT/M$ and

$$\frac{d \log \rho}{dz} = -\frac{Mg}{RT} \quad \text{yielding} \quad \rho(z) = \rho_0 \exp \left[\frac{Mg}{RT} (z - z_0) \right]$$

Exercise 1.17. Atmosphere are not usually isothermic : temperature goes down as we climb higher. However air mass motion can be treated as isentropic, since heat exchanges are slow. Compute the variation of ρ in this case, assuming a diatomic gas atmosphere.

Archimedes theorem is simply obtained by integrating Pascal law on a control volume V . One has

$$\iiint_V \vec{\nabla} p \, d^3V = \vec{P}_e$$


and using Gauss-Stokes-Ostrogradski-Green theorem this give

$$\oint_S p(-\vec{n}) \, d^2S = -\vec{P}_e \quad (1.17)$$

The l.h.s. of this equation is the resultant of pressure forces acting on V , while the r.h.s. is the opposite of the weight of the fluid contained in V , and this equation expresses the balance of forces acting on V . The trick is now to state that the pressure forces acting on V do not depends of its content. In that way, it is clear that this theorem is valid whether or not the fluid is compressible, the content of V is solid or liquid ...

Note that it is a current use to write $p = p_h + p'$ where p_h is the hydrostatic pressure as defined by Pascal equation (1.16). One then has $\vec{\nabla} p = \rho \vec{g} + \vec{\nabla} p'$ and the fundamental equations for an ideal incompressible fluid write

$$\frac{D\vec{v}}{Dt} = -\frac{1}{\rho} \vec{\nabla} p' \quad , \quad \vec{\nabla} \cdot \vec{v} = 0 \quad , \quad \vec{v} \cdot \vec{n}|_{\text{boundary}} = 0 \quad (1.18)$$

1.7 Bernoulli Theorem

We use the following vector identity :

$$\vec{\nabla} \left(\frac{1}{2} \vec{u}^2 \right) = (\vec{u} \cdot \vec{\nabla}) \vec{u} - (\vec{\nabla} \wedge \vec{u}) \wedge \vec{u}$$

Exercise 1.18. Show this identity. You may use tensor notations, namely show that (remind Einstein's convention !)

$$\partial_i \frac{1}{2} u_j u_j = u_j \partial_j u_i - \varepsilon_{ijk} \varepsilon_{jlm} (\partial_l u_m) u_k$$

where ε_{ijk} is the fully antisymmetric tensor s.t. $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = -\varepsilon_{132} \dots = 1$, and with the property that $\varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}$. You may use this property to replace the $\varepsilon_{ijk} \varepsilon_{jlm}$ term in the above equation and proceed.

Using the above identity, the Euler equation can be written as

$$\partial_t \vec{v} + (\vec{\nabla} \wedge \vec{v}) \wedge \vec{v} = -\vec{\nabla} H \quad (1.19)$$

$$H = \frac{\vec{v}^2}{2} + \frac{p}{\rho} + \chi(\vec{r})$$

where $\chi(\vec{r}) = gz$ here and more generally $\vec{F}/\rho = -\vec{\nabla}\chi$ for conservative forces. The function H is in fact the specific enthalpy $U + PV$, hence the notation used. Let us now consider a stationary flow $\partial_t \vec{v} = 0$. We then have immediately $(\vec{v} \cdot \vec{\nabla})H = 0$, namely H is constant on a current line :

$$\frac{\vec{v}^2}{2} + \frac{p}{\rho} + \chi(\vec{r}) = Cte$$

this is Bernoulli theorem. Be cautious that the constant may be different for different current lines (see exercises below). This theorem is very counter-intuitive : wherever velocity increases on a current line, pressure is decreasing. In fact pressure is due to microscopic thermal motion ($p = 1/3\rho u^2$). Wherever the macroscopic velocity increases the thermal motion must decrease by energy transfer from one form to another. We see that this theorem actually assert energy conservation in a perfect fluid, where dissipation in heat does not exist. We now present classic applications of this theorem.

1.7.1 Venturi

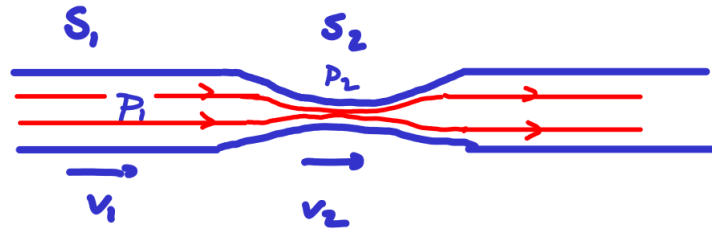


Figure 1.6: Venturi effect.

Let us consider the above stationary flow. By flow conservation one has $v_1 S_1 = v_2 S_2$. From Bernoulli we get

$$p_1 - p_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) = \frac{1}{2}\rho v_1^2 \left((S_1/S_2)^2 - 1 \right) = \frac{1}{2}\rho v_1^2 \left((r_1/r_2)^4 - 1 \right)$$

1.7.2 Clepsydra

From Bernoulli theorem we have the velocity at the outlet $v = \sqrt{2gh}$. You can remark that v is not proportional to the pressure drop in the outlet. The volume exit rate is $Q = vs = -S \frac{dh}{dt}$ so $dh/\sqrt{h} = -\frac{s}{S}\sqrt{2g} dt$, giving finally the law

$$\sqrt{h_0} - \sqrt{h(t)} = \frac{s}{S}\sqrt{\frac{g}{2}} t$$

Exercise 1.19. Find the correct form $S(h)$ to obtain an uniform flow $h \propto t$. Draw the corresponding vessel.

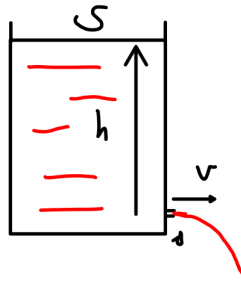


Figure 1.7: a Clepsydra.

1.7.3 Pitot tube

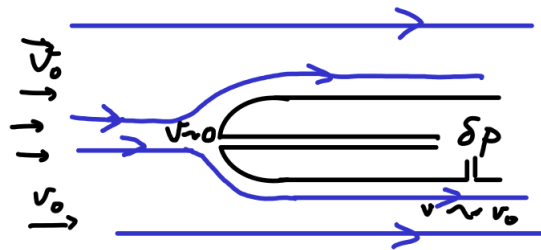
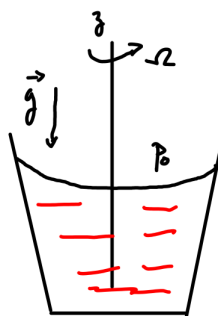


Figure 1.8: a Pitot device in a flow.

Once again from Bernoulli theorem we have the pressure drop $\delta p \propto 1/2 \rho v_0^2$ giving access to the velocity. This is a very cheap and unbreakable device, unless it freezes !

Exercise 1.20. We consider the stationary water flow in a rotating bucket. The goal is to find the shape of the free surface $z(r)$.



- Show that a careless application of Bernoulli theorem gives a nonsensical result (a convex shape !): the constant appearing in the theorem is not the same for different current lines !
- Compute the acceleration of the particles of fluid (see exercise 1.9)
- Write Euler's equation and integrate it from the free surface where $p = p_0$. Show that the result produces a concave shape of the surface as it should.
- Find the pressure anywhere in the fluid and gives the physical meaning of your result.

1.8 Potential flow

We define $\vec{\omega} \equiv \vec{\nabla} \wedge \vec{v}$ as the flow vorticity. This is an important hydrodynamic quantity. Taking the curl of equation (1.19) and using $\vec{\nabla} \wedge (\vec{\nabla} \cdot H) = 0$ we obtain the vorticity equation

$$\frac{\partial \vec{\omega}}{\partial t} + \vec{\nabla} \wedge (\vec{\omega} \wedge \vec{v}) = 0 \quad (1.20)$$

Exercise 1.21. Show the identity $\vec{\nabla} \wedge (\vec{a} \wedge \vec{b}) = \vec{a}(\vec{\nabla} \cdot \vec{b}) - \vec{b}(\vec{\nabla} \cdot \vec{a}) + (\vec{b} \cdot \vec{\nabla})\vec{a} - (\vec{a} \cdot \vec{\nabla})\vec{b}$

Exercise 1.22. Use incompressibility $\nabla \cdot \vec{v} = 0$ and $\nabla \cdot \vec{\omega} = 0$ (since ω is rotational) to show that the vorticity equation can be recast into

$$\frac{D\vec{\omega}}{Dt} = (\vec{\omega} \cdot \vec{\nabla})\vec{v}$$

In either form this seems to tell that a flow such that $\vec{\omega} = 0$ initially stays so for any time. Such a flow, if it exists, is called *irrotational* or *potential flow*. Indeed in that case we have in analogy with electrostatics

$$\vec{\omega} = \vec{\nabla} \wedge \vec{v} = 0 \quad \Longleftrightarrow \quad \vec{v} = \vec{\nabla} \phi$$

ϕ is called the velocity potential. Let us now examine the physical meaning of potential flow.

Consider the following situation : It would seem that every flow is irrotational, since it

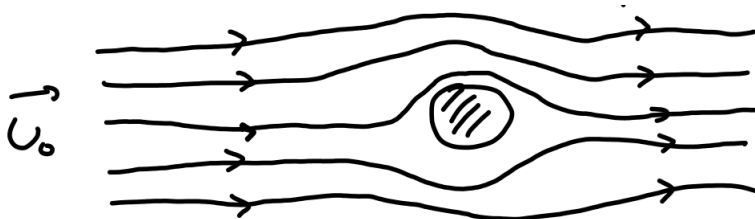


Figure 1.9: a potential flow ?

starts from rest at initial time, or the velocity is uniform upstream, and therefore irrotational far from obstacles, and should stay so from the theorem. This is in sharp contradistinction with what is observed in any brooklet where vortices are omnipresent and the flow is certainly rotational⁵. One could then deduce that Euler's equation is incorrect, since it lacks the viscous forces, and that we should discard it as useless.

This conclusion is however a bit harsh : the Euler equation does describe in a very satisfactory way the flow in a great variety of situations in regions far from the obstacles and the associated wakes : Near obstacles, the Euler equation is indeed incorrect, and we have to take viscosity into account, and use Navier-Stokes equations. This generates vorticity, which then propagates through the wake in a large portion of the fluid. What happens nearby obstacles is therefore of fundamental importance, and brings in the notion of Prandtl boundary layer that we will study in chapter 3.

We will study later in great detail the conditions that make the ideal fluid assumption valid, and we take as an established experimental fact that large regions of many fluid flow can be

⁵as a conclusion beware of theoreticians!



Figure 1.10: a real flow, with a wake.

described by potential flow $\vec{\omega} = 0$, $\vec{v} = \vec{\nabla}\phi$. Let us now examine the consequences of this assumption. Equation (1.19) then gives

$$\vec{\nabla} \left(\frac{\partial \phi}{\partial t} + \frac{v^2}{2} + \frac{p}{\rho} + \chi \right) = 0$$

and this yields

$$\frac{\partial \phi}{\partial t} + \frac{v^2}{2} + \frac{p}{\rho} + \chi = f(t) \quad (1.21)$$

the function f does not depend of position, and can be taken null with a redefinition of the velocity potential. We then obtain Bernoulli identity but with the same constant throughout the flow.

Exercise 1.23. *What prevents the use of this theorem in exercise 1.20 ?*

As a final comment, the conclusions of this paragraph are not correct in a rotating frame such as on the Earth. Indeed vortices are continuously created in the circulating atmosphere and oceans, due to the Coriolis force, see appendix D for a very brief account on the subject.

1.8.1 Potential flow past a sphere

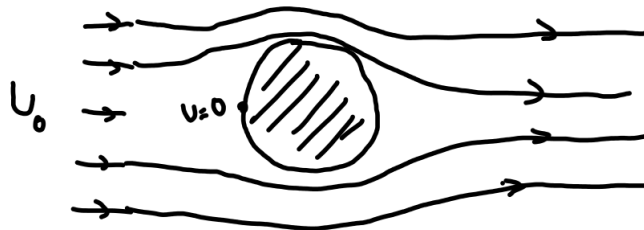


Figure 1.11: potential flow past a sphere.

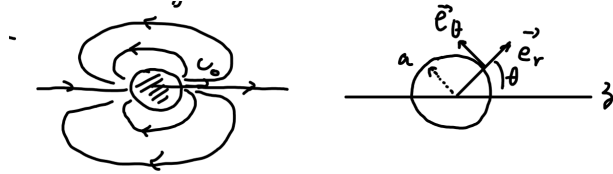
Although this flow can never be observed this exercise is a classic, and actually an useful result, if used in the correct way. The question of its relation to an actual flow past a sphere will be examined in subsequent chapters. It is also useful to gain practice in manipulating flow equations.

From the stationary potential flow assumption we have $\frac{v^2}{2} + p = p_a$ where p_a is the pressure at the stagnation point. This is the only equation for the pressure, that will be deduced from

the velocity field. The main equation is actually the incompressibility condition $\vec{\nabla} \cdot \vec{v} = 0$. Assuming $\vec{v} = \vec{\nabla} \phi$ we get Laplace equation for the velocity potential⁶

$$\Delta \phi = 0$$

All this is a strong reminder of electrostatics, and the flow lines picture in the frame where the fluid is at rest at infinity recalls that of an electric dipole, so we test the following Ansatz solution with $\vec{A}$ an unknown constant vector to be specified :



$$\phi = U_0 z + \frac{\vec{A} \cdot \vec{e}_r}{r^2}$$

From that we have (check that !)

$$\vec{v} = \vec{U}_0 + \frac{1}{r^3} \left(\vec{A} - 3\vec{e}_r(\vec{A} \cdot \vec{e}_r) \right)$$

Using now $\vec{v} \cdot \vec{n} = 0$ for $r = a$ we have the condition $0 = \vec{U}_0 \cdot \vec{e}_r - 2/a^3 \vec{A} \cdot \vec{e}_r$ that should be valid for any $\vec{e}_r$. This yields $\vec{A} = \frac{1}{2} a^3 \vec{U}_0$, and finally

$$\vec{v} = \vec{U}_0 + \frac{a^3}{2r^3} \left(\vec{U}_0 - 3\vec{e}_r(\vec{U}_0 \cdot \vec{e}_r) \right)$$

You can check that $\vec{v}$ cancels on the sphere at $\theta = 0$ and $\theta = \pi$. There pressure is given by the Bernoulli theorem : $p_a = \rho/2 U_0^2 + p_\infty$. You can also check that a condition $\vec{v} = 0$ for $r = a$ could not be satisfied. Let us finally compute the resulting pressure on the sphere. We have

$$\vec{v}_s = \frac{3}{2} \left(\vec{U}_0 - 3\vec{e}_r(\vec{U}_0 \cdot \vec{e}_r) \right) \quad v_s^2 = \frac{9}{4} U_0^2 (1 - \cos^2 \theta)$$

on the sphere. We see that v_s^2 is symmetric under the change $\theta \rightarrow \pi - \theta$, and from Bernoulli p is symmetric in a front-back reflection. The resulting total pressure force therefore cancels, which constitutes the 'd'Alembert paradox' : in the Euler description the total force by the fluid on the sphere is null⁷! Once again this result shows the discrepancy between Euler description and the real world, and that viscosity is essential to describe both the correct flow and the action of this flow on obstacles.

Exercise 1.24. Solve the same problem for the flow past a cylinder. This is the same problem in 2-D. You can try an Ansatz in the form $\phi = U_0 x + \frac{\vec{A} \cdot \vec{e}_r}{r}$

As a conclusion of this subsection we once again want to mitigate the feeling that all this whole Euler approach is useless : the solutions that we have found have indeed explained how birds and planes are flying, and this is enough credential to state they are useful.

⁶As a general remark, although looking rather dull, the incompressibility condition is actually the main equation in hydrodynamics, since it describes almost all the interactions responsible for the structure of matter. In a practical way, this is often the first equation to consider to solve for the flow, as in this exercise.

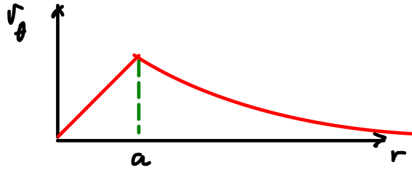
⁷Note that this result is general and not restricted to a flow past a sphere.

1.9 Vortices and eddies

We close this chapter by exploring an essential aspect of fluid dynamics, the existence of localized structures where vorticity is concentrated, namely vortices or eddies, that are notably characteristic of turbulence. We have seen that vortices associated to a non zero value of the vorticity $\vec{\omega}$ can not appear in an Eulerian ideal fluid. We also stated that this conclusion is nonphysical : viscosity always plays a role in the boundary layer where eddies are generated. These eddies are then transported by the fluid and may invade the whole flow. When they exit the BL, vorticity is (almost) conserved, and the eddies stay in principle forever, another approximate and physically questionable result ! This fate of the eddies will be examined in our last chapter about turbulence. For now let us look at a simplified version of these objects.

1.9.1 Rankine Vortex

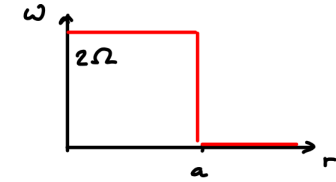
this is a simplified but rather accurate description of a true vortex. It is described in cylindrical coordinates as

$$\begin{cases} v_\theta = \Omega r & r \leq a \\ v_\theta = \frac{\Omega a^2}{r} & r \geq a \end{cases} \quad (1.22)$$


In cylindrical coordinates this gives for the vorticity

$$\vec{\omega} = \vec{\nabla} \wedge \vec{v} = \frac{1}{r} \begin{vmatrix} \vec{e}_r & r\vec{e}_\theta & \vec{e}_z \\ \partial_r & \partial_\theta & \partial_z \\ v_r & rv_\theta & v_z \end{vmatrix} = \frac{1}{r} \partial_r(rv_\theta) \vec{e}_z$$

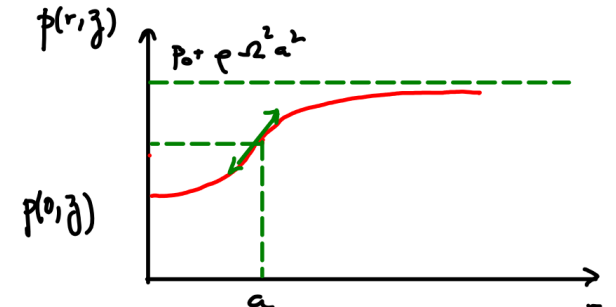
and so

$$\begin{cases} \vec{\omega} = 2\Omega \vec{e}_z & r \leq a \\ \vec{\omega} = 0 & r \geq a \end{cases} \quad (1.23)$$


To obtain the pressure, Bernoulli equation is useless as in exercise 1.20, but Euler's equation yields

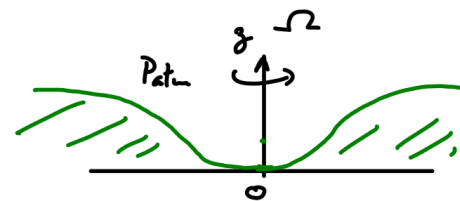
$$\partial_r p = \rho \frac{v^2}{r}$$

a result that we interpret as a balance between the pressure resultant and the centrifugal force. Integrating this equation we obtain, using the continuity of p and $\partial_r p$ at $r = a$,

$$\begin{cases} p(r, z) = p(0, z) + \rho \frac{\Omega^2 r^2}{2} & r \leq a \\ p(r, z) = p(0, z) + \rho \Omega^2 a^2 \left(1 - \frac{a^2}{2r^2}\right) & r \geq a \end{cases}$$


(1.24)

We still have to use the Euler equation in the vertical direction $\partial_z p = -\rho g$. If the vortex is topped by a free surface as in the bucket exercise, with a profile denoted by $z = \zeta(r)$, and we chose the vertical origin at the bottom of the well $\zeta(0) = 0$, we have $p(r = 0, z = 0) = p_{atm}$, $p(r = 0, z) = p_{atm} - \rho g z$ and one can check Bernoulli condition $\frac{1}{2}\rho v^2 + p + \rho g z = Cte$ for $r \geq a$ where the flow is irrotational. Finally the free surface is obtained by stating that $p_{atm} = p(r, \zeta(r))$ giving

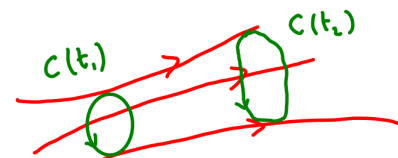
$$\begin{cases} \zeta(r) = \frac{\Omega^2}{2g} r^2 & r \leq a \\ \zeta(r) = \frac{\Omega^2 a^2}{2g} \left(1 - \frac{a^2}{2r^2}\right) & r \geq a \end{cases}$$


(1.25)

We see that a deep is forming at the vortex center, as you can observe in an emptying sink.

1.9.2 Kelvin theorem

As you can see from the names attached to these theorems, vorticity has caught the interest of some great names. This theorem and the following one are concerned with the evolution of vortices in an ideal fluid. Kelvin was fascinated by the conservation of vorticity to the point that he was seeing them as fundamental constituents of matter, so this was particle physics in Kelvin's view ! His theorem states that for an ideal incompressible fluid, circulation of the velocity along a closed loop 'following the flow' is constant. In equation

$$\Gamma \equiv \oint_{C(t)} \vec{v} \cdot d\vec{l} = Cte$$


(1.26)

To prove this result we will need the following lemma

$$\frac{d\Gamma}{dt} = \oint_{C(t)} \frac{D\vec{v}}{Dt} \cdot d\vec{l}$$

(1.27)

Proof : we use the Lagrangian view : the loop is parameterized by an equation $\vec{r}_L(\sigma, t)$ with $0 \leq \sigma \leq 1$. We then write $d\vec{l} = \frac{\partial \vec{r}_L}{\partial \sigma} d\sigma$ and $\vec{v} = \vec{v}_L(\sigma, t) = \partial_t \vec{r}_L(\sigma, t)$ to replace as

$$\frac{d\Gamma}{dt} = \frac{d}{dt} \int_0^1 \vec{v}_L(\sigma, t) \frac{\partial \vec{r}_L}{\partial \sigma} d\sigma$$

using Leibnitz rule to take the derivative under the sum (fixed integral bounds !) the first term is exactly the r.h.s. of equation (1.27) while the second one is an exact derivative taken on a closed loop | **QED**.

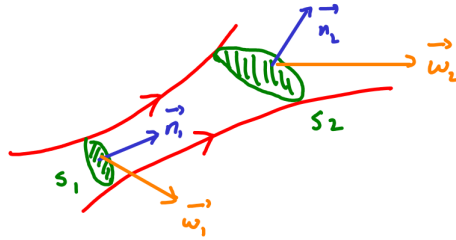
From this lemma and Euler's equation we have

$$\frac{d\Gamma}{dt} = - \oint_{C(t)} \vec{\nabla} \left(\frac{p}{\rho} + gz \right) \cdot d\vec{l} = \left[\frac{p}{\rho} + gz \right]_C = 0 \quad (1.28)$$

From a physical point of view, this theorem states that the circulation associated to a vortex is conserved in an ideal fluid. Note that this theorem is valid wherever viscous forces are negligible, making it useful even for a real fluid flow far from obstacles.

The persistence of irrotational flow can be seen as a consequence of this theorem : if for a portion of fluid $\vec{\omega} = 0$ then the circulation $\Gamma = \oint \vec{v} \cdot d\vec{l} = \iint \vec{\omega} \cdot \vec{n} d^2S = 0$ for any loop drawn in the fluid. When the flow evolves all these loops keep $\Gamma = 0$, and so is $\vec{\omega} = 0$.

In contradistinction this does not states that $\vec{\omega} = Cte$ following the flow, since this will depend on the divergence or convergence of the flow lines . In the following situation we have



$$\vec{\omega}_1 \cdot \vec{n}_1 dS_1 = \vec{\omega}_2 \cdot \vec{n}_2 dS_2$$

In a tornado case $\vec{\omega}$ stays parallel to $\vec{n}$ and we have simply $\omega_1 S_1 = \omega_2 S_2$ showing that the velocity will increase as the vortex shrinks, to attain extremely large velocities and potentially very destructive results.

1.9.3 Helmholtz theorem

We then define *vortex lines* as everywhere tangent to the vorticity field : these are obtained by solving the following equation at each time :

$$\frac{d\vec{r}(\epsilon, t)}{d\epsilon} \propto \vec{\omega}(\vec{r}(\epsilon, t), t)$$

Do not confuses these with current lines : in a 2D flow, current lines are in xy plane but vortex lines are straight lines directed by $\vec{e}_z$ (see e.g. paragraph 1.9.1).

Helmholtz theorem states that for an ideal incompressible flow, vortex lines moves as if 'frozen' in the fluid. A good example of this theorem is given by smoke rings which materialize a vortex tube. these rings move with the fluid, see picture

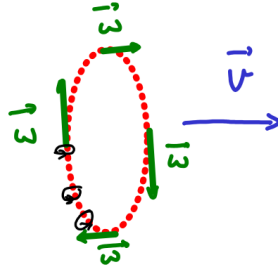
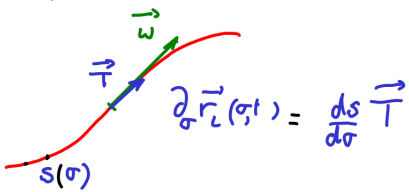


Figure 1.12: black : motion of particles inside the ring, blue : ensemble motion of the ring. The vortex lines move with velocity $\vec{v}$.

Proof : consider a vortex line at time t given in parametric form by $\vec{r}_L(\sigma, t)$, σ being the material index of particles along the line. One has

$$\partial_\sigma \vec{r}_L(\sigma, t) \wedge \vec{\omega}(\vec{r}(\sigma, t), t) = 0$$

(1.29)

This condition has to be valid at any later time t , expressing that the same material particles stay on the vortex line. So the proof is complete if we can show that

$$\frac{\partial}{\partial t} [\partial_\sigma \vec{r}_L(\sigma, t) \wedge \vec{\omega}(\vec{r}(\sigma, t), t)] = 0$$

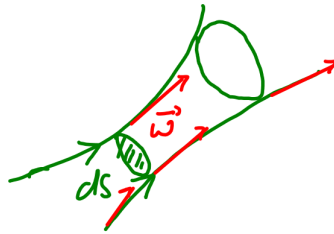
We develop this expression using Leibnitz rule and commuting the derivatives to obtain

$$\partial_\sigma \vec{v}_L(\sigma, t) \wedge \vec{\omega} + \partial_\sigma \vec{r}_L(\sigma, t) \wedge (\vec{\omega} \cdot \vec{\nabla}) \vec{v}$$

where we have used the result of exercise 1.22. Rearranging the last term as $\frac{ds}{d\sigma} \vec{T} \wedge \omega \frac{\partial}{\partial s} \vec{v}_L(\sigma, t)$ and using once again the variable change $s(\sigma)$ and $\vec{\omega} = \omega \vec{T}$ yields 0 immediately ! **QED.**

Defining a vortex tube as formed by vortex lines (see figure) and the strength of the vortex tube as :

$$\Gamma = \iint_S \vec{\omega} \cdot \vec{n} d^2S$$
(1.30)



Γ does not depend on the chosen section (Stokes theorem) and is constant in time (Kelvin theorem), hence it is a constant of the motion of a vortex tube. As stated before all these conclusions are valid wherever viscosity can be neglected. It is now amply time to introduce these forces, and we do so in the next chapter.

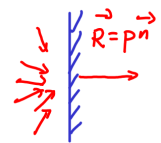
Chapter 2

Navier Stokes Equation

2.1 Introduction

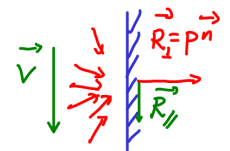
2.1.1 Newtonian Fluid

Euler's equation was established as a balance of forces taking into account gravity and internal pressure constraints. In this balance viscosity forces were neglected. Their physical origin is of the same nature as pressure forces. We recall that for an ideal gas, elementary kinetic theory shows that pressure is given by $p = 1/3 m n u^2$, with m the particles mass, n the number of particles per unit volume, and u the mean quadratic velocity, and that the resulting force is normal to the wall due to the isotropy of the velocity distribution.



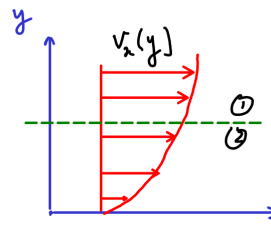
If the fluid has a mean relative velocity $\vec{V}$ parallel to the wall, the associated momentum transfer produces a force with a component parallel to the wall, the viscous force.

In the *Newtonian fluid model* these forces have a simple expression as a gradient of velocity. This model describes water and air correctly, but fails for *complex fluids* that are mid-way between liquids and solids, and in which constraints are much more complicated, including non-locality, non-linearity, retardation and memory ... See M. Nobili M2 Lectures.



For a Newtonian fluid, that will be considered exclusively in the sequel of these lectures, we have in the situation described here

$$\vec{F}_{1 \rightarrow 2} = \eta \frac{\partial v_x}{\partial y} S \vec{e}_x$$



(2.1)

where η is called the *dynamic viscosity* of the fluid, that we will assume constant for a given fluid. Actually η depends on the temperature but we will neglect this in our lectures. η is expressed in *Pa.s* or *Poiseuille*. Another quantity that will be used in these lectures is the *kinematic viscosity* defined as $\nu = \eta/\rho$ and whose unit $m^2.s^{-1}$ is that of a diffusion coefficient. The dynamic viscosity gives access to the forces exerted on obstacles, while the kinematic viscosity is related to how the liquid flows, our main concern in this course. The values for

fluids of interest are given below in C.G.S. units (to convert in S.I. units, just multiply by 10^{-4}). Note that for what concerns the flow pattern, air is more viscous than water !

fluid	value of ν ($cm^2 s^{-1}$)
Water	0.01
Air	0.15
Olive Oil	1
Glycerin	18

2.1.2 The role of viscosity : Boundary Layers

As we said in many occurrences we have neglected viscous forces. These will be important wherever the velocity gradient is large, and this will occur near obstacles. This region whose characteristics we will derive in the next chapter, is called a boundary layer (BL).

Outside this layer, whenever velocity gradients are small (to be precised quantitatively later) the Euler equation is a very good approximation of the laws of flow motion. However, an essential feature is that eddies, that are generated in the BL because of viscosity, can separate from obstacles, generating a turbulent flow far from obstacles quite different from what is predicted by the Euler equation. To understand this phenomenon qual-

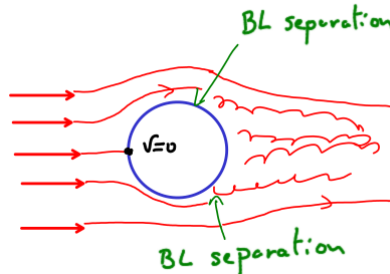
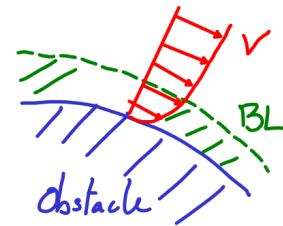


Figure 2.1: Boundary layer separation.

itatively we follow a fluid particle who impinges the obstacle near the stopping point. At this point pressure is maximum and then increases since v is increasing¹ until the maximal section of the obstacle. Then v decreases along the obstacle's narrowing side (mass flux conservation) and p increases. This adverse pressure gradient is responsible for separation. This effect can be mitigated by making this adverse pressure gradient as small as possible using streamlined profiles. For these, separation occurs near the drop edge and Euler description is valid almost everywhere. This explains why airfoil theory is based almost exclusively on the Euler equation.

¹To justify the use of Bernoulli theorem in this context, you need to wait until next chapter !



Figure 2.2: Profiled obstacle.

2.2 The Navier Stokes Equation

2.2.1 Viscous stress tensor

Consider again the Euler equation written as a momentum balance, equation(1.11) :

$$\partial_t \rho v_i + \partial_j \Pi_{ij} = \rho g_i \quad \Pi_{ij} = \rho v_i v_j + p \delta_{ij} \quad (2.2)$$

In order to take into account viscous forces, while respecting symmetries (rotational invariance notably) and using fluid incompressibility, one can show that it is necessary to add to Π_{ij} an additional term :

$$\Pi_{ij} = \rho v_i v_j + p \delta_{ij} - \eta \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right) \quad (2.3)$$

2.2.2 Navier Stokes (NS) Equation

Using the same computation as in chapter 1 we obtain the NS equation :

$$\partial_t \vec{v} + (\vec{v} \cdot \vec{\nabla}) \vec{v} = -\frac{1}{\rho} \vec{\nabla} p + \vec{g} + \nu \Delta \vec{v} \quad (2.4)$$

completed of course by the incompressibility condition $\vec{\nabla} \cdot \vec{v} = 0$

Exercise 2.1. Do this computation to get NS !

Exercise 2.2. One should realize what is hidden behind notations : write in detail all the terms of x component of NS. Do the same for y and z .

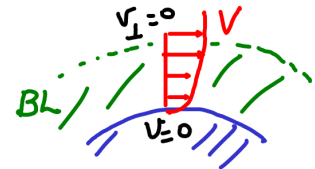
2.2.3 NS Boundary conditions.

The NS equation is second order because of the Laplacian, where Euler was first order. One therefore should impose stronger BC to completely specify a solution This condition turns out to be

$$\vec{v}|_{\text{boundary}} = 0 \quad (2.5)$$

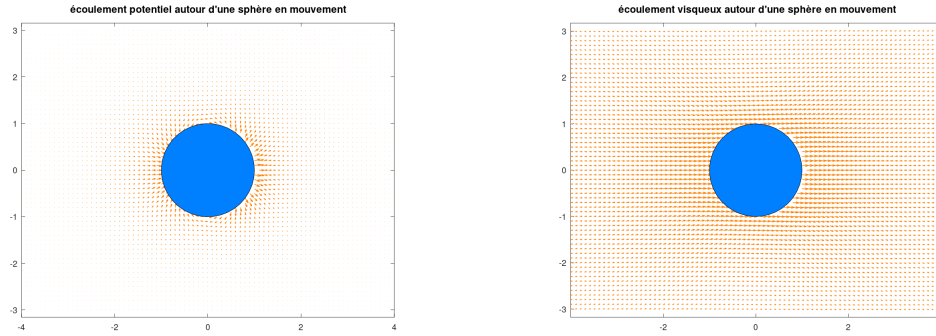
for fixed boundaries, and more generally $\vec{v}$ should be continuous at the interface between two media. One henceforth understand that Euler's equation describes what happens outside the boundary layer, and can not properly adhere to the obstacle, while NS do so nicely.

Why in that case do we still insist to use Euler equation ? Because of it's relative simplicity, the many associated theorems (Bernoulli, vorticity conservation, potential motion ...) and the fact that it very often provides an excellent description of what is observed, as in the case of water waves that we will study in chapter 4.



2.3 Reynolds Number

Let us give a quantitative criterium for the validity of Euler versus Navier-Stokes equations, whose flows are qualitatively very different as can be seen in figure 2.3. For that, let us com-



(a) Potential flow through a moving sphere. (b) Viscous flow through a moving sphere.

Figure 2.3: Euler versus NS flow past a sphere

pare in NS the order of magnitude of the 'inertial term' $(\vec{v} \cdot \vec{\nabla})\vec{v}$ compared to the viscous one $\nu \Delta \vec{v}$. If we note U the typical order of magnitude of the velocity in the flow, and L the typical value over which this velocity varies (usually the obstacle size) we can write

$$\frac{\text{Inertial forces}}{\text{viscous forces}} = \frac{|(\vec{v} \cdot \vec{\nabla})\vec{v}|}{|\nu \Delta \vec{v}|} \sim \frac{U^2/L}{\nu U/L^2} = \frac{UL}{\nu}$$

We then define the Reynolds number as

$$\mathcal{Re} = \frac{UL}{\nu} \quad (2.6)$$

and we then have the different cases :

- $\mathcal{Re} \gg 1$ this is Euler regime, except very near the obstacle and in the wake .
- $\mathcal{Re} \ll 1$ this is the very viscous case of so called 'creeping flow' where one can neglect the inertial term $(\vec{v} \cdot \vec{\nabla})\vec{v}$. The flow equations then becomes linear, and have a lot of simplifying properties related to this regime.
- $\mathcal{Re} \sim 1$ this is the complicated case where both inertia and viscosity compete, and only few exact solutions are known. We will present some of these in this chapter.

Note that we can have $\mathcal{Re} \gg 1$ in some part of the fluid and $\mathcal{Re} \sim 1$ in others, depending of your local choice of characteristic velocity and length scale. As an example, the boundary layer will be defined as the region where the local Reynolds number is ~ 1 .

Exercice 2.3. Estimate the Reynolds number $\mathcal{Re}$ in the following situations

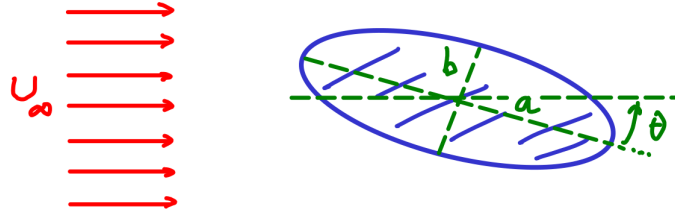
- you are swimming in a pool;
- you are walking (in air !);
- you are driving a car;

- you are flying a plane;
- you are sailing a yacht;
- you look at water pouring out from a water tap;
- you get an injection of a water serum;
- you are a paramecia swimming in water;
- you are a bacteria crawling in water.

From this exercise you see that Re spans many order of magnitude, and that Euler equation will be adequate for a great majority of macroscopic flows.

2.3.1 Adimensionalization of N.S.

Consider the following flow : It is often very useful to describe the problem in an dimensionless



form, for example before attempting a numerical resolution. Let us explain how this is done. The flow is described by

$$\partial_t \vec{v} + (\vec{v} \cdot \vec{\nabla}) \vec{v} = -\frac{1}{\rho} \vec{\nabla} p + \nu \Delta \vec{v} \quad , \quad \vec{\nabla} \cdot \vec{v} = 0 \quad , \quad \vec{v}|_{boundary} = 0$$

Let us introduce a typical length scale of the problem, e.g. a or b or the cross section $2a \sin \theta$, that we will denote by L . All the geometry of the problem (the figure above) can be stated using L and dimensionless quantities, here the angle θ and the aspect ratio b/a . Let us also introduce a typical velocity U , in the present case this could be U_∞ , and a typical pressure p_0 (e.g. pressure at infinity or at the stopping point). Note that this choice is not unique, and depends partly of your skill, experience, and available data. Let us then define

$$\vec{v}' \equiv \frac{\vec{v}}{U} \quad , \quad \vec{r}' \equiv \frac{\vec{r}}{L} \quad , \quad t' \equiv \frac{tU}{L}$$

and replace in the above equations. After defining $p' \equiv \frac{p-p_0}{\rho U^2}$ we obtain the dimensionless equations

$$\left(\partial_{t'} + (\vec{v}' \cdot \vec{\nabla}') \right) \vec{v}' = -\vec{\nabla}' p' + \frac{1}{Re} \Delta' \vec{v}' \quad , \quad \vec{\nabla}' \cdot \vec{v}' = 0 \quad , \quad \vec{v}'|_{boundary'} = 0$$

As a conclusion we see that the flow is not described by five parameters p_0 , L , ν , ρ , U but only one Re plus the geometrical (dimensionless) features. This fact has the following practical consequences :

- for simulations and experiments this is a time saver : you don't have to replicate your computation with many parameters, only look for what happens at different values of Re .
- two flows with the same (a-dim) geometry and same Re are said *similar*. They are homothetic and share the same flow pattern (if one is turbulent, the other is also).
- You have the possibility to build models at reduced scales for boats, planes, cars... and this has been done a lot in the past, before the era of numerical simulations.
- Forces on structures can be expressed as $\vec{F} = \rho U^2 L^2 \vec{f}(Re, \text{geometry})$ and we can define various empirical dimensionless coefficients such as $C_x = F_x / \rho U^2 L^2$

One sees that formally we have

$$\begin{cases} N.S. \longrightarrow \text{Euler} \\ Re \longrightarrow \infty \end{cases} \quad (2.7)$$

however this limit is *singular* since the neglected term is that of higher derivation order. As a consequence solutions of the first equation does not provide a straightforward limit to those of the second. This is shown in the following exercise :

Exercise 2.4. Solve the ODE $\epsilon u''(x) + u'(x) = 1$, $u(0) = 0$, $u(1) = 2$ on the interval $0 \leq x \leq 1$. Plot this solution for $\epsilon = 0.01$. Examine then the limit $\epsilon \rightarrow 0$. Can you solve this with the same boundary conditions ? Why ? Which one of the two BC has to be discarded to obtain a solution close to the solution of the initial problem?

The value of Re will therefore distinguish two otherwise geometrically identical flows. It was observed by Reynolds (1883) that for values $Re \gtrsim 2000 - 3000$, flows are unstable with respect to small perturbations and become turbulent. We will devote our last two chapters to these questions, and for now we turn to classical viscous flows, in order to get better intuition of their physics.

2.4 Navier-Stokes Classical flows

2.4.1 Planar Couette flow

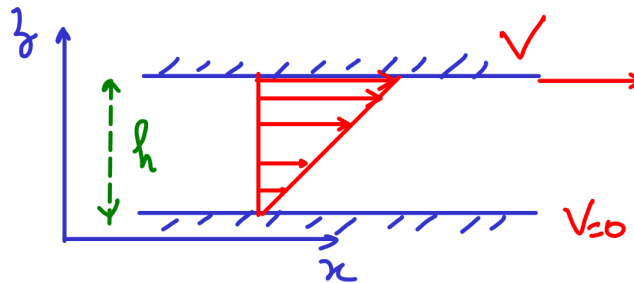


Figure 2.4: Planar Couette flow

In this flow the fluid lays between two parallel plates, one of them being fixed, and the other moving at a constant velocity V in the plane direction. We call this direction x , the other

one being z as in figure 2.4. There is no forcing of the fluid, meaning that pressure p is uniform. We first look for a stationary solution. Because of the symmetries we assume a velocity flow along x and depending only on z : $\vec{v} = (u(z), 0, 0)$, where the function $u(z)$ is still unknown. We check that incompressibility is immediately verified $\vec{\nabla} \cdot \vec{v} = 0$ and x NS component yields

$$0 = -\frac{1}{\rho} \partial_x p + \nu \partial_z^2 u$$

Since there is no forcing, $\partial_x p = 0$ and we obtain $u = az + b$ where the constants a and b must be chosen to fulfill the boundary conditions $u(z = 0) = 0$, and $u(z = h) = V$. We henceforth obtain

$$u = \frac{V}{h} z$$

Exercise 2.5. Check that the fluid exerts on the lower plate a viscous constraint $\vec{F} = \eta \frac{V}{h} \vec{e}_x$. What is the force exerted on the upper plate ?

2.4.2 Non Stationary Couette flow

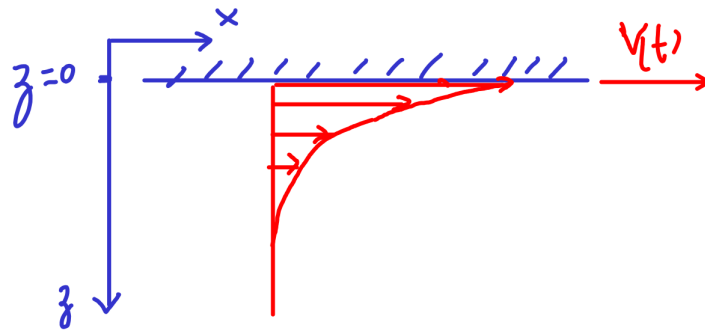


Figure 2.5: Non stationary Couette flow

Let us for a moment deviate from the usual routine, in order to better understand the physics of viscosity. In the figure 2.5 the fluid is initially at rest, and is carried by a plane moving at a constant velocity V along its direction. We look for a solution in the form $\vec{v} = (u(z, t), 0, 0)$, which leads to

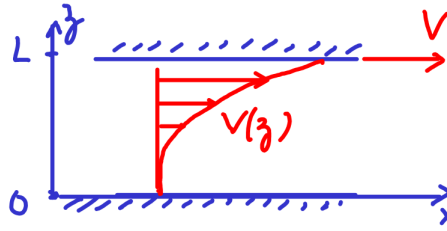
$$\begin{cases} \partial_t u &= \nu \partial_z^2 u \\ u(z > 0, t = 0) &= 0 \\ u(z = 0, t > 0) &= V \end{cases} \quad (2.8)$$

We recognize the diffusion equation with Dirichlet BC, and the classical thermal shock problem or diffusion problem with a stationary source. The solution is well known, see the Appendix E on the diffusion equation.

$$u(z, t) = V \left[1 - \frac{2}{\sqrt{\pi}} \int_0^{\frac{z}{\sqrt{4\nu t}}} e^{-s^2} ds \right] = V \operatorname{Erfc} \left(\frac{z}{\sqrt{4\nu t}} \right) \quad (2.9)$$

One sees the physical interpretation of this situation : the tangential momentum ρv_x diffuses in the z direction in the medium with a diffusion coefficient ν . After a time t has elapsed, one has a layer of thickness $l \propto \sqrt{\nu t}$ that has been set in motion. We will use this fundamental result in the next chapter.

Exercise 2.6. Using results in Appendix E, solve the same problem between two planes. Initially fluid is at rest. You can plot an animated GIF of your solution with Python.



2.4.3 Inclined plane flow

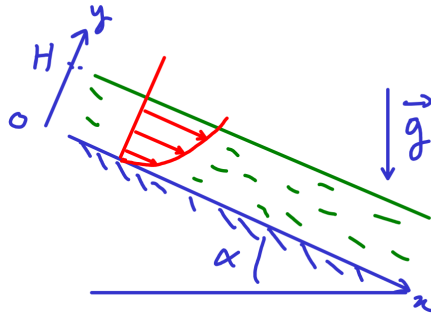


Figure 2.6: Inclined plane flow

This is our first 'free surface' flow. We consider the stationary flow of a liquid on an inclined plane with a surmounting pressure p_0 . We note that the inclination should not be too large otherwise the flow will be non stationary and turbulent. We will come back to this point at the end of this section. Let us find the velocity profile $(u(y), v(y), 0)$. Using incompressibility we obtain $\partial_y v = 0$ and then $v = 0$ using boundary conditions on the plane. We see that there is no acceleration in the flow, and NS write

$$\begin{cases} 0 &= -\frac{1}{\rho}\partial_x p + \nu \partial_y^2 u + g \sin \alpha \\ 0 &= -\frac{1}{\rho}\partial_y p - g \cos \alpha \end{cases} \quad (2.10)$$

We now state the free surface condition which is that the constraints are continuous there². We can then integrate the second equation using $p(y = H) = p_0$ as

$$p = p_0 - \rho g \cos \alpha (y - H)$$

from that we have $\partial_x p = 0$ and the first equation simply writes $\nu u''(y) = -g \sin \alpha$. We use the BC $u(y = 0) = 0$ and $u'(y = H) = 0$, the last condition because there is no constraint in the x direction above the fluid, so the viscous constraint has to vanish at the surface. We finally obtain a parabolic profile

$$u = \frac{g \sin \alpha}{\nu} y(H - y/2)$$

²More about the free surface BC in the next chapter 4 on water waves.

and from that we compute the volume flow

$$Q = \int_0^H u dy = \frac{gH^3}{3\nu} \sin \alpha \quad \text{m}^2/\text{s}$$

We now come to the condition for which this flow is stable. This is given by the condition $Re \lesssim 1000$ with here $U = u(H)$ and $L = H$. So we obtain

$$Re \sim \frac{u(H)H}{\nu} = \frac{gH^3}{2\nu^2} \sin \alpha = \frac{3Q}{2\nu} \lesssim 1000$$

This condition shows that α and H have to be small, and the fluid has to be viscous enough, so this might be ok for honey but not for water !

2.5 Forced flow and Poiseuille Formula

2.5.1 Planar Poiseuille flow

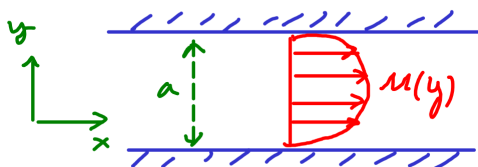


Figure 2.7: Planar Poiseuille flow

We consider here a stationary laminar flow in presence of a pressure gradient in the x direction³, see figure 2.7. Symmetry suggest an Ansatz $\vec{v} = (u(x, y), 0, 0)$ and incompressibility shows that $u(y)$. Reporting in NS gives

$$\begin{cases} \partial_y p &= 0 \\ -\frac{1}{\rho} \partial_x p + \nu \partial_y^2 u &= 0 \end{cases} \implies p(x) \quad (2.11)$$

and each of these terms has to be a constant. We denote $K = -\partial_x p > 0$ the pressure drop by unit length, and use the BC $u(0) = u(a) = 0$, to find a parabolic profile

$$u = \frac{K}{2\eta} y(a - y)$$

Once again we compute the volumetric flow (per unit of width z) as

$$Q = \int_0^a u(y) dy = \frac{Ka^3}{12\eta} \quad \text{m}^2/\text{s}$$

Finally the constraint on the wall is given by

$$\sigma = \eta \left. \frac{\partial v}{\partial y} \right|_{y=0} = \frac{Ka}{2}$$

This result is a bit counter-intuitive : it is independent on viscosity and proportional to the width of the channel, why ?

³Stability conditions for this flow will be examined in chapter 5

2.5.2 Cylindrical Poiseuille flow

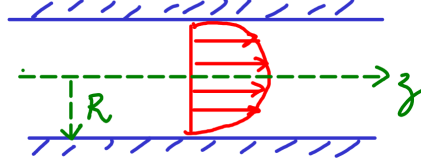


Figure 2.8: Poiseuille flow

This case is very similar to the previous one, but in a cylindrical geometry. Poiseuille was a doctor who was interested in the flow of blood in veins. He derived his formula experimentally. We have here $\vec{v} = (0, 0, u(r))$ in cylindrical coordinates (r, θ, z) ,

$$\begin{cases} \partial_r p &= 0 \\ \partial_z p &= \eta \frac{1}{r} \partial_r (r \partial_r u) = K \\ u(R) &= 0 \end{cases} \implies p(z) \quad (2.12)$$

whence

$$u(r) = \frac{K}{4\eta} (R^2 - r^2)$$

The volumetric flow is

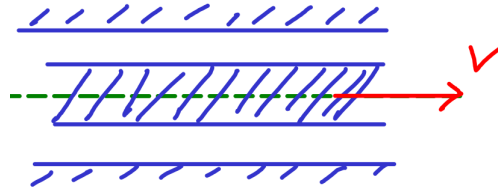
$$Q = 2\pi \int_0^R u(r) r dr = \pi R^2 u(0)/2 = \frac{\pi K R^4}{8\eta} \quad \text{m}^3/\text{s}$$

This is the Poiseuille formula, and this may be used to measure viscosity. We see that the volumetric flow is proportional to the pressure drop, in contradistinction with the Clepsydra example in chapter 1 where it was proportional to $\sqrt{gh}$. Here the flow is governed by viscosity (low Reynolds), while in the clepsydra case it was inertia dominated (high Reynolds). Finally the constraint on the wall is given by

$$\sigma_{rz} = \frac{1}{2} \eta \left. \frac{\partial u}{\partial r} \right|_{r=R} = \frac{KR}{8}$$

Exercise 2.7. Solve for the forced flow in a ring section pipe : $R_1 < r < R_2$.

Exercise 2.8. Solve for the flow in a ring section pipe : $R_1 < r < R_2$ with the inner wall moving at velocity V . In both exercises compute the constraint on walls.



2.6 Rotating flows

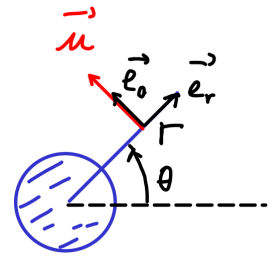
2.6.1 Orthoradial solutions

You may use the formula given in Appendix A of these lectures, related to cylindrical coordinates. We consider plane cylindrical flows :

$$\vec{u} = u(r, t) \vec{e}_\theta$$

One can immediately check that $\vec{\nabla} \cdot \vec{u} = 0$ and one is left with

$$\begin{cases} \frac{u_\theta^2}{r} = \frac{1}{\rho} \partial_r p \\ \partial_t u_\theta = \nu \left(\partial_r^2 u_\theta + \frac{1}{r} \partial_r u_\theta - \frac{u_\theta}{r^2} \right) \end{cases} \quad (2.13)$$



The first of these equations is the barometric cylindrical equation that we already met in the Rankine Vortex section, see chapter 1. It enables to compute p once the velocity flow is determined. The second one is linear, and expresses the diffusion equation in radial coordinates. We look for solutions in the form $u_\theta \propto r^\alpha$ and this gives the condition $\alpha(\alpha - 1) + (\alpha - 1) = 0$ so $\alpha = \pm 1$. Finally all the stationary solutions write

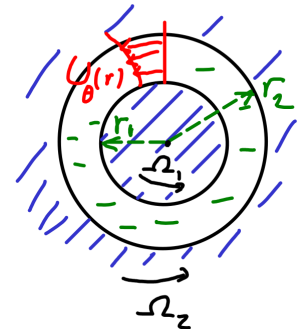
$$u_\theta(r) = Ar + \frac{B}{r} \quad (2.14)$$

with A and B some constants.

2.6.2 Taylor-Couette flow

A standard configuration, both theoretically and experimentally, is the flow between two rotating cylinders. This has been used to model the flow of atmosphere in a rotating Earth. The limiting case $\Omega_2 = 0$ correspond to the experimental set-up of a rheometer in the study of complex fluids. In order to satisfy no-slip condition one should have

$$\begin{cases} Ar_1 + \frac{B}{r_1} = \Omega_1 r_1 \\ Ar_2 + \frac{B}{r_2} = \Omega_2 r_2 \end{cases} \Rightarrow \begin{cases} A = \frac{\Omega_2 r_2^2 - \Omega_1 r_1^2}{r_2^2 - r_1^2} \\ B = (\Omega_1 - \Omega_2) \frac{r_2^2 r_1^2}{r_2^2 - r_1^2} \end{cases} \quad (2.15)$$



Exercise 2.9. Check these results in detail.

We will see in chapter 5 that this solution is unstable if Ω_1 is too large in comparison with Ω_2 . In particular the situation where $\Omega_1 > 0$ while $\Omega_2 = 0$ is always unstable since the fluid is subjected to a centrifugal force, with tendency to eject it. This gives a hint as to why atmospheric flow is unstable.

2.6.3 Non-stationary orthoradial flow - Vortex decay

We suppose at initial time $t = 0$ a rotational flow given by

$$\vec{u}(r, t = 0) = \frac{\Gamma_0}{2\pi r} \vec{e}_\theta$$

Note that this solution is nonphysical at $r = 0$ since shear becomes infinite, but it is a correct description at $r \neq 0$. One then has to solve

$$\begin{cases} \partial_t u_\theta = \nu \left(\partial_r^2 u_\theta + \frac{1}{r} \partial_r u_\theta - \frac{u_\theta}{r^2} \right) \\ u_\theta(r \rightarrow \infty) = 0 \\ u_\theta(r, t = 0) = \frac{\Gamma_0}{2\pi r} \end{cases} \quad (2.16)$$

This equation can be solved by separation of variables and Fourier transform, but there is a neat trick in the present case : we define the circulation on a circle of radius r as $\Gamma(r, t) \equiv 2\pi r u_\theta(r, t)$ and find that it satisfies

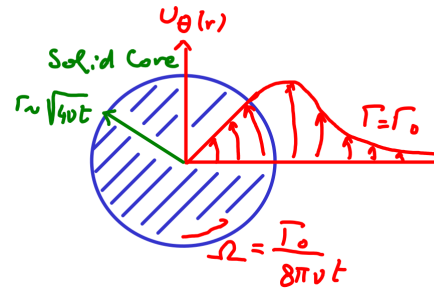
$$\begin{cases} \partial_t \Gamma &= \nu \left(\partial_r^2 \Gamma - \frac{1}{r} \partial_r \Gamma \right) \\ \Gamma(r, t=0) &= \Gamma_0 \end{cases} \quad (2.17)$$

with the regularity condition $\Gamma(r=0, t>0) = 0$ to ensure the regularity of the flow at $r=0$. In the spirit of the diffusion equation, we look for a self similar solution in the form $\Gamma(r/\sqrt{\nu t})$. Replacing and using the initial condition, we finally obtain

$$\Gamma(r, t) = \Gamma_0 \left(1 - e^{-\frac{r^2}{4\nu t}} \right)$$

- for $r \gg \sqrt{4\nu t}$ we have the outside vortex unchanged, since viscosity has not diffused outside where gradients are small.
- for $r \ll \sqrt{4\nu t}$, the solution $u_\theta = \Gamma_0 \frac{r}{8\pi\nu t}$ describes a solid rotation $\propto r$ which is decaying with an angular velocity $\Omega(t) \propto 1/t$.

One finally sees that the vortex decay is slow, since it proceeds by diffusion of the orthoradial momentum u_θ : to have an effective damping at a radius R one has to wait a time $\tau \propto R^2/\nu$.



Exercise 2.10. Check the above computation in details.

Exercise 2.11. For $R = 10$ cm compute the time it takes to damp the vortex. This is surprising and does not correspond to what you may observe in a mug of tea ! This result suggests that there exist other mechanisms of damping, as we will see in the next chapter.

2.7 2-dimensional flow and vorticity equation

Until now all the problems solved were actually one dimensional, and this enables to solve stationary flows using ordinary differential equations (ODE). In this section we consider what can be solved assuming a 2D flow (and overlooking stability considerations). For 2D flows, the generic solution $\vec{u} = (u(x, y), v(x, y), 0)$ and the incompressibility condition $\vec{\nabla} \cdot \vec{u} = 0$ imply the existence of a vector potential $\vec{A} = \psi(x, y)\vec{e}_z$ perpendicular to the flow plane, so that we can write

$$u_x = \partial_y \psi \quad \text{and} \quad u_y = -\partial_x \psi$$

ψ is called the *stream function* because one has readily $\vec{u} \cdot \vec{\nabla} \psi = \partial_y \psi \partial_x \psi - \partial_x \psi \partial_y \psi = 0$. So ψ is constant in the direction of $\vec{u}$ and therefore also along streamlines. Reciprocally, if we know the function ψ , streamlines are obtained by setting $\psi = Cte$.

Let us now see how to use this in NS equation : in order to get rid of pressure, we use the vorticity $\vec{\omega} \equiv \vec{\nabla} \wedge \vec{u}$. Using algebraic manipulations already seen in chapter 1, section 1.8, one can show that

$$\frac{\partial \vec{\omega}}{\partial t} + (\vec{u} \cdot \vec{\nabla}) \vec{\omega} = (\vec{\omega} \cdot \vec{\nabla}) \vec{u} + \nu \Delta \vec{\omega} \quad (2.18)$$

The above *vorticity equation* (2.18) is valid in any dimension, however in the 2D case in the xy plane, $\vec{\omega} = \omega \vec{e}_z$ and $(\vec{\omega} \cdot \vec{\nabla})\vec{u} = 0$. In that case we obtain the convection-diffusion equation :

$$\frac{D\vec{\omega}}{Dt} = \nu \Delta \vec{\omega}$$

moreover we have immediately $\omega = \partial_x u_y - \partial_y u_x = -\Delta \psi$ and the vorticity equation can be recast as the 4-th order ODE for one single function $\psi(x, y)$

$$\partial_t \Delta \psi + \partial_y \psi \partial_x \Delta \psi - \partial_x \psi \partial_y \Delta \psi = \nu \Delta \Delta \psi$$

Finally in all the previous solvable examples we had $(\vec{u} \cdot \vec{\nabla}) = 0$ because of a simplifying geometry. In such case, ω satisfies a simple diffusion equation which makes the problem tractable.

2.8 Creeping flows

Creeping flows correspond to a very low Reynolds number regime $Re \ll 1$. They are everywhere in biology and microfluidics (see examples in section 2.3 of this chapter). At our scale this will be swimming in heavy mud or tar. In this case, the convective term in NS drops, all equations are linear and we end up with a diffusion equation for vorticity

$$\frac{\partial \vec{\omega}}{\partial t} = \nu \Delta \vec{\omega}$$

However this is not solvable directly because :

- the boundary conditions for $\vec{\omega}$ are not known !!
- we have not used the essential information that $\vec{\nabla} \cdot \vec{u} = 0$, so that the above equation should be seen as a constraint not sufficient to solve for the flow.

However dropping the inertial terms has the consequence that a linear combination of solution flows is also a solution of NS, and we may combine fields as in Electromagnetism in vacuum. This fundamental and counter-intuitive property provides many interesting results for the study of such flows.

In the 2D stationary case, we can use the ψ function introduced in the previous section to state the fluid equation in the following form

$$\Delta \Delta \psi = 0$$

with BC given in term of $u_x = \partial_y \psi$ and $v_y = -\partial_x \psi$. We will see examples of this approach in many subsequent problems accompanying this course, and in particular a variant for axisymmetric flows such as the famous Stokes flow past a sphere (see appendix F).

We end here our tour of some classic simple solutions of the Navier Stokes equation, and turn now to the articulation between Euler and Navier Stokes equations : where and when do we use which one ?

Chapter 3

Boundary Layer

3.1 Introduction

3.1.1 Context

We already stated that the relation between Euler and Navier Stokes equations was a subtle one, and we now examine this point in details. We saw that Reynolds number gives an estimate of the ratio

$$\mathcal{Re} = \frac{\text{Inertial Forces}}{\text{Viscous Forces}} = \frac{UL}{\nu} \quad (3.1)$$

Hence whenever $\mathcal{Re}$ is large we can neglect the effects of viscosity¹. Actually this result is based on the estimation $|\nabla u| \sim U/L$ where u is a typical velocity and L the typical length over which U changes appreciably. We took for L the typical object size, but other typical length scales can show up in a flow, for instance

- in turbulence it might be the eddies size
- near obstacles velocity has to change over very small distances to accommodate the boundary condition on the wall.

This last situation is what interest us here. The region where velocity is rapidly varying is called the *Boundary Layer*, (BL) and its theoretical elucidation was given by Prandtl in 1905.

3.1.2 Size of the Boundary Layer

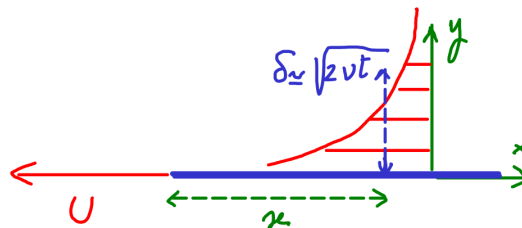


Figure 3.1: A plate driving the fluid

¹Remember that values $\mathcal{Re} \sim 10^4$ are commonplace in macroscopic flows.

The thickness of the BL can be estimated by very simple arguments : let us consider once again the situation shown in figure 2.5, but with the moving plate having a leading edge as in figure 3.1. We saw that the fluid far from the plate is at rest, but for a region of thickness $\delta \simeq \sqrt{2\nu t}$ where t is the time during which the fluid has been carried. For a position at a distance x from the leading edge, this time is $t = x/U$ and we obtain the boundary layer thickness as

$$\delta(x) \simeq \sqrt{\frac{2\nu x}{U}} = \sqrt{\frac{2xL}{Re}} \lesssim \frac{L}{\sqrt{Re}} \quad (3.2)$$

the last result giving a quick way to estimate the maximal width of the BL. In the plate reference frame, the picture is shown in figure 3.2. The result obtained in equation (3.2) shows that BL

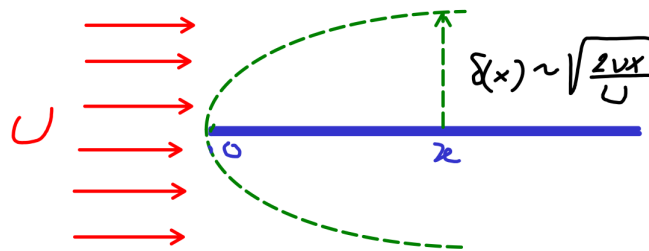


Figure 3.2: BL profile around a moving plate

shrinks as Reynolds number grows. Outside from the BL, Euler equation is valid, but the flow may be turbulent because of the BL separation, a concept that we are going to explore in the following.

Boundary Layer Separation

In case that the obstacle profile forces the flow line to converge, velocity increases because of the fluid incompressibility, and pressure decreases because of Bernoulli theorem. The flow does not meet an adverse pressure gradient and stay close to the obstacle.

On the contrary, in a region where the flow lines are diverging, velocity decreases, and pressure increases. The flow in the BL meets an adverse pressure gradient that will revert its direction, and lead to separation of the flow.

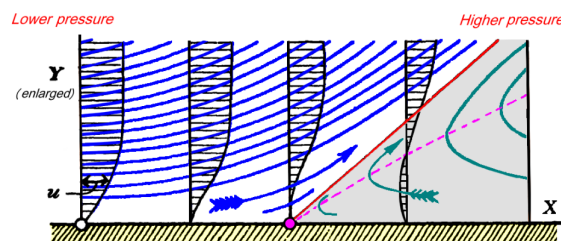


Figure 3.3: BL separation mechanism

In that last case, the flow that is rotational in the BL, will be ejected in the bulk of the flow and will form a *turbulent wake*. It is now evident why the real flow may look totally different from an ideal Euler flow in the wake. Note also that it would be a great achievement to be able to compute the exact position of the separation in a given geometry flow, but this has not been possible up to now even in the simplest configurations.

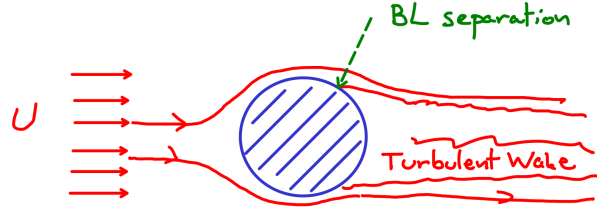


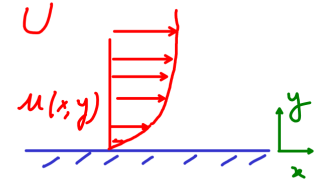
Figure 3.4: The forming of wake after BL separation

3.2 The planar Boundary Layer

3.2.1 Orders of magnitude analysis

We start again from the 2D stationary Navier Stokes equations

$$\begin{cases} u\partial_x u + v\partial_y u &= -\frac{1}{\rho}\partial_x p + \nu(\partial_x^2 u + \partial_y^2 u) \\ u\partial_x v + v\partial_y v &= -\frac{1}{\rho}\partial_y p + \nu(\partial_x^2 v + \partial_y^2 v) \\ \partial_x u + \partial_y v &= 0, \quad \text{and } u = v = 0 \text{ for } y = 0 \end{cases}$$



(3.3)

We denote by L the typical length in the x direction and by $\delta(x)$ the thickness of the BL. We have seen that $\delta/L \sim 1/\sqrt{Re}$, and we will assume that this quantity is small, the condition for this analysis to be valid. We therefore have

- $\partial_x u \quad (\sim U/L) \ll \partial_y u \quad (\sim U/\delta)$
- $\partial_x^2 u \quad (\sim U/L^2) \ll \partial_y^2 u \quad (\sim U/\delta^2)$
- Because of incompressibility $U/L \sim v/\delta$ and so $v \sim \delta U/L \ll u$
- finally comparing the x and y components of NS equations, we have $\partial_y p \sim (\delta/L) \partial_x p$

and because of this last result $p(x)$ does not depend on y in first approximation. Pressure is therefore the same in the boundary layer and in the corresponding Eulerian bulk flow.

$$p(x) + \frac{1}{2}\rho U^2 = Cte$$

It is therefore possible to use Bernoulli equation *outside the BL* to compute $p(x)$, and we indeed have used this property many times before.

At the end of the game, using all the aforementioned estimates we have the Boundary Layer Prandtl equations

$$\begin{cases} u\partial_x u + v\partial_y u &= -\frac{1}{\rho}p'(x) + \nu\partial_y^2 u \\ \partial_x u + \partial_y v &= 0 \\ u = v = 0 &\text{at } y = 0 \end{cases} \quad \text{and } u \xrightarrow{y \gg \delta} U \quad (3.4)$$

One can indeed check that all the remaining terms in the equations are of the same order of magnitude. As an example, the first and last terms of the first line above writes $U^2/L \sim \nu U/\delta^2$, that can be rewritten as $\delta^2/L^2 \sim 1/Re$, a relation that we have shown before. Note also that, for this theory to be valid, Re has to be large but not too large, otherwise the flow will become turbulent and non-stationary.

3.2.2 Planar Boundary Layer velocity profile

We here continue the analysis, supposing an infinite plane. In that case, there is invariance in the x direction and $\partial_x p = 0$. We look for a self similar solution using the change of variable

$$\xi = \frac{y}{\delta(x)} \quad \text{with} \quad \delta(x) \equiv \sqrt{\frac{2\nu x}{U}}$$

After some try and error² we try a velocity profile using a stream function

$$\psi = U \delta(x) f(\xi) \quad u = \partial_y \psi \quad v = -\partial_x \psi$$

Incompressibility is therefore satisfied by this ansatz. Using $\partial_y \xi = 1/\delta(x)$ and $\partial_x \xi = -\xi/2x$ we have

$$\begin{aligned} u &= U f'(\xi) & \partial_x u &= -\frac{\xi}{2x} U f'' & \partial_y u &= \frac{U}{\delta(x)} f'' & \nu \partial_y^2 u &= \frac{\nu U}{\delta^2} f''' = \frac{U^2}{2x} f''' \\ v &= \frac{U \delta(x)}{2x} (\xi f' - f) \end{aligned} \quad (3.5)$$

Using all these results, Prandtl equation miraculously simplifies and we end up with the following equation

$$\begin{aligned} f''' + f f'' &= 0 \\ f'(0) = f(0) &= 0, \quad f'(\xi) \xrightarrow{\xi \rightarrow \infty} 1 \end{aligned} \quad (3.6)$$

One can solve this equation numerically by a shooting method, that is by trying some values of $f''(0)$ until the condition at ∞ is satisfied. One finds $f''(0) = 0.46965$ and the following velocity profile, see figure 3.5.

One can then compute the stress component on the plane as $\sigma = \eta \partial_y u|_{y=0} = \eta U/\delta(x) f''(0)$ that finally yields

$$\sigma = 0.33 \frac{\rho U^2}{\sqrt{Ux/\nu}}$$

²and the input from Prandtl genius !

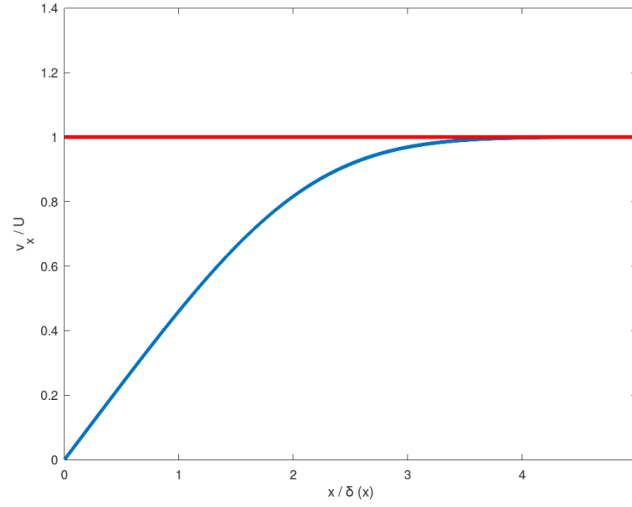


Figure 3.5: Plane BL velocity profile

This result can be analyzed as follows : ρU^2 is expected from dimensional analysis. We see that the stress decreases as the inverse of BL thickness δ , because shear decreases. Neglecting leading and trailing edge problems, we can compute the total force per unit length (in z direction) on the plate :

$$\Sigma = \int_0^L \sigma(x) dx = 0.66 \frac{\rho U^2 L}{\text{Re}^{1/2}}$$

Note that this result is doubled if the fluid is acting on both faces.

Exercise 3.1. Check the computation leading to equation 3.6 in details.

Exercise 3.2. Solve numerically this system and plot the profile of figure 3.5 by yourself.

Exercise 3.3. Check the result for the force Σ acting on the plane.

Let us use these results to determine the inlet zone in a pipe of diameter $2a$ as seen in section 2.5.2. Since mass is conserved, we have $Q = \pi a^2 U_e = \pi a^2 u(0)/2$ and velocity at the

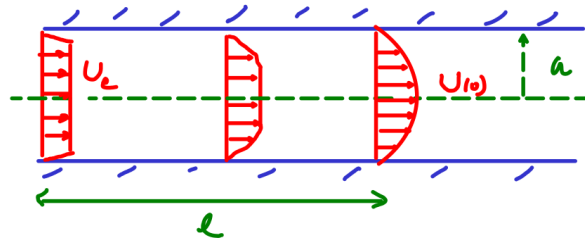


Figure 3.6: Flow at the inlet of a pipe

center of the parabolic profile is twice the inlet velocity. In the inlet region the flow evolves progressively from a flat profile to a parabolic one by growth of the BL until it reaches the pipe diameter $\delta(l) \sim a$, and therefore $l \sim U_e a^2 / 2\nu = a \text{Re} / 2$.

3.2.3 Boundary Layer at a stagnation point

Up to now the argument used to determine the BL thickness was based on a very simple geometry, which led to a diffusion equation for u_x , and actually for vorticity as we saw at the end of the previous chapter. This argument is invalid in case where $\vec{\omega}$ is convected by the flow as we now show in an explicit example. We consider the flow near a stagnation (or rest or stopping) point.

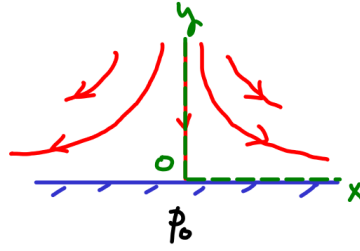


Figure 3.7: Flow near a stagnation point

We already saw the Eulerian solution $u = \alpha x$ and $v = -\alpha y$ leading to hyperbolic stream lines. Use of Bernoulli theorem gives $p = p_0 - 1/2\rho\alpha^2(x^2 + y^2)$ showing that p decreases along a streamline near the plane, so there is no BL separation here.

An exact Navier Stokes solution leads to a constant BL thickness $\delta = \sqrt{\nu/\alpha}$ as we now show : we use the variable $\xi = y/\delta$ and an ansatz stream function

$$\psi = \sqrt{\alpha\nu}xf(\xi) \quad u = \alpha xf'(\xi) \quad v = -\sqrt{\alpha\nu}f(\xi) \quad \partial_x v = 0 \quad \partial_x^2 v = 0 \quad \partial_x^2 u = 0$$

With these conditions, NS turns into

$$\begin{cases} \alpha^2 x(f''' + ff'' - f'^2) &= \frac{1}{\rho} \partial_x p \\ \alpha\sqrt{\alpha\nu}(ff' + f'') &= -\frac{1}{\rho}\sqrt{\frac{\alpha}{\nu}} \partial_\xi p \end{cases} \quad (3.7)$$

while the boundary conditions transcribe to

$$f'(0) = f(0) = 0 \quad f'(\xi) \xrightarrow{\xi \rightarrow \infty} 1$$

the last condition ensuring a proper convergence to the Euler flow at large y . Integrating the second equation with respect to the variable ξ we get

$$p = \rho\alpha\nu(g(x) - f^2/2 - f')$$

and inserting in the first one gives

$$\alpha x(f''' + ff'' - f'^2) = \nu g'(x)$$

for $\xi \rightarrow \infty$ this equation turns into $-\alpha x = \nu g'(x)$ and finally

$$p = p_0 - \frac{\rho\alpha^2}{2}x^2 - \rho\alpha\nu(f^2/2 + f')$$

We are left with the equation

$$f''' + ff'' - f'^2 + 1 = 0$$

together with the aforementioned BC. This is solved by a numerical shooting method as before, and produces the profile :

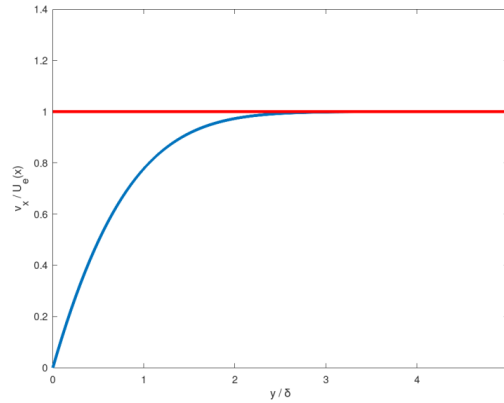
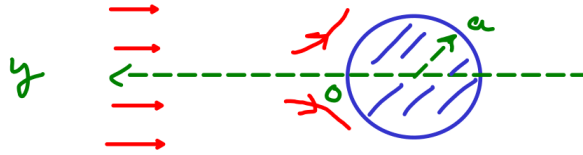


Figure 3.8: Velocity profile near a stagnation point

Exercise 3.4. Do by yourself the above computations in details, and do the numerical. You should find $f''(0) \simeq 1232588$.

Exercise 3.5. Find the constant α for the stagnation point on a sphere of radius $a = 2$ cm with a velocity at infinity $U_\infty = 1$ m/s. What is the thickness of the BL for water ?



You may use Euler solution on the y axis : $v_y = -U_\infty(1 - a^3/r^3) = -U_\infty(1 - a^3/(a + y)^3)$. Expanding for $y \sim 0$ gives $v_y \sim -U_\infty/3a$ and hence the constant $\alpha = U_\infty/3a$. We then obtain

$$\delta = \sqrt{\frac{3a\nu}{U_\infty}} \sim \frac{a}{\sqrt{\mathcal{R}e}}$$

The last exercise suggests that $\delta \sim L/\sqrt{\mathcal{R}e}$ is a general result. We will see how to explain this result from a scaling analysis, but before, let us turn to an interesting application, and examine once again the decay of a vortex. We saw at the end of the preceding chapter (exercise 2.11) that the order of magnitude invoking viscosity only was unrealistic. Let us explain why.

3.3 Secondary flow

When you steer a cup of tea, you may observe that the decay of rotation follows rather rapidly, and that the leaves of tea (or whatever particles floating in the liquid) form an upward moving column at the center of the cup. These results are explained by the centripetal flow of the fluid in the boundary layer at the bottom of the cup, and this flow causes the general circulation that you are witnessing.

Outside from boundary layers near the walls of the vessel, fluid flow is described by Euler equation, and in that case we already saw many times that this equation simply states that the

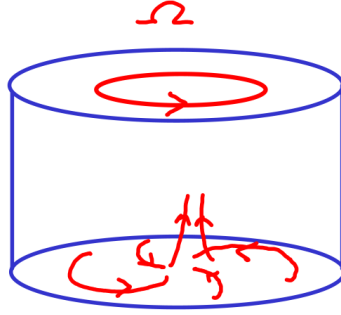


Figure 3.9: A vortex in a cup of tea

centrifugal force balances the pressure gradient :

$$\frac{\partial p}{\partial r} = \rho \frac{v^2}{r}$$

As we also saw, this radial pressure gradient extends to the BL at the bottom of the vessel $z = 0$, otherwise a vertical flow would exist there, and that is prevented by the bottom wall. However in the BL velocity is small $v \sim 0$ (no slip condition) and the pressure gradient is no more balanced there. As a consequence there exists an inside flow at the bottom of the vessel, that is aptly called *secondary flow*, since this is induced by the primary rotational flow.

Let us see some practical applications of these ideas :

- water in rotation draining out from a sink do so by a bottom flow, the bulk being essentially in rotational motion.
- This is also the mechanism responsible for the fast deceleration of the vortex in a circulating vessel : damping at the bottom reduces the angular momentum of the in-flowing fluid to zero, and then this $L = 0$ fluid is sent in the bulk by the general circulation, a very effective damping. This is also why a vortex in a draining sink lasts longer (check it out !).
- sand deposits *inside* meander in rivers, not outside as you would expect from centrifugal forces argument.
- the upward movement of a tornado results from circulation toward the center of the tornado at ground level and volume conservation.

You may find other situations in every-day's life where this rather subtle combination of effects comes into play.

3.4 Scaled adimensional equations of the BL

We saw that quite generally the BL thickness was given by $\delta \sim L/\sqrt{Re}$. Let us obtain this result by a scaling argument similar to what we did in section 2.3.1 of the previous chapter : we consider once again the 2D stationary NS equations in their generality and use the following definitions

$$u' = \frac{u}{U} \quad , \quad x' = \frac{x}{L} \quad , \quad p' = \frac{p}{\rho U^2}$$

This is the obvious scaling for the Euler 1D flow far from the wall. Then we define $y' = \frac{y}{\delta}$ with an unspecified δ that we will choose later to 'simplify' our equations. Mass conservation writes then

$$\frac{U}{L} \partial_{x'} u' + \frac{1}{\delta} \partial_{y'} v = 0$$

and this leads us to define $v' = v / (U\delta/L)$. Inserting all these definitions in NS, we finally end up with

$$\begin{cases} u' \partial_{x'} u' + v' \partial_{y'} u' &= -\partial_{x'} p' + \frac{1}{\mathcal{R}e} (\partial_{x'}^2 u' + \frac{L^2}{\delta^2} \partial_{y'}^2 u') \\ u' \partial_{x'} v' + v' \partial_{y'} v' &= -\partial_{y'} p' + \frac{1}{\mathcal{R}e} (\partial_{x'}^2 v' + \frac{L^2}{\delta^2} \partial_{y'}^2 v') \end{cases} \quad (3.8)$$

Up to now the content of these equations is strictly the same as the original ones. Now we want to take the limit of large $\mathcal{R}e$ while keeping viscous terms³. This suggests to take $\frac{L^2}{\mathcal{R}e \delta^2} = 1$ or $\delta \equiv L/\sqrt{\mathcal{R}e}$. Taking then the limit and dropping the primes, we obtain Prandtl equations in adimensional form

$$\begin{cases} u \partial_x u + v \partial_y u &= -\partial_x p + \partial_y^2 u \\ u \partial_x v + v \partial_y v &= \partial_y^2 v \\ \partial_x u + \partial_y v &= 0 \\ \partial_y p &= 0 \\ u = v = 0 &\text{at } y = 0 \quad \text{and } u \xrightarrow{y \gg 1} 1 \end{cases} \quad (3.9)$$

Since in these equations everything is of the order of magnitude 1, we may expect so for its solution, and this gives the physical scaling of the BL : $\delta \sim L/\sqrt{\mathcal{R}e}$ and $v \sim u/\sqrt{\mathcal{R}e}$.

Exercise 3.6. Check the above computations in detail by yourself.

Exercise 3.7. Find the BL thickness of the flow between a disk rotating at angular velocity Ω and a fixed plane.

3.5 The fate of BL - Turbulence

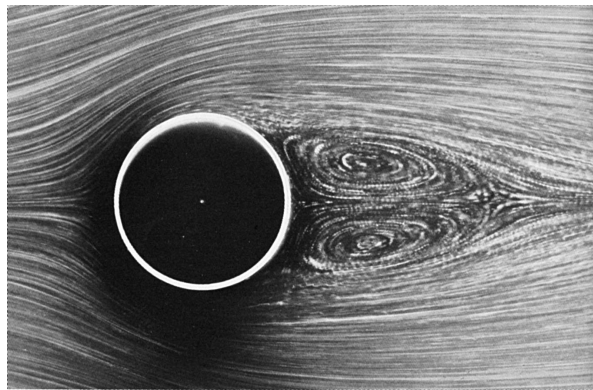


Figure 3.10: Flow past a cylinder at $\mathcal{R}e \sim 40$

³As you see, adimensionalisation can be done in many different ways, depending on your skill and goals. It is generally used to focus on a scale as done here.

We already talked quite a lot about BL separation, vividly illustrated in figure 3.10 for the flow past a cylinder at $Re \sim 40$ where separation leads to the forming of two stationary eddies at the rear of the cylinder.

Increasing the value at about $Re \sim 140$ leads to the alternate unleashing of eddies from left and right, leading to a pattern called a 'Von Karman street'⁴.

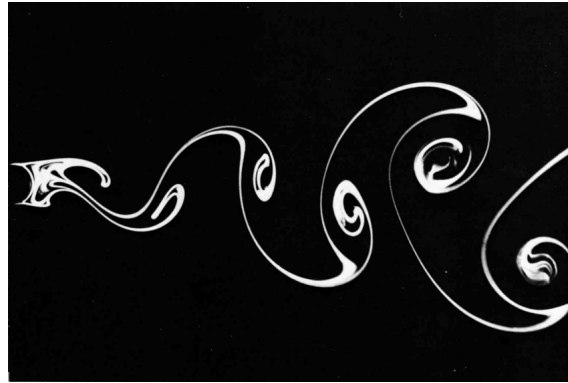


Figure 3.11: Von Karman $Re \sim 140$

Note that the visualization technique is different in the above two pictures : in the first one, one uses a short time photography materializing the motion of markers during a short interval, and streamlines appear. In the second one, ink is pouring from the cylinder into the flow, and this materializes a superposition of trajectories, a pattern called "streak lines".

Increasing further the Reynolds number, $Re \sim 2000$ eddies are replaced by a turbulent region (eddies of many sizes and frequencies) appearing at the BL separation point. The BL in front of the obstacle is still laminar. If we keep going increasing , Re the BL flow eventually

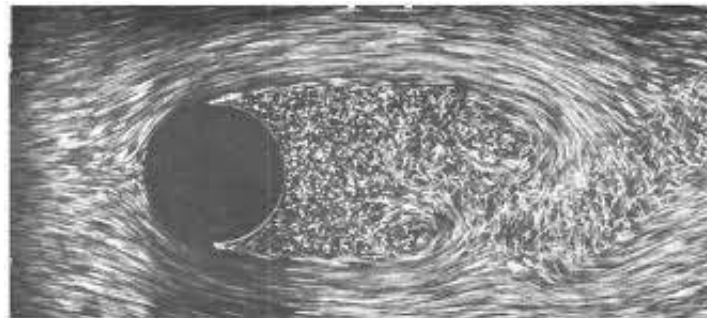


Figure 3.12: Turbulence appears near $Re \sim 2000$

will become unstable and give way to turbulence as shown schematically in the last picture :

With that we close this section, devoting some next chapters to the questions of stability of flows with respect to a change of Reynolds number, and turbulence.

⁴Actually Von Karman did not expect this result and was pressing his student to 'do the experiment correctly and get rid of this erroneous result', see Sommerfeld for an account of this story.

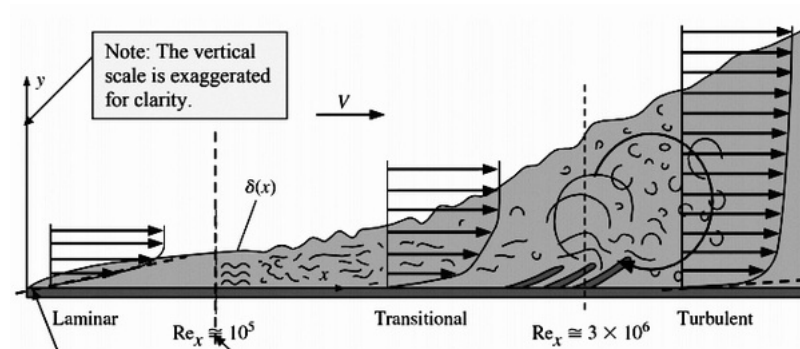


Figure 3.13: The turbulent boundary layer

Chapter 4

Water Waves

4.1 Introduction

Waves are certainly one of the most fascinating and well studied subjects in hydrodynamics. The practical importance of this phenomenon is huge, with innumerable coastal engineering and meteorologic related applications. The greatest names in physics have spent some times to study the subject, and most notably Stokes, Rayleigh and Kelvin, to whom we owe the concept of phase and group velocity. Yet anybody can experience this remarkable phenomenon, specially here in Montpellier.



Figure 4.1: A gentle wave in Palavas.

In this chapter we are interested mainly in gravity waves, where the restoring force is gravity, and we will not speak about sound waves related to the compressibility in a medium. Finally the subject of water waves is vast and we will only scratch the surface of it : one can spend a lifetime studying waves, so this chapter will be introductory (as all of this course) and leave aside many important topics, some of which will be quoted in the conclusion.

4.1.1 General assumptions

In this chapter, the fluid that we call 'water' will be considered as non-compressible, non-viscous and irrotational. As a matter of fact, here the water motion occurs far from the boundaries, so that we can neglect any boundary layer. Actually, while this BL plays an essential

role in the *generation* of waves, once the motion is set its role is weak. Also, for simplicity, we will study the phenomenon in a two dimensional space. Note that the generalization in 3D is simply a matter of more writing. In the beginning we will neglect the role of air surmounting the water : air density is 10^3 smaller than water density, so one may suppose that its role is negligible. Actually, air wind is responsible for the generation of waves, through the formation of a BL at the interface air-water, but this is a diffusive and therefore slow phenomenon as we saw earlier, that we can neglect in a first approximation.

4.1.2 General wave equations

We consider the following geometry, assuming that the bottom is described by a fixed rigid profile (another disputable assumption), and write the following equations for the velocity field $\vec{u} = (u(x, y, t), v(x, y, t))$:

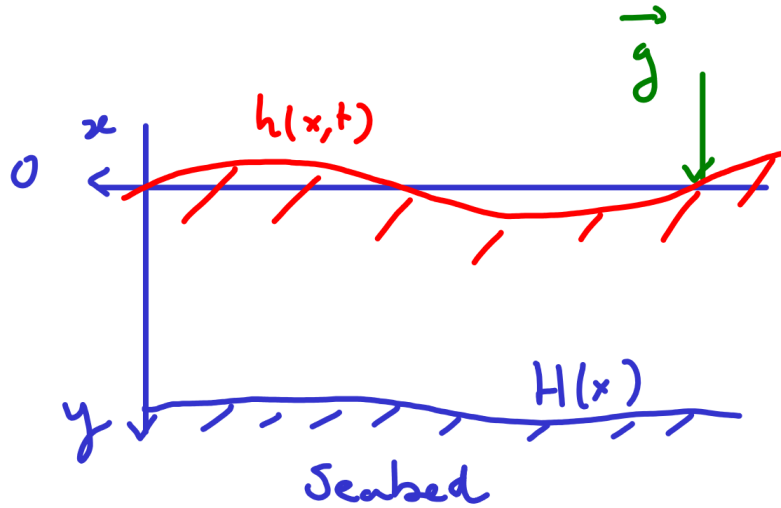


Figure 4.2: Notations and geometry of the flow.

$$\begin{aligned} \vec{\nabla} \cdot \vec{u} &= 0 && \text{Incompressibility} \\ \partial_t \vec{u} + (\vec{u} \cdot \vec{\nabla}) \vec{u} &= -\frac{1}{\rho} \vec{\nabla} p + \vec{g} && \text{Euler} \end{aligned} \quad (4.1)$$

Moreover we have assumed that $\vec{\nabla} \wedge \vec{u} = 0$ since we neglect the effect of viscosity, which plays a crucial role to build the waves, but acts only on a small layer of a few millimeters near the surface. As a consequence there exists a velocity potential $\phi(x, y, t)$ such that $\vec{u} = \vec{\nabla} \phi$. Using this expression for $\vec{u}$ we end up with the following two equations

$$\begin{aligned} \phi_{xx} + \phi_{yy} &= 0 && \text{Laplace equation} \\ \phi_t + \left(\phi_x^2 + \phi_y^2 \right) / 2 + \frac{p}{\rho} - gy &= \frac{p_0}{\rho} && \text{Bernoulli equation} \end{aligned} \quad (4.2)$$

Some remarks are in order

- Here we have used the classical notation with subscripts for differentiation : $u_x = \partial_x u$, $u_{xx} = \partial_x^2 u \dots$
- You may have noticed the unconventional vertical orientation pointing downward. As an exercise, you may rewrite the whole procedure with an upward orientation.
- the constant p_0/ρ in Bernoulli equation has been chosen such that at rest the surface is at $y = 0$. One may change this constant by a shift in the potential : $\phi \rightarrow \phi + Ct$ which does not change the velocity.

The problem that we have to solve seems actually very simple : we solve the Laplace equation for the potential, then compute the velocity field by taking the gradient, and finally find the pressure using the Bernoulli equation. Actually this is not the case, since to solve the Laplace equation we have to specify the boundary conditions on the surface $h(x, t)$ which is actually unknown and a part of the problem. Let us now specify these BC.

4.1.3 Boundary conditions

Fixed boundary

At the bottom of the sea we assume as a first approximation that the boundary is rigid, and that the velocity normal to the boundary cancels, as appropriate for the Euler equation. If we denote by $y = H(x)$ the height of the bottom at position x we have, assuming that the slope is weak, the simple condition $v|_{\text{bottom}} = \phi_y(x, H(x), t) = 0$.

Exercise 4.1. Express the exact BC without any assumption on the slope : $\vec{u} \cdot \vec{n}|_{\text{bottom}} = 0$, where $\vec{n}$ is the normal to the bottom surface. $\vec{n}$ is deduced from $H'(x)$, and $\vec{u}$ is expressed as the gradient of ϕ .

If the bottom is at a sufficient depth, a condition to be precised later, the bottom boundary condition turns into

$$\phi \xrightarrow{y \rightarrow \infty} 0$$

Free surface boundary

Since the surface's height $h(x, t)$ is an unknown function, two conditions are necessary at the free surface :

- A dynamic condition, stating that the normal stress $T_{ij}n_j$ should be continuous across the free surface to prevent for an infinite acceleration. Since viscosity is absent, this simply gives $p(x, h(x, t), t) = p_0$. Using then the Bernoulli equation we obtain at the surface the first condition :

$$\phi_t + \frac{1}{2} \left(\phi_x^2 + \phi_y^2 \right) \Big|_{\text{surface}} - gh(x, t) = 0$$

- A kinematic condition, stating that the particles present at the surface and characterized by the equation $F(x, y, t) = y - h(x, t) = 0$ should stay at the surface. This turns into $DF/Dt = 0$, and written explicitly

$$h_t + \phi_x h_x|_{\text{surface}} = \phi_y$$

See also appendix G to learn more about this condition.

4.1.4 Sakharov's Equations

From the previous sections we can now state exactly the wave problem to solve. For that we will use the restriction of the velocity potential at the surface, that we denote by $\psi(x, t) \equiv \phi(x, h(x, t), t)$. Our problem will be stated in terms of $\psi(x, t)$ and $h(x, t)$. Knowing these two function one can in principle determine the potential completely by solving the Dirichlet problem

$$\begin{aligned} \Delta\phi &= 0 \\ \phi &\xrightarrow{y \rightarrow \infty} 0 \quad \text{and} \quad \phi(x, h(x, t), t) = \psi(x, t) \end{aligned} \tag{4.3}$$

where the time t plays the role of a parameter. We denote this map by $\phi[\psi, h]$. Note that this map is linear in ψ but not in h so this is a first place where some non-linearity enters in the full problem.

From the definition of ψ we have

$$\begin{cases} \psi_x &= \phi_x(x, h(x, t), t) + h_x \phi_y(x, h(x, t), t) \\ \psi_t &= \phi_t(x, h(x, t), t) + h_t \phi_y(x, h(x, t), t) \end{cases} \tag{4.4}$$

Let us denote by $v_s(x, t) \equiv \phi_y(x, h(x, t), t)$ the vertical velocity at the surface. From the above we express the derivatives ϕ_x and ϕ_t and use them in the BC, ending up and after a bit of algebra with the Sakharov equations :

$$\begin{aligned} h_t + h_x \psi_x &= (1 + h_x^2) v_s \\ \psi_t + \psi_x^2/2 - gh &= \frac{1}{2}(1 + h_x^2) v_s^2 \end{aligned} \tag{4.5}$$

These non-linear equations involve only the variables x and t . $v_s = \phi_y(x, h, t)$ has to be obtained from the solution of the Dirichlet problem for ϕ . Once this is done, they yield the time derivatives of ψ and h and we can solve with e.g. an Euler numerical scheme, giving access to ψ and h at a later time, so we see that the exact problem is in principle solvable.

4.2 Deep water small amplitude gravity waves

4.2.1 Linearization

Assuming small amplitude waves first enables to replace the mobile boundary $h(x, t)$ by the rest surface $h = 0$ to solve the Dirichlet problem for ϕ . We will see below that the quantitative condition is $h(x, t) \ll \lambda$ where λ denotes the wavelength.

Then since all the terms are small we can drop all the non linear terms in the Sakharov equations (4.4). Doing so we end up with the following linear problem

$$\begin{aligned} \Delta\phi &= 0 \\ h_t &= \phi_y(x, 0, t) \quad \text{with the BC :} \quad \phi(x, y, t) \xrightarrow{y \rightarrow \infty} 0 \\ \phi_t(x, 0, t) &= gh \end{aligned} \tag{4.6}$$

4.2.2 Solution

Since the problem is now linear and invariant by translation with respect to x and t , we tackle this with the usual Fourier method, seeking for solutions in separated form : $\phi = f(y) \exp i(kx - \omega t)$. The Laplace equation writes then $f'' - k^2 f = 0$ with solutions $f = A \exp(-ky) + B \exp(ky)$. Because of deep water BC we have $B = 0$, and dropping as usual the Fourier term we have

$$\begin{aligned} \phi &= A \exp(-ky) & \phi_y(y=0) &= -kA & \phi_t(y=0) &= -i\omega A & (4.7) \\ h &= -i\frac{\omega}{g}A & h_t &= -\frac{\omega^2}{g}A \end{aligned}$$

4.2.3 Dispersion relation

Eliminating the amplitude A , the dispersion relation writes

$$\omega = \sqrt{g|k|} \quad , \quad v_\phi = \frac{\omega}{k} = \text{sign}(k)\sqrt{g/|k|} \quad , \quad v_g = \frac{d\omega}{dk} = v_\phi/2 \quad (4.8)$$

Recall that the phase velocity v_ϕ is the velocity of the crests, while the group velocity v_g is the velocity of wave packets.

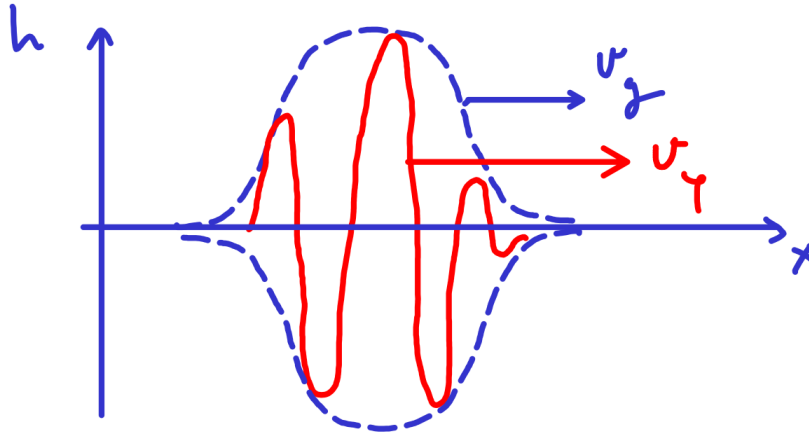


Figure 4.3: A wave packet, moving at velocity v_g while the crests move at v_ϕ .

The gravity waves are dispersive : the waves with longer wavelength propagate at a higher velocity. We also see that the phase velocity is twice the group velocity, all this having profound theoretical and practical consequences that will be studied in section 4.4.

Exercise 4.2. Do the above computations in detail. You have to master it to get the exam !

Exercise 4.3. Compute v_ϕ and the period T for $\lambda = 100$ m. Note that for water waves the phase velocity is the velocity of the crests, and is of great importance, contrarily to electromagnetism.

Exercise 4.4. Show that the linear approximation is acceptable for $ka \ll 1$ where a is the amplitude of the surface h . To do so, estimate the magnitude of the neglected terms in the Sakharov equations

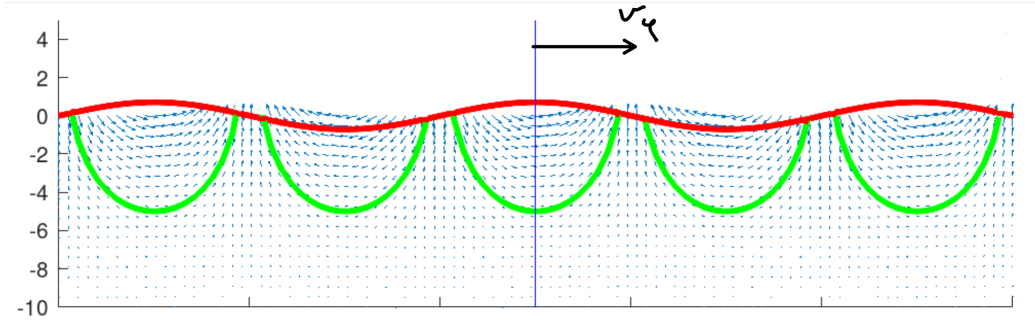


Figure 4.4: A wave profile : the wave surface (red), the velocity field (blue) and the current lines (green). Arbitrary units and different x and y scales.

(4.4), using the linear solution. For the Dirichlet problem, the exact BC $\psi(x, t) = \phi(x, h(x, t), t)$ may be approximated by a Taylor expansion $\psi(x, t) = \phi(x, 0, t) + h\phi_y(x, 0, t) + \dots$. Show that under the above condition the neglected terms are indeed small.

Exercise 4.5. Plot the wave profile and the velocity field with Python. You may animate your plot using frames. One frame should resemble the profile shown in figure 4.4. Observe carefully the relationship between the direction of propagation, the velocity field and the surface profile. Note in particular that the velocity is horizontal on the crests and the troughs, and vertical at the inflection points.

Exercise 4.6. Find the current lines for this wave flow. You have to compute the velocity component, v_x and v_y and integrate the differential equation $dx/v_x = dy/v_y$. These are plotted in figure 4.4

Exercise 4.7. Find the particle trajectories for this wave flow. The system of ordinary differential equations to solve writes $\frac{d\vec{r}(t)}{dt} = \vec{v}(\vec{r}(t), t)$ and is non linear. Linearize it around the mean particle position $\vec{r}_0$, under the assumption $ka \ll 1$. Show that trajectories are circles traveled in the anticlockwise sense when the wave moves right ($v_\phi > 0$) as in the figure above.

4.3 Finite Depth

The analysis above can be transposed easily to the case of a finite depth L , with the geometry shown in the figure below : One still has the same equations to solve as in (4.6) but instead of the indicated BC one has to impose the cancellation of the vertical velocity at the bottom : $\phi_y(x, L, t) = 0$. Still looking for a separable solution $\phi = f(y) \exp i(kx - \omega t)$, and imposing the condition at $y = L$ gives, dropping the Fourier dependence

$$\phi = A \frac{\cosh k(L - y)}{\cosh kL}, \quad \phi_y = -kA \frac{\sinh k(L - y)}{\cosh kL}, \quad \phi_t = -i\omega A \frac{\cosh k(L - y)}{\cosh kL}$$

and using these expressions we end up with the finite depth dispersion relation

$$\omega^2 = gk \tanh kL \quad (4.9)$$

For $kL \gg 1$ this gives back the deep water dispersion relation so that the limit between the two regimes corresponds to $\lambda \sim L$:

- For $L > \lambda$ we are in the deep water regime and the waves are dispersive.

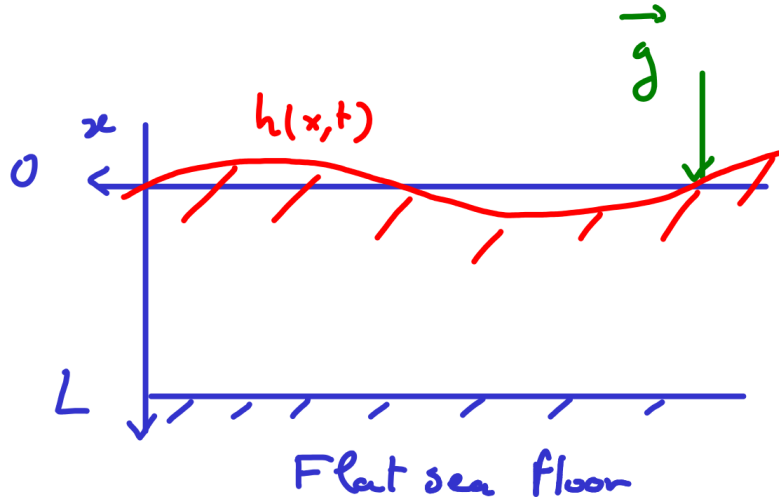


Figure 4.5: Geometry with a finite depth.

- At the other extreme, when $L \ll \lambda$ one can use $\tanh kL \sim kL$ which results in $\omega = \pm\sqrt{gL}k$. We see that in this case the waves are non-dispersive : $v_g = v_\phi = \sqrt{gL}$. This is what happens in the laguna ('étangs') near Montpellier, where the depth is about 1 m. If you have experienced windsurfing in the sea and in the laguna, you know that the waves do not feel the same at all !

Exercise 4.8. Compute v_ϕ for $L = 1$ m. This is the typical wave velocity in the nearby laguna.

4.4 Propagation and Dispersion

Water waves are strongly dispersive. Moreover contrarily to Electromagnetism, phase propagation is perceptible and important (try to tell a sailor that 'crest propagation is not that important' !). Finally waves are not generated 'monochromatically' but at the opposite, wind excites waves in a broad range of wavelength from ~ 1 m to about 10^3 m, so we have to adapt the ideas and techniques that you already studied in electromagnetic waves propagation to the present situation. Historically this analysis is due to Lord Kelvin, and this section owes a lot to Whitham's monography "Linear and non-linear waves" [8].

4.4.1 Phase velocity

We consider a mode solution of the wave equation, in the form

$$h(x, t) = Ae^{i(kx - \omega(k)t)} + c.c. \quad (4.10)$$

where c.c. stands for *complex conjugate*¹. Here $\omega(k)$ stands for the dispersion relation (DR). In this paragraph we will use the deep water DR $\omega(k) = \sqrt{g|k|}$ as an example, but this choice

¹This notation is very often used in the water waves literature, where non-linearity may play a role, so a foolproof notation is useful.

is not restrictive. This wave is traveling to the right or to the left according to the sign of k . We denote by $\theta(x, t) \equiv kx - \omega(k)t$ the *phase* of this wave, which is constant on the lines $x = v_\phi t + x_0$, where

$$v_\phi = \frac{\omega}{k} = \pm \sqrt{g/|k|}$$

For water waves, long wavelength are traveling faster than short ones : the swell is observed before the storms strikes the west coast of Brittany, and Tsunami are traveling even faster : $\lambda \simeq 10^3 \text{ m}$ gives $v_\phi \simeq 40 \text{ m.s}^{-1} \simeq 140 \text{ km/h}$!

4.4.2 Group velocity

Because of linearity, we now consider a superposition

$$h(x, t) = \int_{-\infty}^{\infty} A(k) e^{i(kx - \omega(k)t)} dk + c.c. \quad (4.11)$$

The amplitude $A(k)$ is determined by the initial conditions. As an example, we will use $A(k) = 1/4\pi$ which gives $h(x, 0) = \delta(x)$ corresponding to a localized initial disturbance, as 'a stone in a 1D pond', quite opposite to a quasi-monochromatic wave packet. We will ask two type of (related) questions :

- What is the asymptotic behaviour of $h(x, t)$ at large x and t with x/t fixed ?
- What sequence of events a distant observer at x will observe during the course of time t ?

A good way to visualize wave propagation is by the use of *space-time diagrams* as in the figure below².

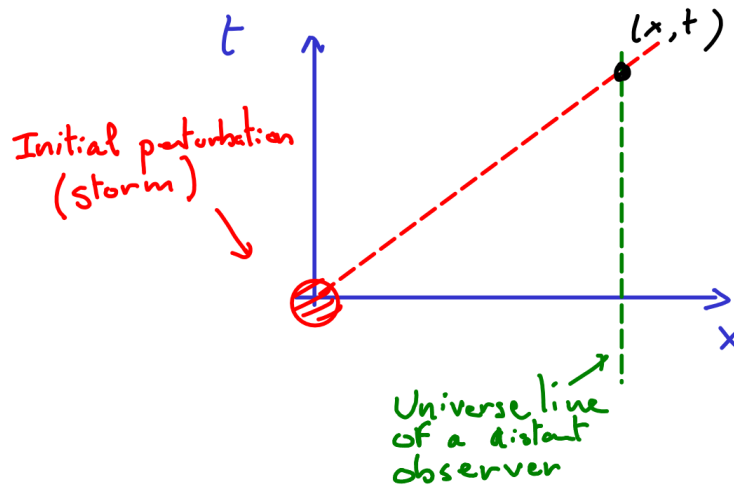


Figure 4.6: A space-time view of a storm.

Let us denote $\chi(k) \equiv \omega(k) - kx/t$ so that we write

$$h(x, t) = \int_{-\infty}^{\infty} \frac{1}{4\pi} e^{-it\chi(k)} dk + c.c.$$

²Although popularized by the theory of the Relativity, these diagrams were invented well before 1905 precisely to describe wave propagation.

For large t the term under the integral oscillates very rapidly with k and the sum cancels, except when $\chi(k)$ is stationary. This is therefore called the *stationary phase* analysis. To make it quantitative let us Taylor expand $\chi(k)$ at an (yet unspecified) value k_0 . We have

$$\chi(k) \sim \chi(k_0) + (k - k_0)\chi'(k_0) + \frac{(k - k_0)^2}{2}\chi''(k_0) + \dots$$

and

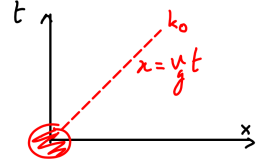
$$h(x, t) = \frac{1}{4\pi} e^{i(k_0 x - \omega(k_0)t)} \int_{-\infty}^{\infty} e^{-it\chi'(k_0)(k-k_0)} e^{-it\chi''(k_0)(k-k_0)^2} dk + c.c.$$

As said before, the phase in $\exp[-it\chi'(k_0)(k - k_0)]$ oscillates wildly with k unless $\chi'(k_0) = 0$, which writes using the expression of χ :

$$\frac{x}{t} = \left. \frac{d\omega}{dk} \right|_{k=k_0} \equiv v_g(k_0) = \frac{1}{2} \sqrt{\frac{g}{|k_0|}} \quad (4.12)$$

This equation should be interpreted as follows : far from the initial perturbation, and in a space time direction $x/t \equiv v_g$ fixed, one will observe waves with a wave vector given by the above condition, that we can rewrite as

$$k_0 = \frac{1}{4} g \left(\frac{t}{x} \right)^2, \quad \omega_0 = \sqrt{gk_0} = \frac{1}{2} g \frac{t}{x}$$



One then observes a fan of asymptotic waves with different values of k_0 , with larger wavelength preceding smaller ones, as in figure 4.7, so that the distant observer will first observe large wavelength (the swell) followed by the main disturbance at shorter wavelengths.

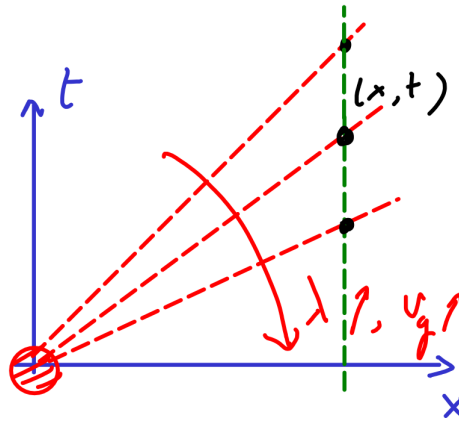


Figure 4.7: Asymptotic view of a storm.

Going on with the computation, we have

$$\begin{aligned} \chi'(k_0) &= 0 \\ \chi''(k_0) &= -\frac{2}{g} \left(\frac{t}{x} \right)^3 \\ h(x, t) &= \frac{e^{i(k_0 x - \omega(k_0)t)}}{4\pi} \int_{-\infty}^{\infty} e^{-it\chi''(k_0)(k-k_0)^2} dk + c.c \end{aligned}$$

The Gaussian integral over k evaluates as $\sqrt{2\pi/it|\chi''(k_0)|}$ finally giving

$$h(x, t) = \frac{\cos(k_0 x - \omega(k_0)t - \pi/4)}{\sqrt{2\pi t|\chi''(k_0)|}} = \sqrt{\frac{g}{4\pi x}} \frac{1}{v_g} \cos\left(\frac{gt^2}{4x} + \pi/4\right) \quad (4.13)$$

We see that this asymptotic analysis, valid for x/t fixed gives an oscillating phase at $\omega_0/2$ and a damping term $\propto 1/\sqrt{x}$.

Lastly, following a wave crest at velocity v_ϕ one will observe waves of increasing wavelength (and group velocity) as shown in the diagram below. This behavior can be observed on

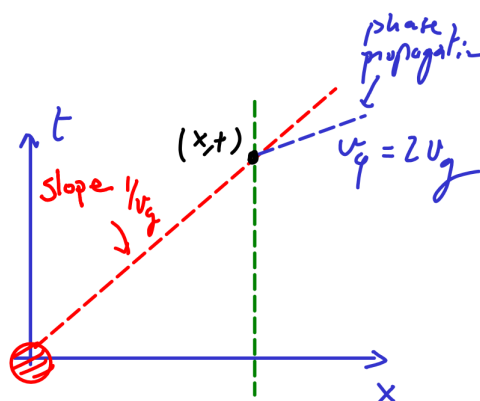


Figure 4.8: Following a wave crest at celerity v_ϕ .

the animation display on the Moodle pages of this course, showing the above explicit solution without the damping factor.

As a practical application of these considerations, we can use the succession of the different wavelengths with time on the Brittany coast to estimate the distance at which the waves originated from the storm in north Atlantic sea : we write

$$\begin{cases} x - x_0 &= v_g(k_1)(t_1 - t_0) \\ x - x_0 &= v_g(k_2)(t_2 - t_0) \end{cases} \quad (4.14)$$

where t_1 and t_2 are the observing times at position x of waves originated at x_0 at a time t_0 , with a wave vector k_1 and k_2 respectively. Eliminating t_0 yields

$$x - x_0 = \frac{t_1 - t_2}{1/v_1 - 1/v_2}$$

with $v_1 = v_g(k_1) = \sqrt{g/k_1}/2$ and $v_2 = v_g(k_2) = \sqrt{g/k_2}/2$. Finally using the observed frequencies ω_1 and ω_2 at the different times instead of the wave vectors, we end up with the simple result :

$$D = \frac{g(t_1 - t_2)}{2(\omega_1 - \omega_2)}$$

an expression that you may use for yourself if you happens to stay near the Atlantic sea³.

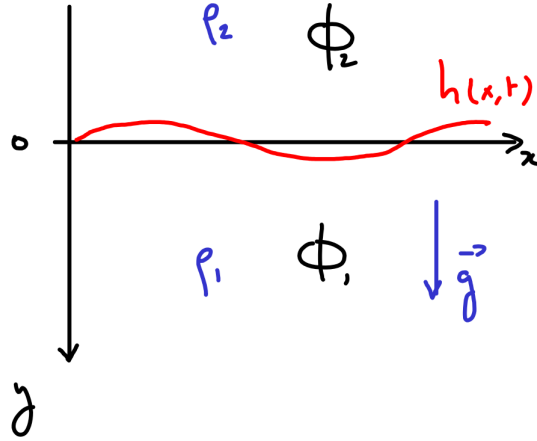


Figure 4.9: Superposed fluids with different densities.

4.5 Waves between two fluids

Let us now turn to a more realistic situation where the fluid below at volumetric mass ρ_1 is surmounted by another lighter fluid such that $\rho_2 < \rho_1$. We still assume an inviscid incompressible irrotational 2D flow, and follow the same linearized description as before, with now two velocity potential ϕ_1 and ϕ_2 for the two parts of the fluid, with now the conditions at infinity :

$$\phi_1 \xrightarrow{y \rightarrow \infty} 0 \quad \text{and} \quad \phi_2 \xrightarrow{y \rightarrow -\infty} 0$$

Still assuming a Fourier dependence $\propto \exp i(kx - \omega t)$ we solve Laplace equation separately for ϕ_1 and ϕ_2 and find

$$\begin{cases} \phi_1 = Ae^{-ky}e^{i(kx-\omega t)} \\ \phi_2 = Be^{+ky}e^{i(kx-\omega t)} \end{cases} \quad (4.15)$$

with two yet unspecified constants A and B . Let us now impose the kinematic and dynamic conditions at the interface $y = 0$:

- the continuity of the velocity $h_t = \phi_{1y}(y=0) = \phi_{2y}(y=0)$ gives $-kA = kB = -i\omega h$,
- the continuity of pressure in the Bernoulli equation gives $\rho_1\phi_{1t} - \rho_1gh = \rho_2\phi_{2t} - \rho_2gh$,

and eliminating the constants A , B and h results in the dispersion relation

$$\omega^2 = \frac{\rho_1 - \rho_2}{\rho_1 + \rho_2} g k \quad (4.16)$$

When the surmounting fluid is very light as in the case of water waves where $\rho_2/\rho_1 \sim 10^{-3}$ we are back to the results obtained in section 4.2, and this justify the assumption of neglecting the air above.

Another interesting case appears when $\rho_2 > \rho_1$ and the surmounting fluid is heavier. Intuition suggests that this situation is unstable, and this is traduced by the fact that $\omega = \pm i\Omega$ is a pure imaginary number. In that case the Fourier modes have a time dependence $\propto \exp(\Omega t)$ and grow until the linear approximation breaks. We will study this 'Rayleigh -Taylor instability' together with other instabilities in the next chapter.

³The Mediterranean sea is to small to observe such effects.

4.6 Capillary Waves

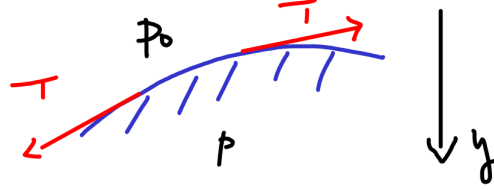


Figure 4.10: The effect of surface tension.

We end up this chapter with the practically important subject of capillary waves. What you have seen in a small wave tank during your practical sessions on waves are capillary shallow water waves ! They appear when the surface curvature is large, at small values of the wavelength λ . Capillarity and the resulting modification of the dispersion relation also explain the old and effective practice of leaking oil in small quantities into the sea to calm the waves temporarily.

Capillarity effects are caused by the surface tension T . In a 2D geometry, where the surface is 1D, its effect is strictly analog to that of the string tension that you already met when studying the vibrating string. Following figure 4.10, we compute the normal projection of the force by unit length as

$$\sigma = T \frac{\partial h}{\partial x}(x + dx) - T \frac{\partial h}{\partial x}(x) = T \frac{\partial^2 h}{\partial x^2} dx$$

and this constraint σ which is positive in the situation above, will increase the internal pressure under the surface :

$$p_1 = p_0 + T \frac{\partial^2 h}{\partial x^2}$$

and this constitutes the appropriate dynamic BC in this situation. Using this as before together with the Bernoulli theorem gives

$$\rho \phi_t - \rho g h + T h_{xx} = 0$$

and leads, following the same line of reasoning as before, to the DR

$$\omega^2 = gk + \frac{T}{\rho} k^3 \quad (4.17)$$

Exercise 4.9. Show this result. Also check its homogeneity.

Exercise 4.10. Adapt these computations to take into account a finite depth and a superposed fluid.

We see that capillary effects are negligible for $k^2 T / \rho g \lesssim 1$ and we define the critical wavelength as

$$\lambda_c = 2\pi \sqrt{\frac{T}{\rho g}}$$

For the air-water interface at 20°C, $T \simeq 0.074 \text{ N.m}^{-1}$, and $\lambda_c \simeq 1.7 \text{ cm}$.

- for $\lambda > \lambda_c$ we are in the gravity waves regime and this is what you see on the seaside.

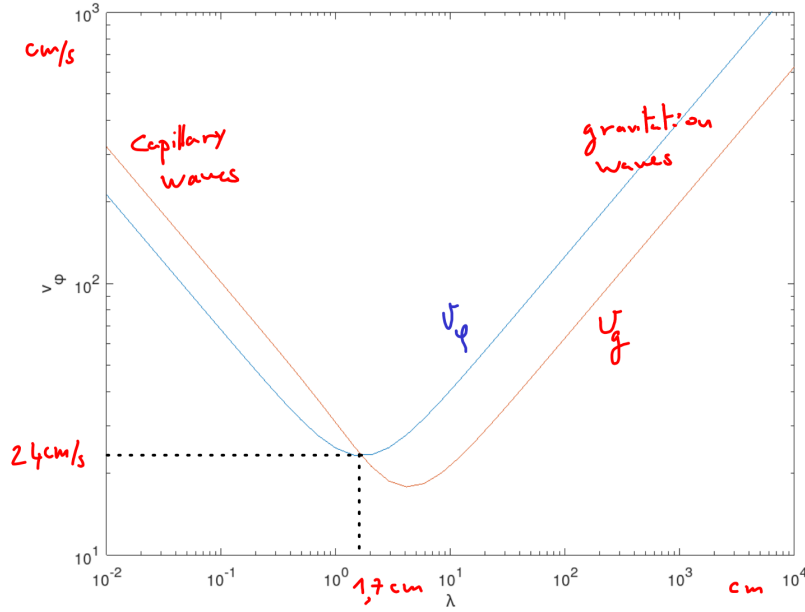


Figure 4.11: Log-log diagram of the phase (blue) and group (red) velocities of gravity and capillary waves.

- for $\lambda < \lambda_c$ we are in the capillary waves regime, and this is what you observe in your mug.

We can now compute the phase and group velocities :

$$v_\phi = \frac{\omega}{k} = \sqrt{\frac{g}{k} \left(1 + \frac{T}{\rho g} k^2 \right)^{1/2}} \quad \text{and} \quad v_g = \frac{d\omega}{dk} = \frac{1}{2} \sqrt{\frac{g}{k} \frac{\left(1 + \frac{3T}{\rho g} k^2 \right)}{\left(1 + \frac{T}{\rho g} k^2 \right)^{1/2}}} \quad (4.18)$$

and visualize these results on the graphic shown in figure 4.11.

- In the gravity regime $v_g = v_\phi/2$ and crest are traveling faster than packets.
- In the capillary regime $v_g = 3v_\phi/2$ and crest are lagging behind packets.

The critical regime separating gravity from capillary effects at $\lambda = \lambda_c$ corresponds to the minimum of the phase velocity, in the air-water interface at 20°C this minimal value is $v_{\phi c} = 23 \text{ cm.s}^{-1}$, leading to a funny fact : around a fishing line in a stream at velocity $U > 23 \text{ cm.s}^{-1}$ you can observe both type of waves around the line simultaneously. Changing to the stream frame of reference, the line is traveling at celerity $-U = v_\phi$, and you observe upstream the capillary waves whose packets go faster, and downstream the gravity waves lagging behind.

4.7 Conclusion

As said in the introduction, we have only scratched the surface of this very vast subject. Particularly we have not discussed

- The effects of non-linearity,
- the geometrical patterns originating from dispersion in 3D,
- the generation of waves, and the Miles theory, see appendix ??,
- solitary waves,
- the evolution of the wave field in a slowly changing environment (depth or gravity variations),
- waves in a rotating frame,
- ...

The interested reader should consult Whitham [8] and all the specialized literature about these many related subjects.

Chapter 5

Stability of flows

5.1 Introduction

5.1.1 Short historical outline

Since the beginning of these lectures we have met a few stationary analytical solutions of flows in the laminar regime, and you may think that this is the typical situation : laminar flows would occur in a majority of situations. Actually a view on a flowing brook will persuade you of the opposite : laminar flows are generally *unstable* and are not observed in a majority of cases. This is related to the non-linearity of the Euler and Navier-Stokes equations. The link between non-linearity and instability, and then turbulence, has slowly emerged during the XX-th century, while ideas have progressed on these subjects :

- During the XIX-th century the general view on Physics and Science is that Nature is deterministic, and instabilities are related to pathological academical examples in mathematics.
- At the turn of the XX-th century, Poincaré study dynamical non linear systems related to planetary motions, and introduces the notion of sensibility to initial conditions, and that of bifurcation : a radical change of dynamical behaviour when a parameter crosses a critical value.
- On an experimental side, the realization around 1910 that the so called Von Karman street of vortices was an intrinsic phenomenon that would appear independently of the precise experimental setup, whatever efforts are made to eliminate it, came as a total surprise even to the eyes of Prandtl himself.
- The general belief during the first part of the XX-th century is that chaos (unpredictability) and turbulence are related to the fact that the corresponding dynamical systems have an infinite number of degrees of freedom that contribute with an infinity of frequencies. This is exactly the view stated in the older editions of the Landau and Lifshitz course [4].
- It therefore came as a surprise when in 1963 Edward Lorenz, a meteorologist theoretician, studying a reduction of the Couette cylindrical flow to simulate the global atmospheric motion, came out with a 3 degrees of freedom system presenting a chaotic behaviour. This flow is now named the Lorenz attractor, the first known "strange attractor". The surprise was increased by the fact that the dynamical system is dissipative : it does not

converge to a fixed point although the phase space volume is shrinking during the flow, which is made possible by the fractal structure of the flow, see references [10, 11].

- During the 70', the "chaos theory" has emerged from different contributions in mathematics, physics, chemistry, engineering with the fact that very simple and deterministic rules could lead to totally unpredictable behaviour. A paradigmatic example is the so called "Feigenbaum cascade" : a discrete dynamical system with only one control parameter governing an universal transition from a deterministic to a chaotic behaviour with a cascade of period doubling, see figure 5.1.

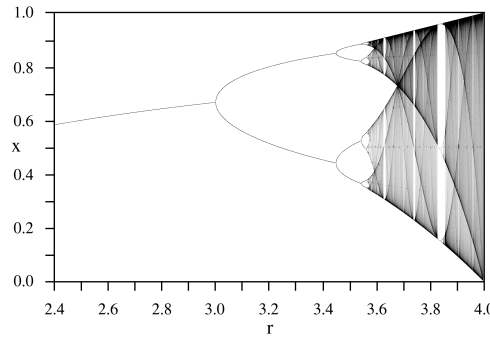


Figure 5.1: The Feigenbaum cascade of bifurcations.

A part of these ideas stem from hydrodynamics, and in particular the *linear stability analysis* : the idea is very simple and consist in studying the evolution of very small perturbations of stationary laminar solutions. Since these perturbations are supposed to be small in the early phase of their development, one can linearize the Euler or Navier Stokes equations and eventually come to a tractable analytical analysis of stability.

5.1.2 Linear stability analysis

As said previously the idea is to start from a (generally stationary) solution of the Euler or Navier Stokes equations for the flow, $\vec{U}, P \dots$, and then suppose that a small perturbation appears $\vec{U} + \delta\vec{u}, P + \delta p \dots$. One then develops the equations to the first order in the perturbation, and study how the solutions evolve in time. Since the dynamic is generally autonomous (external conditions are stationary) it is invariant in time and one may look for Fourier modes $\propto e^{-i\omega t}$. Instability corresponds to the existence of growing modes, obtained with dispersion relations where $\Im m(\omega) > 0$.

This is very similar to what we have already done in the previous chapter on waves : a linearized water wave is a small perturbation of the air-water interface at rest, which happens to be stable. However if the surmounting fluid is heavier than the bottom one the situation is unstable, and leads to the Rayleigh-Taylor instability that will be our first example. As in the previous chapter on waves, we will use this approach using the Euler equation of motion for the fluid, since the corresponding analysis for the Navier-Stokes equation leads generally to qualitatively similar results, but is much harder to perform.

5.2 Rayleigh-Taylor instability

Let us come back to the situation studied in section 4.5, but with the above fluid heavier than the bottom one $\rho_2 > \rho_1$, and with surface tension taken into account. It is a simple exercise to show that this leads to the following dispersion relation :

$$(\rho_1 + \rho_2) \omega^2 = k \left[k^2 T - (\rho_2 - \rho_1) g \right] \quad (5.1)$$

We immediately see that for

$$k < k_{cr} \equiv \sqrt{\frac{(\rho_2 - \rho_1) g}{T}}$$

the Fourier mode $\propto \exp(i(kx - \omega t))$ is unstable, and will lead to a digitation of the heavier fluid in the lighter one, as in figure 5.2 The physical interpretation of the stability result is that

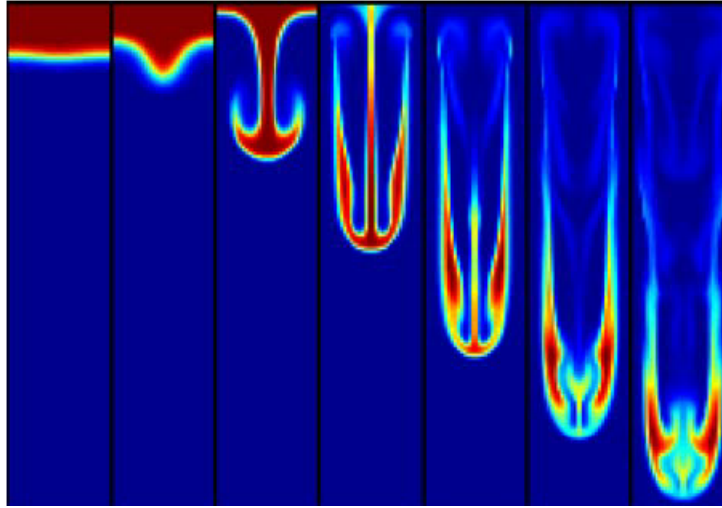


Figure 5.2: The Rayleigh-Taylor instability (simulation).

long wavelength will be unstable since surface tension limits short wavelength, the larger T the lower the critical value, and that the difference of weight $(\rho_2 - \rho_1) g$ is the cause of the phenomenon.

Exercise 5.1. *Use the methods of chapter 4 to show the above dispersion relation.*

This instability has been revived during the Cold War thermonuclear assays, where a heavy fluid (the thermonuclear blast) accelerates into a lighter one (the atmosphere), leading to a Rayleigh-Taylor instability which is annoying to the military since it reduces the power of the blast , leading to "Huh, huh, this is not the big one I expected !" kind of situations.

Another very simple view at this instability is the following buoyancy argument : In the stratified fluid, a fluid particle is at equilibrium under the effect of its weight and of the pressure gradient exerted by the rest of the fluid, as stated by the Pascal equation

$$\rho(z) g + \frac{\partial p}{\partial y} = 0$$

Moving a heavy particle from the region ② to the region ①, the pressure gradient (Archimedes force) will be insufficient to balance the weight and the particle will sink. This effect is symmetric and a lighter particle will ascend in the heavier fluid. However this simple approach does not take the stabilizing role of surface tension into account.

5.3 Kelvin-Helmholtz instability

Let us now consider a slightly more complicated situation, where the surmounting lighter fluid is moving parallel to the interface with an uniform velocity U . The instability mechanism can be explained as a kind of Venturi situation, where a compressed part of the moving fluid accelerates and aspirates the other fluid, as in figure 5.3. One may hope that this instability is

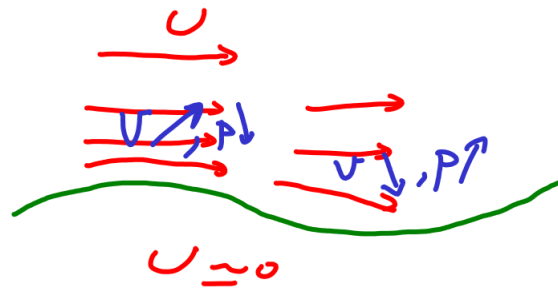


Figure 5.3: The Kelvin -Helmholtz instability mechanism.

responsible for the generation of water waves by the wind. Unfortunately this is not the case as we will see below.

5.3.1 General equations

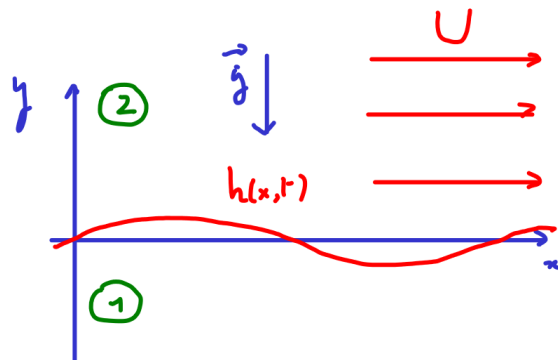


Figure 5.4: The Kelvin -Helmholtz geometry.

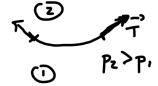
These are almost identical to the water waves equations : we will treat the fluids as ideal non viscous, incompressible and irrotational, and use the Euler equation, neglecting the boundary layer between the two fluids. We will take the y axis orientation up, see figure 5.4 above.

Since the fluid is irrotational, there is a velocity potential ϕ such that $\vec{v} = \vec{\nabla}\phi$ and the fluid equations are

$$\begin{cases} \Delta\phi &= 0 & (\text{Incompressibility}) \\ \phi_t + \frac{1}{2}||\vec{v}||^2 + p/\rho + gy &= Cte & (\text{Bernoulli equation}) \end{cases} \quad (5.2)$$

with a different constant in both parts of the fluid to account for the infinite shear localized at the interface. We now state the BC at the interface. These are :

- the dynamic condition $p_2 - p_1 = T h_{xx}$, see figure right for the correct sign :



- the kinematic condition $h_t + u h_x = v$

5.3.2 Unperturbed solution

- Fluid ① is at rest : $\vec{v} = 0$ for $y < 0$.
- Fluid ② is moving at constant velocity : $\vec{v} = U\vec{e}_x$ for $y > 0$.
- The interface is flat $h = 0$ and the dynamic condition gives $p_1 = p_2 \equiv p_0$ at the interface.
- all the BC are trivially satisfied, together with the incompressibility conditions.

One see that there are indeed two values for the Bernoulli constant corresponding to an infinite shear at $y = 0$:

$$\begin{aligned} \frac{1}{2} \rho_2 U^2 + p_0 &= \rho_2 C_2 & \text{for } y > 0 \\ p_0 &= \rho_1 C_1 & \text{for } y < 0 \end{aligned}$$

hence $\rho_2 C_2 - \rho_1 C_1 = \frac{1}{2} \rho_2 U^2$ and since these are constants during the flow, this relation will remain valid for the perturbed flow.

5.3.3 Perturbed solution

We inject in the equations the perturbation $U \rightarrow U + u$, v , $P \rightarrow P + p$ and look for modes $\phi = A(y) \exp(i(kx - \omega t))$. Solving for the Laplace equations with the BC at infinity we obtain

$$\begin{cases} A(y) &= A_1 \exp(ky) & y < 0 \\ A(y) &= A_2 \exp(-ky) & y > 0 \end{cases}$$

and we use this in the linearized Bernoulli relation at the interface, with the cancellations given by the unperturbed solution :

$$\begin{cases} -i\omega A_1 + \cancel{\frac{\rho_2 U^2}{2}} + gh &= \cancel{\rho_1 C_1} & y < 0 \\ -i\omega A_2 + \cancel{\frac{\rho_2 U^2}{2}} + U u_2 + \cancel{\frac{\rho_2 U^2}{2}} + gh &= \cancel{\rho_2 C_2} & y > 0 \end{cases}$$

- the dynamic condition writes $p_2 - p_1 = -k^2 T h$
- the kinematic condition on both sides gives $-i\omega h = k A_1$ and $-i(kU - \omega)h = k A_2$

Using these in the Bernoulli conditions at the interface give

$$\begin{cases} -\frac{\omega^2}{k} \rho_1 h + p_1 + \rho_1 g h & = 0 & y < 0 \\ (U - \frac{\omega}{k})(kU - \omega) \rho_2 h + p_2 + \rho_2 g h & = 0 & y > 0 \end{cases}$$

Finally by taking the difference and using the dynamic condition we eventually end up with the Kelvin-Helmholtz dispersion relation¹ :

$$(\rho_1 + \rho_2) \omega^2 - 2\omega k U \rho_2 + k^2 U^2 \rho_2 = k [k^2 T + (\rho_1 - \rho_2) g] \quad (5.3)$$

Exercise 5.2. Do the computation in detail to obtain the above dispersion relation (5.3). You should be able to reproduce it without reading the notes ! This is an excellent resume of the techniques of chapters 4 and 5.

5.3.4 Stability condition

The above relation is an algebraic equation of power two in ω , hence if its discriminant δ is positive the solutions are real and the mode is stable. On the contrary if $\delta < 0$ the two solutions are complex conjugates, and there is an unstable solution $\Im m(\omega) > 0$. Writing this condition and after some trivial simplifications we end up with the instability condition

$$\delta < 0 \iff \delta' \equiv k^2 T - k U^2 \frac{\rho_1 \rho_2}{\rho_1 + \rho_2} + (\rho_1 - \rho_2) g < 0$$

Once again we have a second order polynomial this time in k . In figure 5.5, we have plotted δ' as a function of k in the case of a water-oil interface : $\rho_1 \simeq 1 \text{ g/cm}^3$, $\rho_2 \simeq 0.9 \text{ g/cm}^3$, $T = 0.02 \text{ N/m}$, for different values of U . We see that there is a critical value U_{cr} above which this quantity can become negative and the flow unstable. The corresponding value k_{cr} , which corresponds to the wavelength of the first instability to appear when $U = U_{cr}$, is computed as the value of k corresponding to the double root of the red curve.

For δ' to eventually become negative the discriminant of this quantity has to be positive, and this leads to the instability absolute condition

$$U^2 \frac{\rho_1 \rho_2}{\rho_1 + \rho_2} > 2 \sqrt{T(\rho_1 - \rho_2) g}$$

When this condition is fulfilled an instability grows first at a wave vector $k = k_{cr}$. In the case of the water-air interface $\rho_1 \simeq 1000 \text{ kg.m}^{-3}$, $\rho_2 \simeq 1.3 \text{ kg.m}^{-3}$, $T \simeq 0.074 \text{ N.m}^{-1}$ giving finally $U_{cr} \simeq 6.5 \text{ m.s}^{-1}$ and a corresponding critical value for k :

$$k_{cr} = \sqrt{\frac{(\rho_1 - \rho_2) g}{T}} \quad \lambda_{cr} = \frac{2\pi}{k_{cr}} \simeq 1.7 \text{ cm}$$

These values do not correspond to what is observed in the waves generation by the wind : waves are created with a much lower value of U and with much higher values of the wavelength. A calm day at the seaside with almost no wind will persuade you of that.

As a conclusion this mechanism is not responsible for the generation of water waves by wind, but can be observed in many planetary atmospheres, as seen the picture 5.6. The mechanism for the generation of water waves was understood by Rayleigh and explained in details by Miles in 1959. It is termed the Miles Mechanism, see appendix ?? and Ref [9] for more information.

¹Note that taking $U = 0$ yields the Rayleigh-Taylor relation, which appears as a special case of this relation.

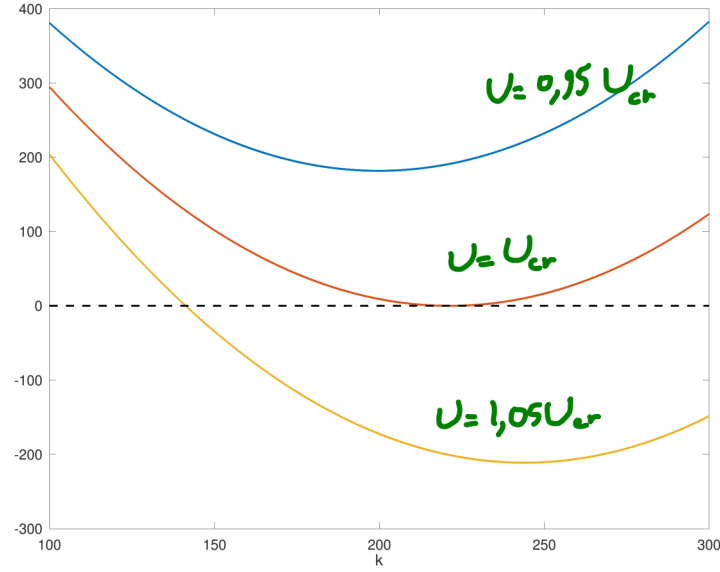


Figure 5.5: The criticality condition for the Kelvin-Helmholtz instability for a water-oil interface. $U_{cr} \simeq 13.6 \text{ cm/s}$, $k_{cr} \simeq 2.2 \text{ cm}^{-1}$.

5.4 Stability of the planar Couette and Poiseuille flows

In this section we will consider forced flows in channels and pipes, and we start with the simplest of all : the planar Couette flow that we have already met in chapter 2. As before we start from a stationary flow whose geometry is displayed in figure 5.7. We denote by $U(y)$ the stationary profile solution of the NS equations, that we analyze in the Euler framework : actually the above profile is also a solution of the Euler equation ! The logic behind this approach is that once the stationary flow is established, viscosity will hopefully only plays a minor role in the stability of the flow. The full NS analysis can be done, but it is much more technical to perform as we will explain below. Finally we use units such that $\rho = 1$ to simplify the writing.

As before, we write a perturbed solution as $\vec{u} = (U(y) + u, v)$ and write the 2D Euler equations in the first order of perturbation :

$$\begin{cases} u_t + U(y)u_x + vU'(y) &= -p_x \\ v_t + U(y)v_x &= p_y \\ u_x + v_y &= 0 \end{cases} \quad v = 0 \text{ at the boundary}$$

Looking for a Fourier solution $\propto \exp(i(kx - \omega t))$, $\partial_t \rightarrow -i\omega$, $\partial_x \rightarrow ik$, gives

$$\begin{cases} -ik(\frac{\omega}{k} - U)u + U'v &= -ikp \\ -i(\omega - kU)v &= -p_y \\ ik u + v_y &= 0 \end{cases}$$

Using the last equation we can eliminate u

$$(\frac{\omega}{k} - U)v_y + U'v = -ikp$$

and inject this value of p in the second equation. All this produces the *Rayleigh equation* :

$$(c - U(y)) [v_{yy} - k^2 v] + U'' v = 0 \quad (5.4)$$



Figure 5.6: Fully developed Kelvin -Helmholtz instability.

where $c \equiv \omega/k$ is the phase velocity of the wave. This equation is a linear second order differential equation, and it has to be solved with the appropriate boundary conditions $v = 0$ on both sides of the domain. If we fix a real value for k , it is similar to the Schrödinger equation², leading to quantized values for ω . If these turn out to have a positive imaginary part, the flow is unstable.

Exercise 5.3. *Perform the same analysis starting from Navier-Stokes equation. You will end up with a 4-th order differential equation for v that is called the Orr-Sommerfeld equation.*

In our case the non perturbed laminar profile is linear and $U'' = 0$ leading to a very simple equation $v_{yy} - k^2 v = 0$, which do not admit solutions satisfying the BC $v(0) = v(L) = 0$. The surprising result is that the planar Couette flow is stable in this Euler analysis, since perturbations satisfying the linearized equations do not exist.

We may enlarge a bit this analysis by assuming a general stationary velocity profile $U(y)$, as would occur in other 2D channel flows that we met in chapter 2 (e.g. a planar Poiseuille flow with a parabolic profile). Multiplying the Rayleigh equation by the c.c. of the vertical velocity v^* we obtain

$$v^* v_{yy} + \left[\frac{U''}{c - U} - k^2 \right] |v|^2 = 0 \quad (5.5)$$

Integrating this between 0 and L , using an integration by part and the BC for v , we have

$$\int_0^L |v_y|^2 dy = \int_0^L \left[\frac{U''}{c - U} - k^2 \right] |v|^2 dy \quad (5.6)$$

whose imaginary part writes

$$\Im m(c) \int_0^L \frac{U''(y) |v(y)|^2}{|c - U|^2} dy = 0 \quad (5.7)$$

For this equation to admit a non zero $\Im m(c)$ value signaling an instability, the integral has to cancel, and this is only possible if the function under the sum has positive and negative values, which is only possible if U'' changes of sign, that is, if the stationary profile $U(y)$ presents

²this is called a Sturm-Liouville problem by mathematicians.

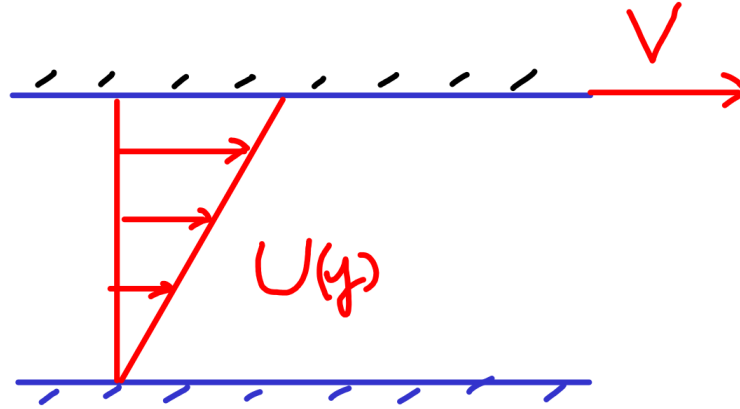
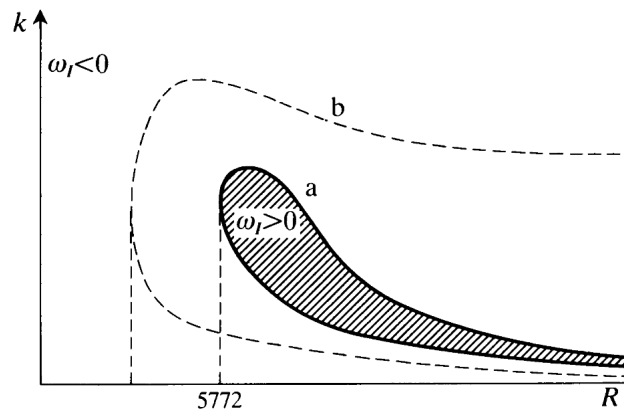


Figure 5.7: Geometry of a planar Couette flow.

an inflection point. We therefore state the very nice theorem : a velocity profile whose concavity does not change across the channel is stable in linear perturbation analysis of the Euler equation.

As an example the Poiseuille plane flow that was discussed in 2.4.1 is stable in this framework.

If we now perform this linear stability analysis starting from the NS equation, the principle is the same but computations are much more involved, since one has to deal with a 4-th order Orr-Sommerfeld equation instead of the Rayleigh equation. This analysis has been done for the planar Poiseuille flow and leads to the results summarized in figure 5.8. The unstable region



1. Marginal stability curves for (a) plane Poiseuille flow and (b) a typical velocity profile having an inflection point.

Figure 5.8: Stability of the planar Poiseuille flow

corresponds to the dashed part of the diagram, and appears for $Re \sim 5772$. The unstable modes are restricted to a finite domain of wavelength that increases with Re .

5.5 Other instability mechanism

The absence of linear instability that we have mentioned in the previous section does not mean that stability is granted, since there are other ways to develop an instability. Let us explain this with a simple model :

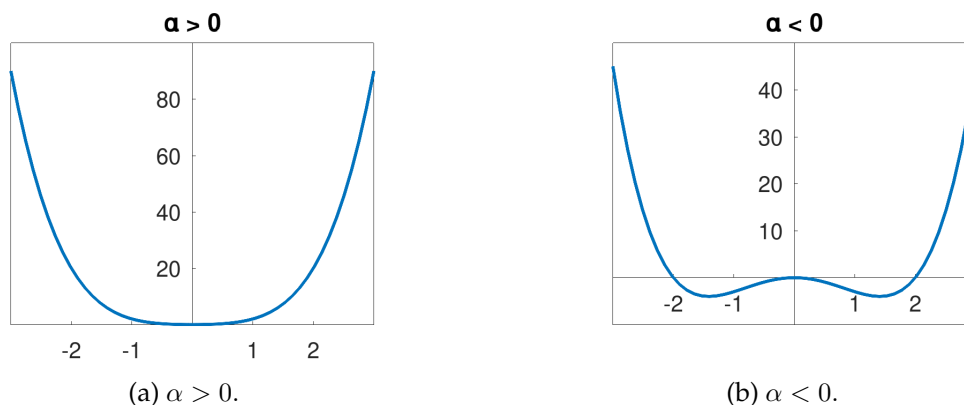


Figure 5.9: A double well potential.

Consider a 1D particle $x(t)$ moving in a potential $V(x) = \alpha x^2 + x^4$, see figure 5.9. For $\alpha > 0$ there is a stationary solution at $x = 0$. A perturbation analysis as we have done in this chapter will reveal that this fixed point becomes unstable³ when $\alpha < 0$, as the symmetric potential becomes a double well, and this corresponds to a linear loss of stability as we already discussed.

However another situation is possible : consider now the potential $V(x) = -x^2/\beta + x^4$, with $\beta > 0$ for "physical" reasons. Note that this is only a relabeling $\alpha = -1/\beta$ and that $\beta > 0$ corresponds to the double well situation of figure 5.9b. If the system sits at one of the two equilibrium points, say $x_+ = +\sqrt{1/2\beta}$, a linear stability analysis will reveal that this equilibrium is always linearly stable for any value of $\beta > 0$. However when β increases, the barrier between the two minima becomes very flat, and any exterior perturbation can make the system flip from x_+ to x_- .

This last situation is what seems to happens for the Poiseuille flow in a pipe studied by Reynolds in 1883, and that we discussed in 2.5.2. Analytical results and careful experiments point to the fact that this flow is linearly stable : it has been possible, by reducing all the mechanical noises due to the environment, to obtain stable flows at very high values of the Reynolds number $Re \sim 10^5$. However in an everyday situation with external perturbations, vibrations ... this flow becomes unstable around $Re \sim 2000$. Note that the subject of the loss of stability and transition to turbulence is still a very active field of fundamental research.

5.6 Taylor-Couette instability

The Couette flow between two cylinders has been the subject of an intense research activity during the XX-th century, culminating with the works of Taylor in 1920-30, and eventually leading to the discovery of the Lorenz attractor in 1963. One of the reasons for this activity is that this flow serves as a model for the global flow of the terrestrial rotating atmosphere. In the rotating frame, a Coriolis force is present and leads to many interesting instability behaviors.

³A particle physicist will speak of "spontaneous symmetry breaking".

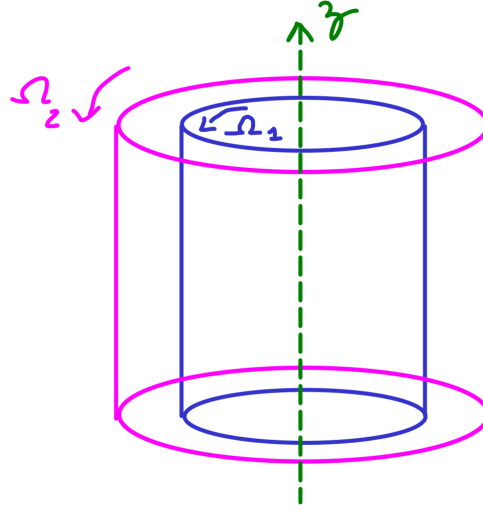


Figure 5.10: Geometry of the Taylor Couette flow.

We recall the classical geometry that we studied in chapter 2.6.2 as shown in figure 5.10. The fluid is moving between the two cylinders, and we study the stability of the laminar flow as we change the values of the angular velocities Ω_1 and Ω_2 . The laminar flow is given by

$$\begin{aligned}
 u_\theta(r) &= Ar + \frac{B}{r} \\
 A &= \frac{\Omega_2 r_2^2 - \Omega_1 r_1^2}{r_2^2 - r_1^2} \\
 B &= (\Omega_1 - \Omega_2) \frac{r_2^2 r_1^2}{r_2^2 - r_1^2}
 \end{aligned} \tag{5.8}$$

Here we will use a simple buoyancy argument similar to the one we presented at the end of section 5.2, to avoid the complicated analysis developed by Taylor. In the stationary flow, a particle is in equilibrium between the centrifugal force and the pressure gradient forces

$$\frac{1}{\rho} \partial_r p = \omega^2 r = \frac{L^2}{r^3}$$

Suppose now that this particle moves outside to a larger $r + dr$. Since all the forces are central, its angular momentum will be conserved : $L(r) = Cte$ during the displacement. In order for this particle to be pulled back to its original position, the pressure gradient at $r + dr$ has to be larger than the centrifugal force acting on the particle, leading to the condition

$$\frac{L(r)^2}{(r + dr)^3} < \frac{L(r + dr)^2}{(r + dr)^3}$$

Stated otherwise $|L(r)|$ has to be growing for the flow to be stable. Applying this condition to the internal and external cylinders leads to

$$|\Omega_2| r_2^2 > |\Omega_1| r_1^2 \tag{5.9}$$

for the flow to be stable. We see in particular that when the outside cylinder is fixed, $\Omega_2 = 0$ and the flow is unstable.

In 1923 G.I. Taylor has done the complete analysis using the Navier Stokes equation to take viscosity into account. In order to be tractable he assumed that the gap between the cylinders was very small $(r_2 - r_1)/r_1 \ll 1$, as is the case for the terrestrial atmosphere. His results are summarized in figure 5.11.

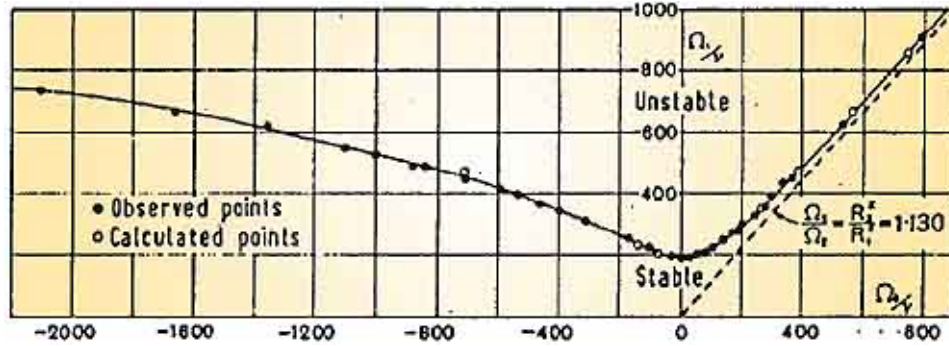


Figure 5.11: Taylor stability diagram for the Couette flow.

The x axis corresponds to Ω_2 , the y axis to Ω_1 . In our analysis without viscosity, the stable region is in the lower right corner of the graph under the dashed line. As expected, viscosity stabilizes the flow, as the unstable region situated above the solid line is reduced. The agreement between the experimental and theoretical values on the diagram is impressive and validates the theory.

If the shear is increased starting from a laminar stationary flow, toroidal Taylor vortices rotating in alternating direction and forming convection cells will form between the two cylinders, as shown in figure 5.12.

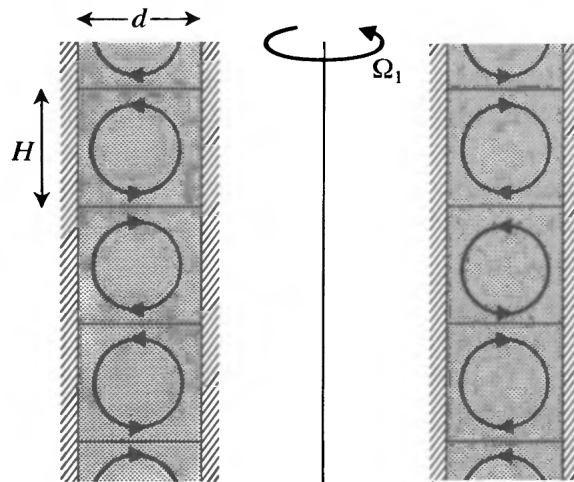


Fig. 9.8. Taylor vortices

Figure 5.12: Taylor vortices.

Increasing the shear further, these vortices will lose stability and undulate in the orthoradial direction as in figure 5.13.

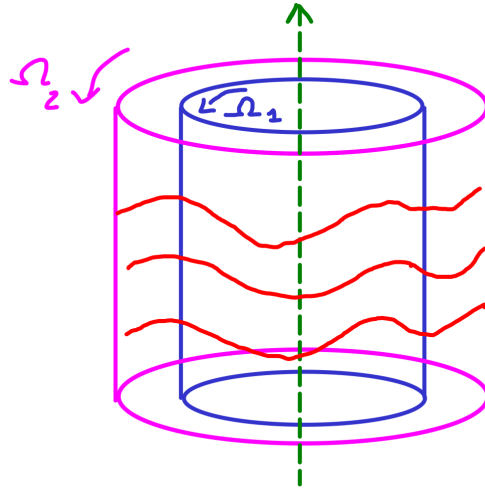


Figure 5.13: Undulating Taylor vortices.

Further increasing the shear leads to a chaotic behavior : turbulence, and we now conclude this course with an introductory chapter on this fascinating subject.

Chapter 6

Turbulence

6.1 Introduction - What is turbulence ?

One of the most complex and important subject in hydrodynamics, turbulence is still the subject of an intense research. This field is experiencing a renewal of interest since this new century, with contributions coming from statistical physics among others.

'When I meet God, I am going to ask him two questions: Why relativity ? And why turbulence ? I really believe he will have an answer for the first.'

Often attributed to W. Heisenberg, although it might be from Lenz, this quote illustrates the difficulty of the subject.

However in a sense, we nowadays have the 'why?' and what we need is the 'how?' As a matter of fact, we now have a satisfactory explanation for turbulence as we saw in the last chapter 5 on instabilities : the non-linearity of the Navier-Stokes equations enables unstable modes to appear and interact in a chaotic way. What is now lacking is a general operational description of turbulence : are there some general laws describing how turbulence arises and develops, and are there some general laws to describe its dynamical behavior ? The practical applications of these questions are tremendous : almost every real flow is turbulent, starting from the atmospheric global flow and the onset of depressions and hurricanes, to any pipe flow in an industrial plant



Figure 6.1: A turbulent flow.

Let us first give an operational definition of what is turbulence, as it is not so simple to characterize, and is often used as a misnomer for perfectly regular flows.

A flow is turbulent when the following characteristics are met :

- Irregular unpredictable dynamics at many spatial and temporal scales.
- Rotational flow.
- Mixing in the usual sense.
- Intermittency of regular and irregular patterns.

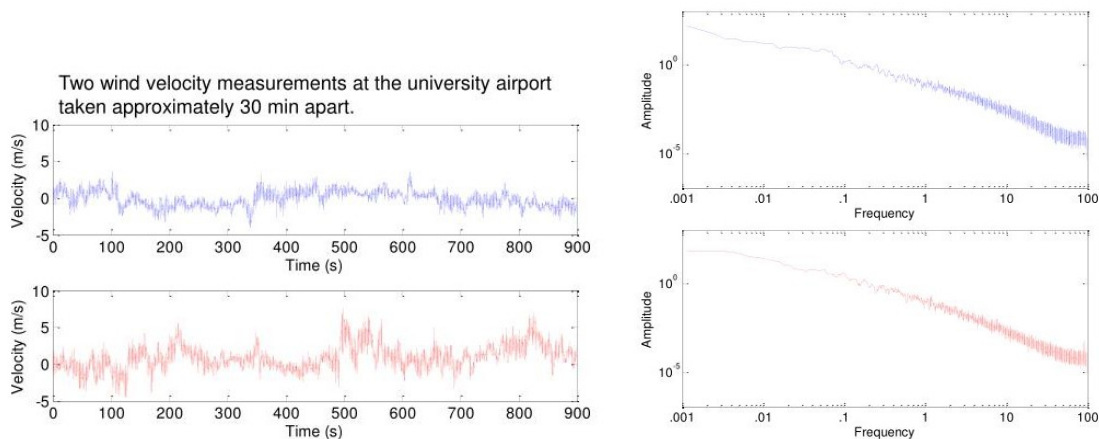
When these characteristics are met, the flow is almost certainly turbulent, as in figure 6.1. Quite at the opposite a Google™ search 'Turbulence' will provides you the following pictures 6.2, that do not share the previous characteristics : absence of many scales, no mixing, no vorticity for the second picture ... these are not turbulent flows !



Figure 6.2: Non turbulent flows !

Let us now briefly address the different characteristics that we have mentioned before, in order to better understand what we are speaking about.

6.1.1 Irregular flow



(a) Record of one component of the wind velocity.

(b) The corresponding energy spectrum.

In figure 6.3a we see a typical record of the wind velocity on a plane surface (an airport). Observe that this flow is clearly not regular and unpredictable in its details. A Fourier analysis

of its energy spectrum will reveal the presence of many frequencies contributing to this velocity component, see figure 6.3b.

From the above recording one can extract many important informations by taking mean values in time of the signal or of its powers : $\langle v_x \rangle$, $\langle v_x^2 \rangle$, $\langle v_x v_y \rangle$, $\langle v_x(t) v_x(t + \tau) \rangle$... as shown in figure 6.4. This will enable to decompose the velocity into a mean part and a fluctuating part

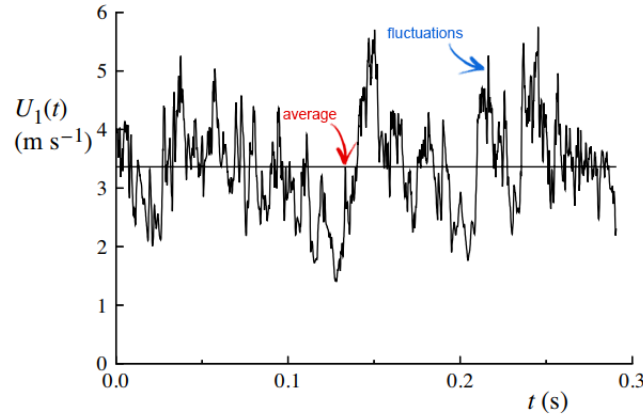


Figure 6.4: Time mean value of a turbulent velocity component.

that we may try to describe separately :

$$v(t) = \bar{v} + \delta v(t)$$

As usual, time mean value is the best option for the experimental physicist who has not the possibility¹ to 'repeat the experiment with exactly the same conditions, at least 1000 times to have a sufficient statistic'.

The assumption usually done is that when conditions are not changing drastically, this *time* mean value is equivalent to an hypothetical *ensemble* mean value.

This statistical ensemble is a very useful tool for the theoretician, although it rarely exists in practice. The equivalence between the two means is called the *Ergodic hypothesis* and this supposes that the conditions for the flow are stationary. In this way we can define mean values that may depend on position and time :

$$\vec{U}(\vec{r}, t) = \langle \vec{v}(\vec{r}, t) \rangle$$

One of the most important question that emerges then, both theoretically and practically, is to assert in what measure these measurements, mean values, spectrum, profiles ... depend or not on the precise experimental setup, or on the contrary if they are generic ? Has turbulence always the same look and feel ? Are there some general laws to describe it independently on the precise experimental setup ? Are there general laws for turbulence ?

6.1.2 Intermittency

One of the difficulties met when one tries to derive general laws for hydrodynamic turbulence is the fact that the dynamic is random, but not extremely random, as would be for example

¹Unless you are working with bacteria, bless you !



Figure 6.5: Coexistence of laminar and turbulent flow at intermediate Reynolds number in a planar forced flow.

the case with the molecules in a gas at equilibrium. This is shown in figure 6.5 where a planar forced flow exhibits coexistence between turbulent and laminar phases. It also occurs in a similar way in the Poiseuille flow where bursts of turbulence coexists with laminar phases as shown below



Indeed in turbulence, spatial and temporal correlations exist at many scales not unlike in phase transitions, and methods used to describe these correlations (e.g. renormalization group) are now being used to describe the onset of turbulence in certain situations.

6.1.3 Rotational flow and mixing

In turbulence, the flow is rotational, with eddies at all the scales. This shows that the fundamental underlying equation to describe it are the Navier Stokes equations. As a matter of fact we saw that the Euler equation can not explain the formation of vortices. This also show

that the turbulence problem is a 3D one. Turbulence in two spatial dimensions is trivial since vorticity is then only one dimensional. This is also annoying for theoreticians, since one of our favorite tricks is to reduce the dimension to make the study simpler, bad luck there ! As a matter of fact, turbulence has resisted all the favorite tricks of the theoretician guys : you can not linearize the equation, you can not go to lower dimensions, you can not average to obtain simple results . . . , man ain't it hard !

One essential aspect of turbulence, of utmost practical importance and much used in everyday's life, is the fact that it provides a very strong mixing in the fluid, due to the presence of eddies at all scales. As a matter of fact this mechanism is much more effective than diffusion to obtain homogeneous temperatures, concentrations, momentum . . . in a fluid. This is what you do when you stir your cup of tea, and this is also massively used in industrial pipe flows, as e.g. in heat exchangers in industrial plants.

6.1.4 Structures (eddies) at all scales

A very good example of such structures is given by the flow of a low pressure atmospheric system creating storms on the Atlantic coast. The largest vortices are caused in part by the Earth rotation, see appendix D.



- The full system has a size of some 1000 km and more, and its typical time scale would be about 1 week.
- A Squall such as is hitting a boat would last about half an hour and its size would be about $10^4 - 10^5$ m.
- A Breeze or a short increase in the wind may last about 10 seconds and the corresponding size is about 100 m.

then some smaller structures, often not relevant to the sailor-man but more to the physicist, exist until a microscopical scale that we will determine soon.

6.1.5 A more precise statement of the question

One of the standard setup to study turbulence is shown in figure 6.6. A grid is positioned

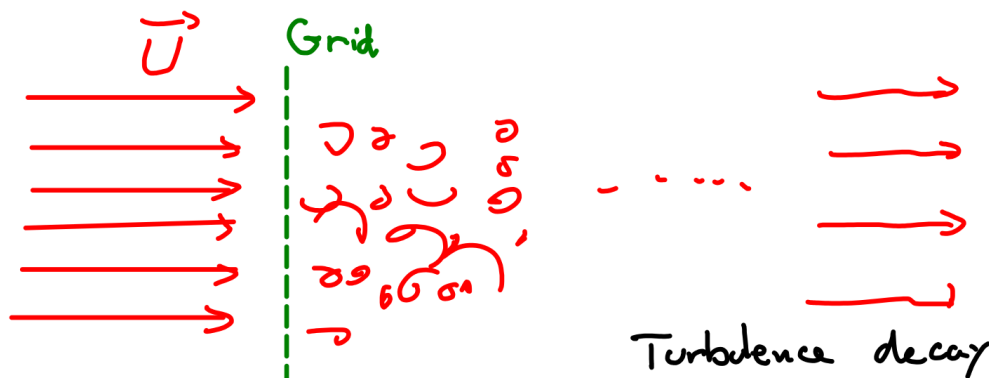


Figure 6.6: A wind tunnel with a grid is a simple setup to study turbulence.

across a wind or water tunnel at large Reynolds number. In this device we can record the velocity downstream of the grid, and it can be decomposed as

$$\vec{u} = \vec{U}_{mean} + \delta\vec{u}_{turbulent}$$

It is then remarkable that the statistical distribution of the turbulent velocity $\delta\vec{u}_{turbulent}$ not far from the grid is homogeneous, isotropic and stationary. It is also slowly decreasing with time as we follow the main flow. In this situation of *fully developed turbulence*, the main question that has been asked by physicists, both on theoretical and experimental side, is to know if these statistical properties are intrinsic and do not depend on the details of the experiment (e.g. the grid size or its geometry), and try to determine them as precisely as possible.

6.2 Richardson-Kolmogorov cascade

The idea of eddies at multiple scales is actually very ancient and can be traced back to Leonardo, who probably coined the word 'turbulence' in his description of the phenomenon. Around 1930 Richardson suggested that energy is injected in the flow at the scale of the largest structures, and then transferred toward vortices of smaller and smaller scales, to be ultimately dissipated in heat by viscosity. This dissipation appears when the viscous terms in the Navier Stokes equation become relevant, and this corresponds to velocity and length scales such that

$$Re = \frac{UL}{\nu} \sim 1$$

In fully developed turbulence, when the regime is stationary, the energy flow from larger scales towards smaller ones must be constant across the different scales. If we denote by ε the rate of transfer of energy by mass unit of fluid this quantity has the dimensions

$$[\varepsilon] = J/s/kg = [L]^2[T]^{-3}$$

and this visually corresponds to what is shown in figure 6.7.

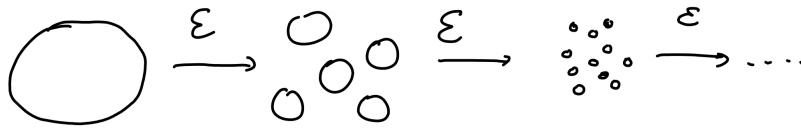


Figure 6.7: The Richardson Kolmogorov cascade.

Kolmogorov, in a seminal paper of 1941 argued that at the intermediary scales of the cascade, the energy transfer could not depend on how the energy is injected at the largest scale and is dissipated at the smaller one. The transfer mechanism and the properties of the vortices at intermediary scales are the same for every fully turbulent flow : they are universal. This is denoted as 'the inertial regime', since viscosity plays no role at these scales.

In this inertial regime, at a scale where vortices have a typical size L and a typical velocity U , the specific energy transfer rate ε can only depend on these two quantities (this is the universality hypothesis) and homogeneity ensures that up to a numerical factor

$$\varepsilon = \frac{U^3}{L} = Cte \quad (6.1)$$

This is the fundamental result of Kolmogorov. The value of ε is determined by the energy injection at the larger scale. For example when I stir my cup of tea,

$$L_{in} \simeq 0.05 \text{ m}, \quad U_{in} \simeq 0.2 \text{ m/s}, \quad \varepsilon \simeq 0.16 \text{ J/s/kg}$$

At a scale of $L = 1 \text{ mm}$ the typical velocity of the eddies in their rest frame is then given by

$$\frac{U}{U_{in}} = \left(\frac{L}{L_{in}} \right)^{1/3} \quad \text{so that} \quad U \simeq 0.05 \text{ m/s}$$

In this way, one can relate order of magnitudes at different scales. The smallest scale is the one at which dissipation by viscosity occurs :

$$\frac{U_o L_o}{\nu} = 1, \quad U_o = \frac{\nu}{L_o}, \quad \varepsilon = \frac{\nu^3}{L_o^4} \quad (6.2)$$

and finally the scale at which dissipation occurs and terminates the cascade is

$$L_o = \left(\frac{\nu^3}{\varepsilon} \right)^{1/4}, \quad U_o = (\varepsilon \nu)^{1/4}, \quad T_o = \left(\frac{\nu}{\varepsilon} \right)^{1/2} \quad (6.3)$$

with ε given by equation (6.1) at the injection scale. We see that the less the viscosity, the smallest the dissipation scale. In the previous example with $\nu = 10^{-6} \text{ m}^2/\text{s}$, we find

$$L_o \simeq 50 \text{ } \mu\text{m}, \quad U_o \simeq 2.10^{-2} \text{ m/s}, \quad T_o \simeq 2.5 \text{ ms}$$

and this gives an idea of the effectiveness of the mixing process in turbulence.

Exercise 6.1. Check the above results. In the case of a very large system as a cyclone, what is the ultimate scale of dissipation ?

Let us present this result in an equivalent but more useful formulation for experimentalists. We use the wave vector $k \sim 1/L$. From the Kolmogorov hypothesis, we then have $U^3 k = Cte$. Differentiating this relation we obtain $3 dU/U + dk/k = 0$ and hence the variation of scale and velocity in the cascade are related by

$$|dU| = \frac{1}{3} U \frac{dk}{k}$$

The specific energy of the fluid at this scale is given by $E = U^2/2$ and its density in a scale interval is $dE = U dU = \frac{1}{3} U^2 \frac{dk}{k}$ and the spectral density of energy is then given by $dE/dk = \frac{1}{3} U^2/k = \frac{1}{3} \varepsilon^{2/3} k^{-5/3}$ or

$$\rho_E(k) \sim \varepsilon^{2/3} k^{-5/3} \quad (6.4)$$

This is the famous Kolmogorov cascade energy spectrum. This result is well observed experimentally as you can check in the following figure 6.8.

Exercise 6.2. In the previous experimental results, the horizontal axis is given in frequencies f and not in wave vector k . What is the relation between these two quantities ? You have to think about the experimental setup to answer this question : how the experiment is done, and what is measured by the experimentalist ?

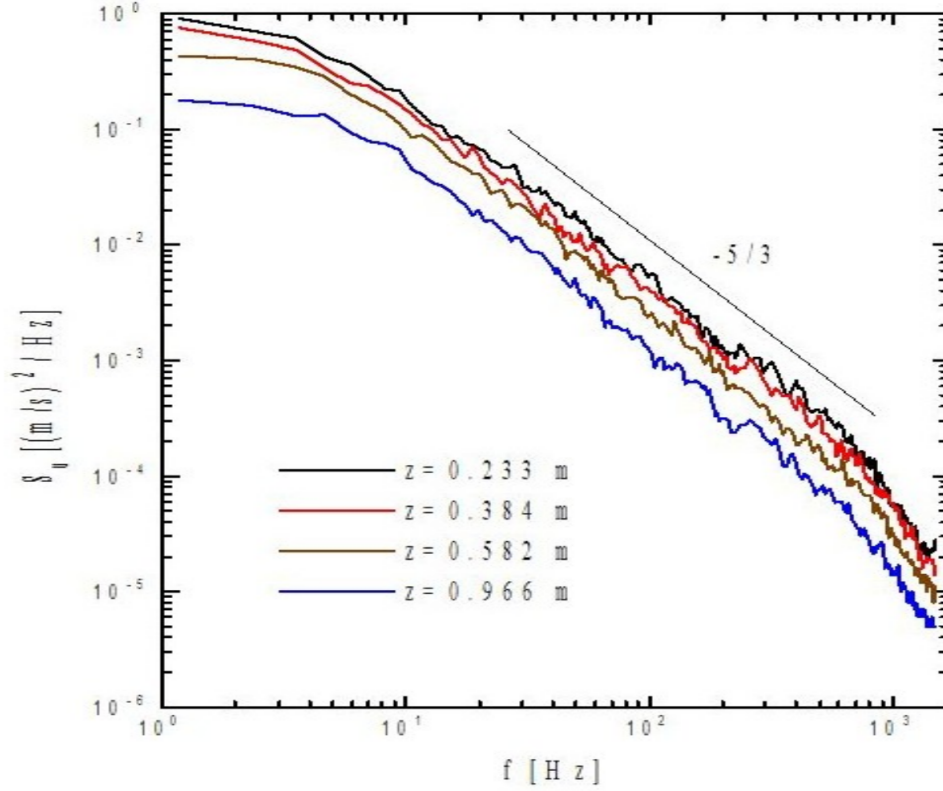


Figure 6.8: The Kolmogorov energy spectrum.

6.3 Reynolds stress tensor

We already stated that the mixing caused by turbulence causes an important transfer of linear momentum. This is made obvious by a result due to Reynolds. We start again from the NS equations

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{v} = -\frac{1}{\rho} \vec{\nabla} p + \nu \Delta \vec{v} \quad , \quad \vec{\nabla} \cdot \vec{v} = 0 \quad (6.5)$$

where we have incorporated the hydrostatic pressure $-\rho g z$ in the p term as usual. We may write this in tensorial form :

$$\partial_t v_i + v_j \partial_j v_i = -\frac{1}{\rho} \partial_i p + \nu \partial_j \partial_j v_i \quad (6.6)$$

Let us now decompose the velocity field in a turbulent random part and a non turbulent laminar one as seen before : $u_i = U_i + u'_i$ with by definition $\langle u'_i \rangle = 0$. In the same way we write $p = P + p'$ with $\langle p' \rangle = 0$. We use this decomposition in the Navier Stokes equation, take the statistical mean of that equation, and use the fact that $\partial_i u'_i = 0$ obtained by taking the mean of the incompressibility equation. This results in

$$\partial_t U_i + U_j \partial_j U_i = -\frac{1}{\rho} \partial_i P + \nu \partial_j \partial_j U_i - \langle u'_j \partial_j u'_i \rangle \quad (6.7)$$

we can write this in the form of a conservation equation

$$\rho (\partial_t U_i + U_j \partial_j U_i) = \partial_j T_{ij}^t \quad (6.8)$$

where T_{ij}^t is the total stress tensor containing a pressure diagonal part, a laminar viscous part and a turbulent statistical one called the Reynolds stress tensor $T_{ij}^R = -\rho\langle u'_i u'_j \rangle$, such that

$$T_{ij}^t = -p\delta_{ij} + \eta(\partial_i U_j + \partial_j U_i) + T_{ij}^R$$

So we have obtained a Navier Stokes equation for the laminar mean part of the flow, with a stress tensor part associated to a transfer of impulsions by vortices in the turbulent flow $-\rho\langle u'_j \partial_j u'_i \rangle$. The hope is now to close this expression by expressing this fluctuating part as an effective function of the mean flow

$$\langle u'_j u'_i \rangle \stackrel{?}{=} f(U_i)$$

In that case we can solve the Navier-Stokes equation for the laminar flow U_i , while taking into account the role of turbulence. This is what is done in any simulation involving turbulence, notably for meteorology simulations. Note in particular that the role played by the Reynolds tensor is crucial to enable dissipation of energy by mixing, since at large scales the mean viscous term is negligible as the mean Reynolds number is large.

Exercise 6.3. Reproduce in detail the chain of calculus leading from the Navier Stokes equation to (6.7).

6.4 The log velocity profile

Stemming from the previous idea, another fundamental result on turbulence obtained during the XX-th century concerns the mean velocity profile above a flat surface in the case of a high Reynolds, turbulent flow. This describes in particular the mean wind profile at sea or on an airport. This was obtained by Prandtl and Von Karman around 1930.

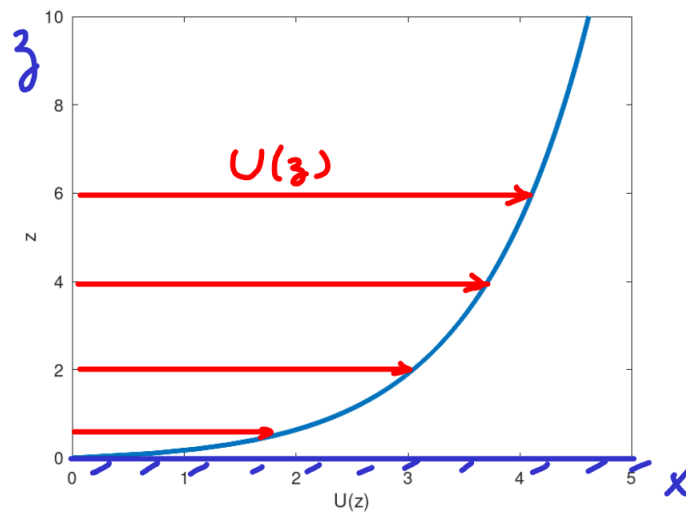


Figure 6.9: The Log mean velocity profile above a flat surface.

This profile will be approximately valid at distances large in comparison with the boundary layer above the ground, and small in comparison with the larger scales responsible for the global flow, such as the distance to the walls in the case of a channel flow. Let us see how to motivate this result, that seems reasonably well established experimentally (see below).

6.4.1 Kinetic model of viscosity

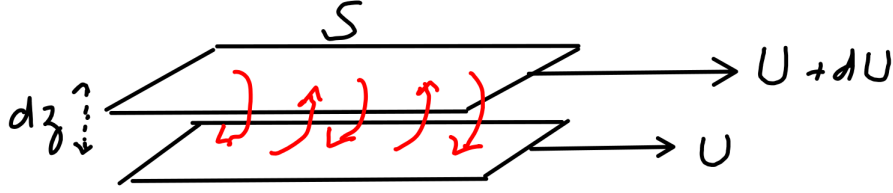


Figure 6.10: A simple microscopic viscosity model.

To understand the line of thought behind Prandtl result, it is useful to come back to a very simple kinetic model of viscosity in a gas, as due to the exchange of momentum by thermal diffusion. Let us consider two laminar layers with different velocities, as in figure 6.10, exchanging momentum by diffusion. An analogy often used is that one of two trains of different speed, exchanging mailbags at a constant rate.

Two surfaces S exchange particles at a rate $\bar{n} \frac{\bar{v}}{3} S$, where $\bar{v}$ is the mean kinetic velocity, $\bar{n}$ the mean density of particles, and the factor $1/3$ comes from the 3 spaces dimensions². The transferred momentum from one slab to the other is then $\bar{n} \frac{\bar{v}}{3} S m dU$ and this corresponds to a stress

$$t = \bar{n} \frac{\bar{v}}{3} S m \partial_z U dz \quad (6.9)$$

dz is the typical distance in which an exchange of momentum takes place, in other words this is the mean free path l such that $l\sigma\bar{n} = 1$, σ being the collision cross-section. Finally using the fact that $\rho = \bar{n}m$, we obtain the kinetic viscosity for a perfect gas :

$$\nu = \frac{\bar{v}l}{3} \quad (6.10)$$

Note that in this model, the viscosity increases with temperature, which is correct for a gas, but not for a liquid for which other transfer mechanisms are at play. The take home message is that in a gas, viscosity is the consequence of a microscopic exchange of momentum by thermal diffusion.

6.4.2 Turbulent viscosity model

We are now in position to adapt these ideas to the case where turbulence is present, and causes a much stronger momentum transfer, due to macroscopic eddies, than the microscopic transfer associated with diffusion. In order to do so we have to give a typical velocity for the transfer (the easy part) and a typical exchange size playing the role of mean free path (the hard part).

Shear velocity

Let us look at the geometry described in figure 6.9. Outside from the boundary layer, and in stationary regime, we use the decomposition of the flow as a laminar and turbulent part : $\vec{v} = (U(z) + u, 0, w)$. The first component of mean NS, equation (6.7) writes

$$0 = \partial_t U = -\partial_z \langle uw \rangle$$

²We already saw this argument in the perfect gas kinetic pressure theory.

which states that in the stationary regime, the momentum transfer is globally constant (no accumulation of momentum somewhere). This enables to define a shear velocity as

$$u^* = \sqrt{\langle uw \rangle} \quad (6.11)$$

with a constant value everywhere in the flow. The only component of the Reynolds turbulent stress tensor is then $t^R = \rho u^{*2}$, and u^* will characterize the typical velocity of eddies, superposed to the main stream velocity U .

Mixing length

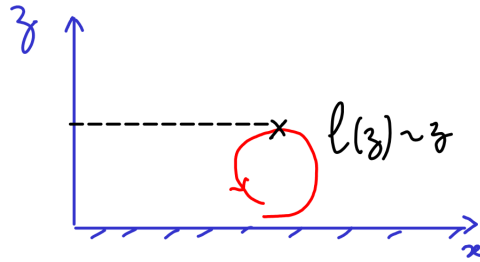


Figure 6.11: The mixing length at an altitude z .

We now have to give the typical length over which the eddies are exchanging momentum. The idea of Von Karman is that the most effective momentum exchange is produced by the largest possible eddies, and these have a typical size of the order z at a distance z from the wall, see figure 6.11 :

$$l = \kappa z$$

κ is called the 'Von Karman' constant, and is supposed to be universal, that is not depending on the macroscopic geometry of the flow. Its experimentally deduced value (see below) is $\kappa \simeq 0.44$.

Turbulent viscosity and the Log profile

Using then the same ideas as for the kinetic viscosity, we write

$$\nu_t = u^* l = u^* \kappa z$$

and use this result to express the turbulent stress tensor as

$$t^R \equiv \rho u^{*2} = \rho \nu_t \frac{\partial U}{\partial z}$$

that can be simplified as

$$u^* = l \frac{\partial U}{\partial z}$$

Using $l = \kappa z$ this equation can be integrated with respect to z resulting in

$$U(z) = \frac{u^*}{\kappa} \log \left(\frac{z}{z_0} \right) \quad (6.12)$$

This is the famous log-profile of the mean velocity above a flat surface.

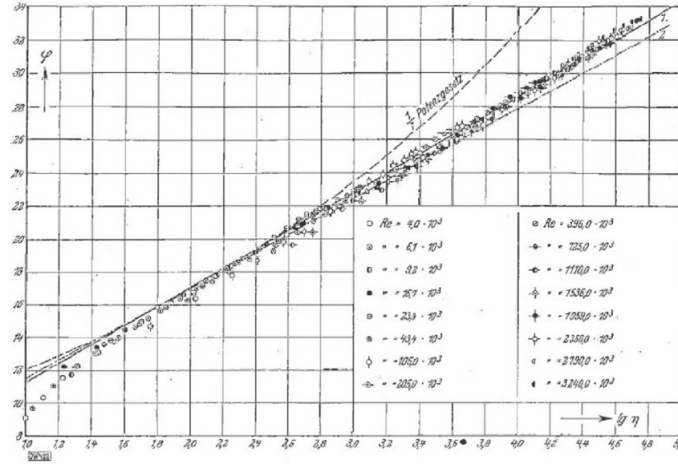


Figure 6.12: Experimental results of Nikuradse (1932).

Experimental status of the Log law

In 1932 Nikuradse, a Georgian-German physicist then student of Prandtl, conducted an impressive series of experiments on turbulent flows in rough pipes, measuring the velocity profiles as a function of the distance from the wall, in the range $Re \sim 10^3 - 10^6$. His results have been completed since and indicates that the log profile law is a good first approximation of what is measured, as shown in figure 6.12. These curves enables the experimental determination of κ by fitting, The value of u^* being deduced from the measurement of the stress component t^R .

Taking the usual viscosity into account

The log profile given in the previous subsection diverges for $z = 0$, since the usual velocity and the boundary layer has not been taken into account. In that case we should write

$$(\nu + \kappa z u^*) \frac{\partial U}{\partial z} = u^{*2}$$

Which results in the modified law

$$U(z) = \frac{u^*}{\kappa} \log \left(1 + \frac{z}{z_0} \right) \quad (6.13)$$

where the 'Roughness length' $z_0 = \frac{\nu}{\kappa u^*}$ measures the thickness of the viscous boundary layer.

6.5 Conclusion

In this chapter we have presented some of the most accepted and universal results about hydrodynamic turbulence, namely the Kolmogorov cascade and the Prandtl - Von Karman log-profile of the mean velocity above a plane surface at large Re . These results are still discussed and criticized, notably in what concerns their universality, and the range of their validity, but they form the basis of more modern theories about turbulence.

With some consecrated words 'this domain is still an active field of research, that has gained a renewal of interest in the recent years'. We advise the interested reader to explore the more

recent literature on the subject, and close here this introductory course to Physical Hydrodynamics.

Appendix A

Some Useful Formula

A.1 operators

A.1.1 Vector identities :

$$\nabla \times (\nabla \rho) = 0$$

$$\nabla \cdot (\nabla \times \mathbf{u}) = 0$$

$$\nabla \times (\nabla \times \mathbf{u}) = \nabla(\nabla \cdot \mathbf{u}) - \Delta \mathbf{u}$$

Derivative of products are obtained using the Leibniz rule and tensorial calculus :

$$\nabla(\rho g) = (\nabla \rho)g + \rho \nabla g$$

$$\nabla \cdot (\rho \mathbf{u}) = (\nabla \rho) \cdot \mathbf{u} + \rho \nabla \cdot \mathbf{u}$$

$$\nabla \times (\rho \mathbf{u}) = (\nabla \rho) \times \mathbf{u} + \rho \nabla \times \mathbf{u}$$

$$\nabla \cdot (\mathbf{u} \times \mathbf{v}) = (\nabla \times \mathbf{u}) \cdot \mathbf{v} - (\nabla \times \mathbf{v}) \cdot \mathbf{u}$$

$$\nabla \times (\mathbf{u} \times \mathbf{v}) = \mathbf{u}(\nabla \cdot \mathbf{v}) - \mathbf{v}(\nabla \cdot \mathbf{u}) + (\mathbf{v} \cdot \nabla) \mathbf{u} - (\mathbf{u} \cdot \nabla) \mathbf{v}$$

$$\nabla(\mathbf{u} \cdot \mathbf{v}) = (\mathbf{u} \cdot \nabla) \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{u} + \mathbf{u} \times (\nabla \times \mathbf{v}) + \mathbf{v} \times (\nabla \times \mathbf{u})$$

A.1.2 Cartesian Coordinates:

$$\mathbf{r} = x \mathbf{e}_x + y \mathbf{e}_y + z \mathbf{e}_z$$

$$d\mathbf{r} = dx \mathbf{e}_x + dy \mathbf{e}_y + dz \mathbf{e}_z$$

$$\nabla p = \frac{\partial p}{\partial x} \mathbf{e}_x + \frac{\partial p}{\partial y} \mathbf{e}_y + \frac{\partial p}{\partial z} \mathbf{e}_z$$

$$\nabla \cdot \mathbf{u} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}$$

$$\nabla \times \mathbf{u} = \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ \partial_x & \partial_y & \partial_z \\ u_x & u_y & u_z \end{vmatrix}$$

$$\Delta p = \nabla \cdot \nabla p = \frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2}$$

$$\Delta \mathbf{u} = \Delta u_x \mathbf{e}_x + \Delta u_y \mathbf{e}_y + \Delta u_z \mathbf{e}_z$$

A.1.3 Cylindrical Coordinates :

$$\begin{aligned}
\mathbf{r} &= r \mathbf{e}_r + z \mathbf{e}_z & d\mathbf{r} &= dr \mathbf{e}_r + r d\theta \mathbf{e}_\theta + dz \mathbf{e}_z \\
\nabla p &= \frac{\partial p}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial p}{\partial \theta} \mathbf{e}_\theta + \frac{\partial p}{\partial z} \mathbf{e}_z & \nabla \cdot \mathbf{u} &= \frac{1}{r} \frac{\partial r u_r}{\partial r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} \\
\nabla \times \mathbf{u} &= \frac{1}{r} \begin{vmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & \mathbf{e}_z \\ \partial_r & \partial_\theta & \partial_z \\ u_r & r u_\theta & u_z \end{vmatrix} \\
\Delta p &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 p}{\partial \theta^2} + \frac{\partial^2 p}{\partial z^2} \\
\Delta \mathbf{u} &= \left(\Delta u_r - \frac{u_r}{r^2} - \frac{2}{r^2} \frac{\partial u_\theta}{\partial \theta} \right) \mathbf{e}_r + \left(\Delta u_\theta - \frac{u_\theta}{r^2} + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} \right) \mathbf{e}_\theta + \Delta u_z \mathbf{e}_z
\end{aligned}$$

A.1.4 Spherical Coordinates :

$$\begin{aligned}
\mathbf{r} &= r \mathbf{e}_r & d\mathbf{r} &= dr \mathbf{e}_r + r d\theta \mathbf{e}_\theta + r \sin \theta d\phi \mathbf{e}_\phi \\
\nabla p &= \frac{\partial p}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial p}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} \mathbf{e}_\phi & \nabla \cdot \mathbf{u} &= \frac{1}{r^2} \frac{\partial r^2 u_r}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial \sin \theta u_\theta}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial u_\phi}{\partial \phi} \\
\nabla \times \mathbf{u} &= \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & r \sin \theta \mathbf{e}_\phi \\ \partial_r & \partial_\theta & \partial_\phi \\ u_r & r u_\theta & r \sin \theta u_\phi \end{vmatrix} \\
\Delta p &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial p}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial p}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 p}{\partial \phi^2} \\
\Delta \mathbf{u} &= \left(\Delta u_r - \frac{2u_r}{r^2} - \frac{2}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (u_\theta \sin \theta) - \frac{2}{r^2 \sin \theta} \frac{\partial u_\phi}{\partial \phi} \right) \mathbf{e}_r \\
&\quad + \left(\Delta u_\theta + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r^2 \sin^2 \theta} - \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial u_\phi}{\partial \phi} \right) \mathbf{e}_\theta \\
&\quad + \left(\Delta u_\phi + \frac{2}{r^2 \sin \theta} \frac{\partial u_r}{\partial \phi} - \frac{u_\phi}{r^2 \sin^2 \theta} + \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial u_\theta}{\partial \phi} \right) \mathbf{e}_\phi
\end{aligned}$$

A.2 Hydrodynamics Equations

A.2.1 Vectorial form:

- Particular Derivative :

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u}$$

- Incompressibility :

$$\nabla \cdot \mathbf{u} = 0$$

- Euler Equation :

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \mathbf{g} \quad \text{BC :} \quad \mathbf{u} \cdot \mathbf{n}|_{\text{boundary}} = 0$$

- Navier Stokes Equation :

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \mathbf{g} + \nu \Delta \mathbf{u} \quad \text{BC :} \quad \mathbf{u}|_{\text{boundary}} = 0$$

- Strain Rate Tensor :

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right)$$

- Stress Tensor : $t_i = T_{ij}n_j$ is the stress i-component on a surface element with normal n_j pointing outside. For a Newtonian fluid one has :

$$T_{ij} = -p\delta_{ij} + 2\eta\epsilon_{ij} = -p\delta_{ij} + \eta \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

and Navier Stokes equations can be written as :

$$\rho \frac{Du_i}{Dt} = \frac{\partial T_{ij}}{\partial x_j} + \rho g_i$$

A.2.2 Cartesian Coordinates:

- Particular Derivative :

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u}$$

with

$$\mathbf{u} \cdot \nabla = u_x \frac{\partial}{\partial x} + u_y \frac{\partial}{\partial y} + u_z \frac{\partial}{\partial z}$$

- Incompressibility :

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = 0$$

- Navier Stokes Equation :

$$\begin{aligned} \left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \right) u_x &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + g_x + \nu \Delta u_x \\ \left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \right) u_y &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + g_y + \nu \Delta u_y \\ \left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \right) u_z &= -\frac{1}{\rho} \frac{\partial p}{\partial z} + g_z + \nu \Delta u_z \end{aligned}$$

- Strain Rate Tensor :

$$E = \begin{bmatrix} \partial_x u_x & (\partial_x u_y + \partial_y u_x)/2 & (\partial_x u_z + \partial_z u_x)/2 \\ (\partial_x u_y + \partial_y u_x)/2 & \partial_y u_y & (\partial_y u_z + \partial_z u_y)/2 \\ (\partial_x u_z + \partial_z u_x)/2 & (\partial_y u_z + \partial_z u_y)/2 & \partial_z u_z \end{bmatrix}$$

- Stress Tensor :

$$T = \begin{bmatrix} -p + 2\eta \partial_x u_x & \eta (\partial_x u_y + \partial_y u_x) & \eta (\partial_x u_z + \partial_z u_x) \\ \eta (\partial_x u_y + \partial_y u_x) & -p + 2\eta \partial_y u_y & \eta (\partial_y u_z + \partial_z u_y) \\ \eta (\partial_x u_z + \partial_z u_x) & \eta (\partial_y u_z + \partial_z u_y) & -p + 2\eta \partial_z u_z \end{bmatrix}$$

A.2.3 Cylindrical Coordinates:

- Particular Derivative :

$$\begin{aligned}\frac{Du_r}{Dt} &= \frac{\partial u_r}{\partial t} + (\mathbf{u} \cdot \nabla)u_r - \frac{u_\theta^2}{r} \\ \frac{Du_\theta}{Dt} &= \frac{\partial u_\theta}{\partial t} + (\mathbf{u} \cdot \nabla)u_\theta + \frac{u_r u_\theta}{r} \\ \frac{Du_z}{Dt} &= \frac{\partial u_z}{\partial t} + (\mathbf{u} \cdot \nabla)u_z\end{aligned}$$

with

$$\mathbf{u} \cdot \nabla = u_r \frac{\partial}{\partial r} + \frac{u_\theta}{r} \frac{\partial}{\partial \theta} + u_z \frac{\partial}{\partial z}$$

- Incompressibility :

$$\frac{1}{r} \frac{\partial}{\partial r}(r u_r) + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0$$

- Navier Stokes Equation :

$$\begin{aligned}\frac{Du_r}{Dt} &= -\frac{1}{\rho} \frac{\partial p}{\partial r} + \nu \left(\Delta u_r - \frac{u_r}{r^2} - \frac{2}{r^2} \frac{\partial u_\theta}{\partial \theta} \right) \\ \frac{Du_\theta}{Dt} &= -\frac{1}{\rho r} \frac{\partial p}{\partial \theta} + \nu \left(\Delta u_\theta + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r^2} \right) \\ \frac{Du_z}{Dt} &= -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \Delta u_z\end{aligned}$$

with the particular derivative given above, and the Laplacian scalar operator in cylindrical coordinates given by :

$$\Delta = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}$$

- Strain Rate Tensor :

$$\begin{aligned}\epsilon_{rr} &= \frac{\partial u_r}{\partial r} & \epsilon_{\theta\theta} &= \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} & \epsilon_{zz} &= \frac{\partial u_z}{\partial z} \\ 2\epsilon_{\theta z} &= \frac{1}{r} \frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z} & 2\epsilon_{zr} &= \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} & 2\epsilon_{r\theta} &= r \frac{\partial}{\partial r} \left(\frac{u_\theta}{r} \right) + \frac{1}{r} \frac{\partial u_r}{\partial \theta}\end{aligned}$$

A.2.4 Spherical Coordinates:

- Particular Derivative :

$$\begin{aligned}\frac{Du_r}{Dt} &= \frac{\partial u_r}{\partial t} + (\mathbf{u} \cdot \nabla)u_r - \frac{u_\theta^2}{r} - \frac{u_\phi^2}{r} \\ \frac{Du_\theta}{Dt} &= \frac{\partial u_\theta}{\partial t} + (\mathbf{u} \cdot \nabla)u_\theta + \frac{u_r u_\theta}{r} - \frac{u_\phi^2 \cot \theta}{r} \\ \frac{Du_\phi}{Dt} &= \frac{\partial u_\phi}{\partial t} + (\mathbf{u} \cdot \nabla)u_\phi + \frac{u_r u_\phi}{r} + \frac{u_\theta u_\phi \cot \theta}{r}\end{aligned}$$

with

$$\mathbf{u} \cdot \nabla = u_r \frac{\partial}{\partial r} + \frac{u_\theta}{r} \frac{\partial}{\partial \theta} + \frac{u_\phi}{r \sin \theta} \frac{\partial}{\partial \phi}$$

- Incompressibility :

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (u_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial u_\phi}{\partial \phi} = 0$$

- Navier Stokes Equation :

$$\begin{aligned} \frac{Du_r}{Dt} &= -\frac{1}{\rho} \frac{\partial p}{\partial r} + \nu \left(\Delta u_r - \frac{2u_r}{r^2} - \frac{2}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (u_\theta \sin \theta) - \frac{2}{r^2 \sin \theta} \frac{\partial u_\phi}{\partial \phi} \right) \\ \frac{Du_\theta}{Dt} &= -\frac{1}{\rho r} \frac{\partial p}{\partial \theta} + \nu \left(\Delta u_\theta + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r^2 \sin^2 \theta} - \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial u_\phi}{\partial \phi} \right) \\ \frac{Du_\phi}{Dt} &= -\frac{1}{\rho r \sin \theta} \frac{\partial p}{\partial \phi} + \nu \left(\Delta u_\phi + \frac{2}{r^2 \sin \theta} \frac{\partial u_r}{\partial \phi} - \frac{u_\phi}{r^2 \sin^2 \theta} + \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial u_\theta}{\partial \phi} \right) \end{aligned}$$

with the particular derivative given above, and the Laplacian scalar operator in spherical coordinates given by :

$$\Delta = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

- Strain Rate Tensor :

$$\begin{aligned} \epsilon_{rr} &= \frac{\partial u_r}{\partial r} & 2\epsilon_{r\theta} &= r \frac{\partial}{\partial r} \left(\frac{u_\theta}{r} \right) + \frac{1}{r} \frac{\partial u_r}{\partial \theta} & 2\epsilon_{r\phi} &= \frac{1}{r \sin \theta} \frac{\partial u_r}{\partial \phi} + r \frac{\partial}{\partial r} \left(\frac{u_\phi}{r} \right) \\ \epsilon_{\theta\theta} &= \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} & 2\epsilon_{\theta\phi} &= \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{u_\phi}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial u_\theta}{\partial \phi} & \epsilon_{\phi\phi} &= \frac{1}{r \sin \theta} \frac{\partial u_\phi}{\partial \phi} + \frac{u_r}{r} + \frac{u_\theta \cot \theta}{r} \end{aligned}$$

Appendix B

Tensor Calculus

Tensor calculus is an extension of vector and matrices calculus that was originally devised in the framework of Elasticity theory. It enables a better understanding and writing for vector computation and is a very useful tool of modern physics. It is important to master simple tensor calculus, and be able to make the correspondence with vector notations. This is the purpose of this simple introduction to help you doing that. Viewed by mathematicians, tensor are only elements of vector spaces formed by tensorial product (see below). Actually tensor theory was invented before vector space theory, and is more suited to physicists computations.

B.1 Basic correspondence

We begin with the basic correspondence in space dimension 3.

$$\begin{aligned} \text{Vector} &\longleftrightarrow \text{Tensor} \\ \vec{v} &\longleftrightarrow v_i, \quad i = 1 \dots 3 \end{aligned} \tag{B.1}$$

In both notation this represents 3 physical quantities that transform in a definite way under rotations, and denotes an intrinsic physical object, even when its components varies under a change of basis. (v_x, T, ρ) are three quantities, but they do not form a tensor ! More subtle $(\lambda_x, \lambda_y, \lambda_z)$ do not form a vector, but (k_x, k_y, k_z) do ! You may ask why (see below).

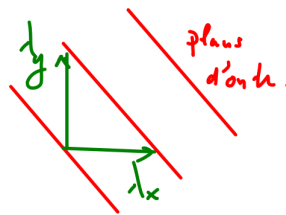


Figure B.1: You may define the pair (λ_x, λ_y) , but they *do not* form a vector, while (k_x, k_y) do !

B.2 Tensor product

From that we can form new objects such as

$$v_i w_j, \quad i = 1 \dots 3, \quad j = 1 \dots 3$$

which denotes 9 quantities transforming in a definite way under rotations. This is a tensor of rank 2. Note also that it is different from $\vec{v} \cdot \vec{w}$ or $\vec{v} \wedge \vec{w}$. Actually there exist a standard notation for this object using a *tensor product* :

$$v_i w_j \longleftrightarrow \vec{v} \otimes \vec{w}$$

It is also possible to represent this object as a matrix

$$v_i w_j \longleftrightarrow A = \begin{pmatrix} v_x w_x & v_x w_y & v_x w_z \\ v_y w_x & v_y w_y & v_y w_z \\ v_z w_x & v_z w_y & v_z w_z \end{pmatrix}$$

where the first index i is (by convention) the line index and j is the column index, but this convention is ambiguous and not very useful : it is given for pedagogical purpose only. Note also that $v_i w_j$ and $v_j w_i$ are different and correspond to transposed matrices A and A^T .

B.3 Scalar product, vector product

It is now useful to construct new objects

- $v_i w_i \left(= \sum_{i=1}^3 v_i w_i \right) \longleftrightarrow \vec{v} \cdot \vec{w}$ is a scalar invariant by rotations and corresponds to the *trace* of the tensor product $v_i w_j$. Here we have used Einstein's convention of summation : all repeated indices are summed and become dummy namely $v_i \neq v_j$ but $v_i w_i = v_j w_j$.
- $(v_i w_j + v_j w_i)/2 \longleftrightarrow (A + A^T)/2$ corresponds to the symmetrized of A .
- $(v_i w_j - v_j w_i)/2 \longleftrightarrow (A - A^T)/2$ corresponds to the anti-symmetrized of A . Written explicitly this object is actually (up to a factor two) equivalent to the vector product :

$$\begin{pmatrix} 0 & v_x w_y - v_y w_x & v_x w_z - v_z w_x \\ v_y w_x - v_x w_z & 0 & v_y w_z - v_z w_y \\ v_z w_x - v_x w_z & v_z w_y - v_y w_z & 0 \end{pmatrix} = \begin{pmatrix} 0 & (\vec{v} \wedge \vec{w})_3 & -(\vec{v} \wedge \vec{w})_2 \\ -(\vec{v} \wedge \vec{w})_3 & 0 & (\vec{v} \wedge \vec{w})_1 \\ (\vec{v} \wedge \vec{w})_2 & -(\vec{v} \wedge \vec{w})_1 & 0 \end{pmatrix}$$

The fact that $\vec{v} \wedge \vec{w}$ transforms as a vector in rotations is only true in dimension 3. It is denoted as a *pseudo-vector* but its true nature is more of an *oriented bi-vector* as in next figure :

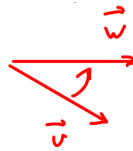


Figure B.2: A vector product actually represents a pair of vector with an orientation, or a bi-vector. The correspondence with a (pseudo)vector only works in 3D.

B.4 Totally antisymmetric tensor

The correspondence between vector product and tensor calculus is very useful and is made explicit as

$$\vec{c} = \vec{v} \wedge \vec{w} \longleftrightarrow c_i = \varepsilon_{ijk} v_j w_k$$

where ε_{ijk} is the signature of the permutation σ_{ijk} or 0 if any index is repeated, namely $\varepsilon_{123} = 1$, $\varepsilon_{121} = 0$, $\varepsilon_{132} = -1 \dots$. This object is called the 3D totally antisymmetric, or Ricci¹ tensor. One of the most useful and important result of this appendix is the double vector product rule :

$$\varepsilon_{ijk} \varepsilon_{ilm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$$

where δ_{ij} denotes the Kronecker symbol.

Exercise B.1. Show that this result enables you to prove that $\vec{a} \wedge (\vec{b} \wedge \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$

B.5 Differential calculus

Differentiation appears through the use of the Nabla operator, which transforms as a vector under rotations :

$$\vec{\nabla} \longleftrightarrow \partial_i, \quad i = 1 \dots 3$$

Applying this operator to a tensor will in general raise its rank. Let us see some examples :

- $\vec{\nabla} p \leftrightarrow \partial_i p$ is the *gradient* of p . It transforms as a vector.
- $\partial_i v_j$ is a rank two tensor with no denomination in usual vector calculus.
- $v_{ij} = (\partial_i v_j + \partial_j v_i)/2$ is a rank two symmetric tensor that we met in the definition of a Newtonian fluid.
- $\omega_{ij} = (\partial_i v_j - \partial_j v_i)/2$ is a rank 2 antisymmetric tensor related to the rotational.
- $\vec{\nabla} \wedge \vec{v} \leftrightarrow \varepsilon_{ijk} \partial_j v_k$ is the rotational.
- $\vec{\nabla} \cdot \vec{v} \leftrightarrow \partial_i v_i$ is the divergence, it is a scalar.

Exercise B.2. Compute $\vec{\nabla} \wedge (\vec{\nabla} \wedge \vec{v})$ and show that it gives back the usual result $\vec{\nabla}(\vec{\nabla} \cdot \vec{v}) - \Delta \vec{v}$.

Exercise B.3. Write the tensor expressions corresponding to $(\vec{v} \cdot \vec{\nabla})\vec{v}$, $\vec{v} \wedge (\vec{\nabla} \wedge \vec{v})$, $\vec{v}(\vec{\nabla} \cdot \vec{v})$.

For higher rank tensors, the Nabla notation is often ambiguous. As an example consider the tensor $\Pi_{ij} \leftrightarrow \tilde{\Pi}$. What is $\vec{\nabla} \cdot \tilde{\Pi}$? $\partial_i \Pi_{ij}$ or $\partial_j \Pi_{ij}$?

¹Actually A. Einstein came to Bologna to meet Ricci and learn about differential geometry and the tensor calculus

B.6 Covariance

One of the main benefits of tensor calculus is to write a priori *covariant relations* between physical quantities. These quantities will transform in a coherent way under rotations². As an example of this use, we seek an empirical linear relation between the viscous stress tensor σ_{ij} and the shear $\partial_i v_j$. A priori the relation writes with a rank 4 tensor as $\sigma_{ij} = \eta_{ijkl} \partial_k v_l$, but because of space isotropy the relation has to be exactly the same with the same coefficients η_{ijkl} in any frame. To do that, one look for irreducible representations in $\partial_i v_j$. These are $\partial_i v_i$ (the divergence) and $(\partial_i v_j + \partial_j v_i - 2/3 \delta_{ij} \partial_k v_k)/2$ the symmetric and traceless rank 2 strain rate tensor. The antisymmetric part can not enter the relation because we know that σ_{ij} is symmetric. Finally a linear relation between these two tensors has to write

$$\sigma_{ij} = \eta(\partial_i v_j + \partial_j v_i - 2/3 \delta_{ij} \partial_k v_k) + \mu \delta_{ij} \partial_k v_k$$

with only two scalar (invariant) coefficients η and μ , the shear and bulk viscosity respectively, instead of 81 coefficients in η_{ijkl} !

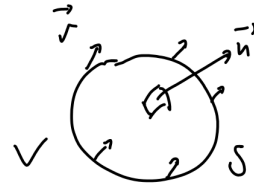
²You may remark that this is very similar to the homogeneity argument in physics : physical relations have to be covariant in a change of units.

Appendix C

Proof of Reynolds Theorem

We examine the integral $Q(t) \equiv \iiint_{V(t)} f_E(\vec{r}, t) d^3r$, where the subscript E is a reminder that we use an Eulerian description of the motion, and $\vec{r}$ refers to fixed points in space. Reynolds theorem states that

$$\frac{dQ(t)}{dt} = \iiint_{V(t)} \frac{\partial}{\partial t} f(\vec{r}, t) d^3r + \oint_{S(t)} f(\vec{r}, t) (\vec{v} \cdot \vec{n}) dS$$



$\iiint_{V(t)} d^3r$ means that we are following particles while they flow in space, hence a Lagrange description is well suited :

$$Q(t) = \iiint_{V_0} f(\vec{r}_L(\vec{r}_0(t), t)) |J| d^3r_0$$

where $J(t) = \det \left(\frac{\partial \vec{r}_L(\vec{r}_0(t), t)}{\partial \vec{r}_0} \right)$ is the Jacobian of the transformation $\vec{r}_L(\vec{r}_0(t), t)$. At $t = 0$ one has $J = 1$ and this show that the Jacobian is always positive since it can not vanish (infinite compression). So taking the time derivative of Q and transforming back to Euler gives

$$\frac{dQ}{dt} = \iiint_{V(t)} \partial_t f d^3r + \iiint_{V(t)} \vec{v} \cdot \vec{\nabla} f d^3r + \iiint_{V_0} f \partial_t J d^3r_0$$

Finally we use the following lemma :

$$\partial_t J d^3r_0 = \vec{\nabla} \cdot \vec{v} d^3r$$

which after use of Ostrogradski theorem produces the expected result.

"Physical proof" of the lemma : $J = d^3r/d^3r_0$ represents the dilatation of the fluid, and $\partial_t J/J = \frac{1}{V} \frac{dV}{dt}$ is the rate of relative volume change as the fluid moves and this is given by $\vec{\nabla} \cdot \vec{v}$, as one may prove in the same fashion as the mass conservation equation.

It is also possible to provide a direct algebraic proof using the chain of relations

$$\begin{aligned}
J &= \varepsilon_{ijk} \frac{\partial r_i}{\partial x_0} \frac{\partial r_j}{\partial y_0} \frac{\partial r_k}{\partial z_0} \\
\frac{\partial J}{\partial t} &= \varepsilon_{ijk} \left(\frac{\partial v_i}{\partial x_0} \frac{\partial r_j}{\partial y_0} \frac{\partial r_k}{\partial z_0} + \frac{\partial x_i}{\partial x_0} \frac{\partial v_j}{\partial y_0} \frac{\partial r_k}{\partial z_0} + \frac{\partial x_i}{\partial x_0} \frac{\partial r_j}{\partial y_0} \frac{\partial v_k}{\partial z_0} \right) \\
&= (\varepsilon_{ljk} \partial_i + \varepsilon_{ilk} \partial_j + \varepsilon_{ijl} \partial_k) v_l \frac{\partial r_i}{\partial x_0} \frac{\partial r_j}{\partial y_0} \frac{\partial r_k}{\partial z_0} \\
&= \partial_l v_l \varepsilon_{ijk} \frac{\partial r_i}{\partial x_0} \frac{\partial r_j}{\partial y_0} \frac{\partial r_k}{\partial z_0} = (\vec{\nabla} \cdot \vec{v}) J
\end{aligned}$$

where we have used the Leibniz rule and commutation of derivatives at the first step, then the chain rule for differentiation of Eulerian variables at the second step,

$$\frac{\partial \vec{v}_E(\vec{r}(\vec{r}_0, t), t)}{\partial x_0} = \left(\frac{\partial \vec{r}}{\partial x_0} \cdot \vec{\nabla} \right) \vec{v}_E$$

and direct inspection to show that $(\varepsilon_{ljk} \partial_i + \varepsilon_{ilk} \partial_j + \varepsilon_{ijl} \partial_k) v_l = \varepsilon_{ijk} \partial_l v_l$.

Altogether this is not a very elegant proof, but it forms a good training in using tensor calculus and the Euler and Lagrange descriptions on motion, try it by yourself !

Appendix D

Hydrodynamics in a rotating frame

D.1 The Coriolis force

In the case when the flow takes place in a rotating frame such as is the case for the general atmospheric circulation on Earth, we have to take into account the effect of the centrifugal and Coriolis forces. As for the usual mechanics of points and solids, the centrifugal force is incorporated in the gravity field modifications of the direction and intensity of $\vec{g}$. The Coriolis force acting on an object of volume d^3V is

$$d\vec{f}_c = -2\rho d^3V (\vec{\Omega} \wedge \vec{v})$$

where as usual ρ is the density of the fluid, and $\vec{\Omega}$ is the angular velocity of the rotating frame. On Earth it is directed to the North and its magnitude is about one revolution per day.

Incorporating this in the Euler equation leads to the Euler rotation in a rotating frame :

$$\frac{\partial}{\partial t} \vec{v} + (\vec{v} \cdot \vec{\nabla}) \vec{v} = -\frac{1}{\rho} \vec{\nabla} p + \vec{g} - 2\vec{\Omega} \wedge \vec{v} \quad , \quad \vec{\nabla} \cdot \vec{v} = 0 \quad (\text{D.1})$$

In the case when viscous forces are not negligible, we have to use the Navier Stokes equations augmented by a Coriolis term, and this writes for incompressible fluids :

$$\frac{\partial}{\partial t} \vec{v} + (\vec{v} \cdot \vec{\nabla}) \vec{v} = -\frac{1}{\rho} \vec{\nabla} p + \vec{g} - 2\vec{\Omega} \wedge \vec{v} + \nu \Delta \vec{v} \quad , \quad \vec{\nabla} \cdot \vec{v} = 0 \quad (\text{D.2})$$

The atmospheric circulation takes place on a very large scale, ensuring that the Reynolds number is very large, and that the Euler equation is generally adequate to describe this global flow. However we will see in the next paragraph that the viscosity forces play an important role in the formation of a stationary water flow profile in the presence of rotation and a steady wind. The reason why viscous forces are not negligible here although the Reynolds number is very large, is that they have acted a very long time to obtain the stationary flow, and they have a crucial role in balancing the Coriolis force. This flow pattern is called the Eckman layer, and will be studied below.

D.2 The vorticity equation in a rotating frame

In chapter 6 we saw that turbulence is generated by the largest vortices of the flow, in a (yet unknown) instability mechanism, leading to the so-called Kolmogorov cascade. If we look at

the global circulation in the atmosphere, which is a very turbulent flow, This seems to be in contradiction with the fact that vorticity can not be created in a flow described by the Euler equation of motion, as we saw in the section on potential flow, see chapter 1.8. As a matter of fact these conclusion where only valid in a non rotating frame. If we develop the same tensor algebra as we did there, it is easy to prove that the vorticity equation becomes¹

$$\frac{D\vec{\omega}}{Dt} = [(\vec{\omega} + 2\vec{\Omega}) \cdot \vec{\nabla}] \vec{v}$$

showing that $\vec{\Omega}$ acts as a source of vorticity as expected.

D.3 The Ekman layer

The Ekman layer is the name given to the stationary horizontal flow in water due to the presence of a constant wind in a rotating frame. We use Cartesian coordinates $0xyz$ in the terrestrial frame of reference, with the vertical along Oz and the wind in the y direction : $\vec{U} = U\vec{e}_y$. Since we are interested in a flow in the horizontal plane, only the vertical component of the angular velocity is needed : $\vec{\Omega} = \Omega_T \cos(\lambda) \vec{e}_z$, where λ is the co-latitude, and Ω_T is one revolution per day.

A reasonable ansatz for the Ekman flow is that it is invariant by translation in the xy plane (all the places see the same flow), so that $\vec{u}(z)$ is a function of the vertical coordinate only. In this case, the rotating frame NS equation boils down to

$$\begin{cases} 0 &= 2\Omega v_y + \nu \partial_z^2 v_x \\ 0 &= -2\Omega v_x + \nu \partial_z^2 v_y \end{cases} \quad (\text{D.3})$$

which express the balance between the Coriolis and the viscous forces as expected. A typical thickness of the Ekman layer emerges from these equations as

$$\lambda_e = \sqrt{\nu/\Omega} \simeq 10 \text{ cm}$$

with a numerical value given for water at a latitude of 45° on Earth. To solve these very simple equations we define $w \equiv v_x + iv_y$, which satisfies

$$w''(z) - 2i/\lambda_e^2 w(z) = 0$$

The solution converging at $z \rightarrow -\infty$ writes $w(z) = A \exp[(1+i)z/\lambda_e]$ with A a complex constant. Separating the real and imaginary part gives

$$\begin{cases} v_x &= A \exp(z/\lambda_e) \cos(z/\lambda_e + \phi) \\ v_y &= A \exp(z/\lambda_e) \sin(z/\lambda_e + \phi) \end{cases} \quad (\text{D.4})$$

where the integration constants A and ϕ have to be determined by the boundary conditions at $z = 0$. At a free boundary, this condition is given by the continuity of the stress tensor², and this writes $\partial_z \vec{u}|_{z=0} \propto \vec{U}$ so that we have the conditions

$$\begin{cases} \partial_z v_x|_{z=0} &= A/\lambda_e (\cos(\phi) - \sin(\phi)) &= 0 \\ \partial_z v_y|_{z=0} &= A/\lambda_e (\cos(\phi) + \sin(\phi)) &= S \end{cases} \quad (\text{D.5})$$

¹The factor 2 there appears due to the different definition of the vorticity in the mechanic of solids and of fluids: in fluids $\vec{\omega} \equiv \vec{\nabla} \wedge \vec{v}$, while in solids $\vec{\omega} \equiv 1/2 \vec{\nabla} \wedge \vec{v}$.

²This is the most subtle aspect in Ekman flow : the wind does not impose direction of the velocity of water at the surface, but rather the direction of the shear.

where S is the shear value of the water flow at the surface, assumed to point in the y direction. The first condition gives $\phi = \pi/4$ and the second one $A = \lambda_e S/\sqrt{2}$ with the final result :

$$\begin{cases} v_x &= \frac{S\lambda_e}{\sqrt{2}} e^{\frac{z}{\lambda_e}} \cos(\frac{z}{\lambda_e} + \frac{\pi}{4}) \\ v_y &= \frac{S\lambda_e}{\sqrt{2}} e^{\frac{z}{\lambda_e}} \sin(\frac{z}{\lambda_e} + \frac{\pi}{4}) \end{cases} \quad (\text{D.6})$$

TO BE CONTINUED

Appendix E

The Diffusion Equation

The diffusion equation is one of the simplest and most important equation in statistical physics and hydrodynamics. If the subject is new to you, here is some basic information.

E.1 The equation

Let us consider particles that diffuse in a fixed medium. These might be for example impurities diffusing in a crystal. We assume that their number is conserved. Let us denote $n(\vec{r}, t)$ and $\vec{j}(\vec{r}, t)$ the associated density and current respectively. The conservation equation is written

$$\frac{\partial n}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0$$

We moreover assume that the current is related simply to the density gradient by Fick's law $\vec{j} = -D\vec{\nabla}n$. D is the diffusion coefficient and its units are $m^2.s^{-1}$. Inserting in the conservation equation we obtain the diffusion (or heat) equation.

$$\frac{\partial n}{\partial t} = D\Delta n$$

Note that it looks like the free Schrödinger Equation but with an imaginary time. We can therefore use some of the techniques used for Schrödinger.

E.2 Fundamental solution

In an infinite d -dimensional media, one shows that the solution decreasing exponentially at infinity and initial conditions $n(\vec{r}, t = 0) = \delta(\vec{r} - \vec{r}_0)$ is

$$G(\vec{r}, t | \vec{r}_0, 0) = \frac{1}{(4\pi Dt)^{d/2}} \exp \left[-\frac{(\vec{r} - \vec{r}_0)^2}{4Dt} \right]$$

This solution satisfies $\int G(\vec{r}, t) d^3r = 1$ and describes the spreading of an ink drop in still water. Below the solution is plotted at different times in 1D. We see that this solution is spreading, but there is no typical front velocity. Instead the traveled distance at a time t is approximately given by

$$l \propto \sqrt{Dt}$$

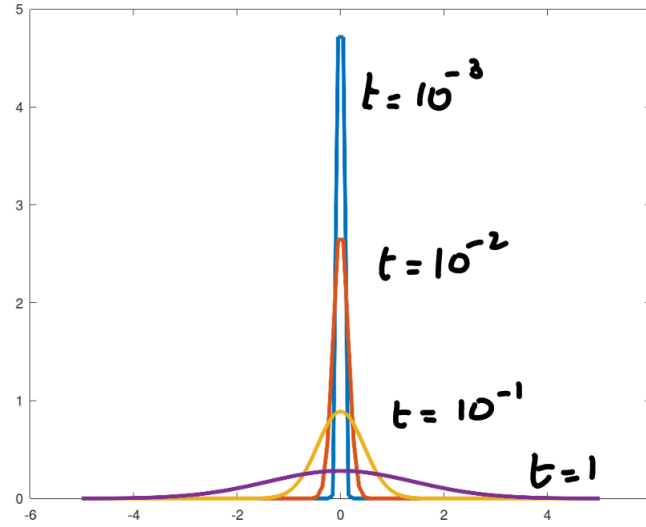


Figure E.1: Diffusion fundamental solution

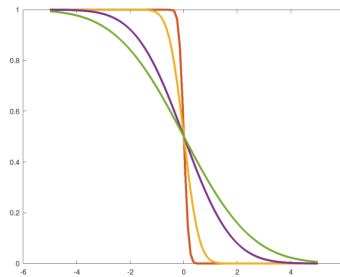
E.3 General solution in unbounded medium

Because of its linearity we can combine any solutions of the diffusion equation. The solution corresponding to an initial condition $n(\vec{r}, t = 0) = n_0(\vec{r})$ and decreasing exponentially at infinity is given by

$$n(\vec{r}, t) = \int G(\vec{r}, t | \vec{r}', 0) n_0(\vec{r}') d^3 r'$$

As an example in 1 D the solution with $n(x, 0) = H(-x)$ is

$$n(x, t) = \int_{-\infty}^0 \frac{1}{(4\pi Dt)^{1/2}} \exp\left[-\frac{(x-x')^2}{4Dt}\right] dx' = \frac{1}{\sqrt{\pi}} \int_{\frac{x}{\sqrt{4Dt}}}^{\infty} e^{-u^2} du \equiv \frac{1}{2} \text{Erfc}\left(\frac{x}{\sqrt{4Dt}}\right)$$



E.4 Solutions 'à la Fourier'

The previous solution is not usable in a bounded box, since it will not satisfy the BC. Indeed, to solve the thermal shock problem in the interval $0 < x < L$ we look for a separable solution

in the form

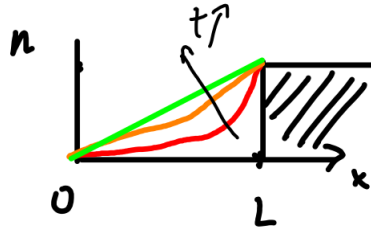
$$n(x, t) = \frac{x}{L} + \sum_{k=1}^{\infty} c_k \sin\left(\frac{k\pi x}{L}\right) e^{-\lambda_k t}$$

Inserting this ansatz in the Diffusion Equation we can easily check that $n(x, t)$ is a solution if the following dispersion relation is satisfied

$$\lambda_k = \left(\frac{k\pi}{L}\right)^2 D$$

This relation shows that the modes with short wavelength (and large k) are very quickly damped. One then has to determine the c_k coefficients by matching the initial condition and this is done by an usual Fourier transformation.

Exercise E.1. Find the coefficients c_k in the case of the thermal shock, where initially $n(x, t = 0) = 0$. Beware of the x/L term : why is it there in the first place ? You may then plot your solution at different times and this should look as :



E.5 Numerical solution in 1D by the 'line method'

To understand better the nature of partial differential equations and the essential role of boundary conditions, we discretize the diffusion equation in the finite interval $a \leq x \leq b$, keeping time as a continuous variable. We replace the continuous variable x by $N + 1$ discrete values $x_i = a + i/N(b - a)$, separated by a step $dx = (b - a)/N$. We then denote $f_i = f(x_i)$ the value of the dependent variable at the point x_i , as in the figure shown below. In this approximation, the derivative is written $f'(x_i) \simeq (f_{i+1} - f_i)/dx$ and the second derivative

$$f''(x_i) \simeq ((f_{i+1} - f_i)/dx - (f_i - f_{i-1})/dx) / dx = \frac{f_{i+1} + f_{i-1} - 2f_i}{dx^2}$$

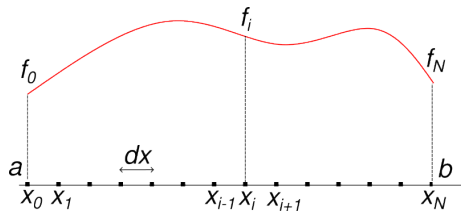


Figure E.2: Discretization of space.

The diffusion equation is then replaced by the set of coupled differential equations for $i = 0 \dots N$:

$$\frac{d}{dt} f_i = \alpha (f_{i+1} + f_{i-1} - 2f_i)$$

with $\alpha \equiv D/dx^2$.

E.6 Boundary conditions

We can now see the role played by boundary conditions: to write the equation corresponding to the index $i = 0$ or $i = N$, we need to know the function outside the interval $[a, b]$, and this is not defined. So a specific recipe should be given for these two equations, which is exactly what boundary conditions do. Two types of BC are commonly used : Dirichlet's and Neumann's.

- Dirichlet BC consist in fixing the value of the function f at the edge of the interval, and we can always reduce to the case $f = 0$ at the boundary by linearity . In the present case, this yields

$$\begin{aligned} \frac{d}{dt} f_0 &= \alpha (f_1 - 2f_0) \\ &\vdots \\ \frac{d}{dt} f_N &= \alpha (f_{N-1} - 2f_N) \end{aligned}$$

- Neumann BC consist in fixing the value of the derivative at the edge of the interval, which in the case of the diffusion equation is equivalent to fixing the value of the current. There are various ways of approximating this condition numerically, but we generally fix values of f symmetrical with respect to the edge $f_{-1} = f_1$ and $f_{N-1} = f_{N+1}$, which here gives

$$\begin{aligned} \frac{d}{dt} f_0 &= 2\alpha (f_1 - f_0) \\ &\vdots \\ \frac{d}{dt} f_N &= 2\alpha (f_{N-1} - f_N) \end{aligned}$$

Boundary conditions can profoundly change the nature of the solution and its physical interpretation, and it is important to specify them in accordance with the physics you wish to describe: in the case of the heat equation, for example, Dirichlet conditions (fixed values) amount to imposing the temperature at the boundary, which requires heat exchange. Neumann conditions (zero flux) correspond to an insulating boundary and adiabatic transformations. Lastly, the value of the function and its derivative at the boundaries cannot be arbitrarily fixed independently, as it may lead to inconsistencies and no solution at all, the problem being over-determinate.

E.7 Stability of the line method

If we use the method presented above and a time discretization to solve the equation numerically by an Euler scheme

$$f_i(t + dt) \simeq f_i(t) + \frac{D dt}{dx^2} (f_{i+1}(t) + f_{i-1}(t) - 2f_i(t))$$

we obtain results that diverge exponentially if $Ddt/dx^2 > 1/2$. The method is *unstable*, which requires very small time steps if we want reasonable spatial accuracy. To avoid this problem, the Crank-Nicholson method replaces the above explicit scheme with an implicit one, which is stable:

$$f_i(t + dt) \simeq f_i(t) + \frac{D dt}{dx^2} (f_{i+1}(t + dt) + f_{i-1}(t + dt) - 2f_i(t + dt))$$

Exercise E.2. Since the equation $f_l(t + 1) \simeq f_l(t) + a (f_{l+1}(t) + f_{l-1}(t) - 2f_l(t))$ is linear, we look for solutions in wave form $f_l(t) = \exp(\omega t) \exp(ikl)$, where ω and k are to be determined.

- Inject this tentative solution into the equation and find the dispersion relation linking ω and k .
- explain why the scheme is unstable when $|\exp(\omega t)| > 1$?
- What value of k leads to the greatest instability, i.e. the greatest value of $|\exp(\omega t)|$.
- What is the critical value of a above which the solution becomes unstable?

Appendix F

Stokes Flow past a sphere

The low Reynolds flow past a sphere or, as it is known, *Stokes flow* is a classic solution of the very viscous NS equations. It leads to the famous Stokes formula for the viscous force $\vec{F} = 6\pi a\eta\vec{U}$, where a is the sphere radius, U the velocity of the sphere relative to the fluid at rest, and η is the dynamic viscosity of the fluid. Note that this formula is obtained under strict assumptions : notably the sphere has to be far from obstacles, and walls in particular. The improved formula to account for the presence of walls is known as the Stokes-Faxen formula. Note also that here we will neglect the inertial terms in NS compared to the viscous ones, an assumption which is not valid far from the sphere, where velocity gradients are weak. The corrected theory to account for these terms and the fact that very far from the sphere the flow should match the irrotational Euler flow seen in chapter 1 is known as Oseen theory.

F.1 Axisymmetric incompressible flow - Stokes current function

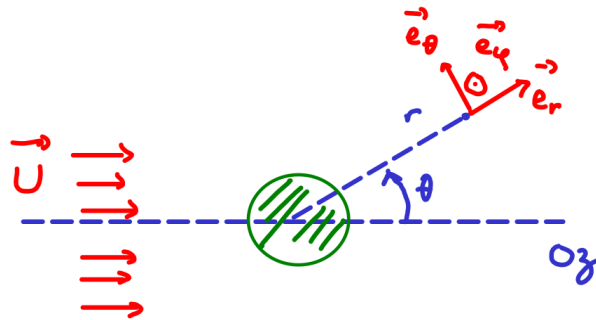


Figure F.1: Geometry of the Stokes flow

Here we study an incompressible, stationary and axisymmetric flow, such that stream lines are contained in axial planes ($u_\phi = 0$) and the flow is invariant by rotation around the $0z$ axis. Henceforth, velocity has the form

$$\vec{u} = u_r(r, \theta)\vec{e}_r + u_\theta(r, \theta)\vec{e}_\theta$$

We then introduce Stokes stream function $\psi(r, \theta)$ such that

$$\vec{u} = \vec{\nabla} \wedge \left(\frac{\psi(r, \theta)}{r \sin \theta} \vec{e}_\phi \right)$$

and from that we obtain

$$u_r = \frac{1}{r^2 \sin \theta} \partial_\theta \psi \quad u_\theta = -\frac{1}{r \sin \theta} \partial_r \psi$$

Incompressibility is then granted :

$$\vec{\nabla} \cdot \vec{u} = \frac{1}{r^2} \partial_r (r^2 u_r) + \frac{1}{r \sin \theta} \partial_\theta (\sin \theta u_\theta) = \frac{1}{r^2} (\partial_r \partial_\theta - \partial_\theta \partial_r) \psi = 0$$

and we have also the property that

$$\vec{u} \cdot \vec{\nabla} \psi = \frac{1}{r^2 \sin \theta} (\partial_r \psi \partial_\theta \psi - \partial_\theta \psi \partial_r \psi) = 0$$

so that ψ is a constant on stream lines. We then compute $\vec{\omega} = \vec{\nabla} \wedge \vec{u}$. Since the velocity $\vec{u}$ is in the axial plane, $\vec{\omega}$ is directed along $\vec{e}_\phi$:

$$\vec{\omega} = \frac{\vec{e}_\phi}{r} (\partial_r r u_\theta - \partial_\theta \theta u_r) = -\frac{\vec{e}_\phi}{r \sin \theta} \left(\frac{\partial^2 \psi}{\partial r^2} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \psi \right) \right) = -\frac{\vec{e}_\phi}{r \sin \theta} E \psi$$

where the operator E , defined as $E \equiv \left(\partial_r^2 - \frac{\sin \theta}{r} \partial_\theta \frac{1}{\sin \theta} \partial_\theta \right)$, is related to the Laplacian :

$$\Delta \left(\frac{\vec{e}_\phi \psi}{r \sin \theta} \right) = \frac{\vec{e}_\phi}{r \sin \theta} E \psi$$

as it may be shown by direct computation.

F.2 Stokes flow

The equations for the flow past a sphere write

$$\begin{cases} -\frac{1}{\rho} \vec{\nabla} p + \nu \Delta \vec{u} &= 0 \\ \vec{\nabla} \cdot \vec{u} &= 0 \\ \vec{u}(r=a) &= 0 \end{cases} \quad \text{and} \quad \vec{u} \xrightarrow{y \rightarrow \infty} U \vec{e}_z$$

As $\vec{e}_z = \vec{e}_r \cos(\theta) - \vec{e}_\theta \sin(\theta)$ the last condition writes

$$\begin{cases} u_r &\rightarrow U \cos \theta \\ u_\theta &\rightarrow -U \sin \theta \\ r &\rightarrow \infty \end{cases}$$

Examining these asymptotic values suggests the following Ansatz for the Stokes function, with separation of the variables : $\psi = U f(r) \sin^2 \theta$ such that the velocity writes

$$u_r = 2U \cos \theta \frac{f(r)}{r^2} \quad u_\theta = -U \sin \theta \frac{f'(r)}{r}$$

The boundary conditions turns into $f(a) = f'(a) = 0$ and $f(r) \sim r^2/2$ at infinity.

Equation for the unknown function f stems from taking the rotational of NS yielding the vorticity equation $\Delta \vec{\omega} = 0$. Using Stokes stream function we obtain after some algebra the equation $EE\psi = 0$. Using the ansatz for ψ we finally have

$$E\psi = \sin^2 \theta \left[f''(r) - 2 \frac{f(r)}{r^2} \right]$$

$$E^2\psi = \sin^2 \theta \left[f^{(4)} - \frac{4f''}{r^2} + \frac{8f'}{r^3} - \frac{8f}{r^4} \right] = 0$$

To solve this linear equation for f we try solutions in the form $f = r^\alpha$ eventually leading to

$$f(r) = \frac{A}{r} + Br + Cr^2 + Dr^4$$

Using the boundary conditions to determine the constants, we end up with $D = 0$, $C = 1/2$, $B = -3a/4$ and $A = a^3/4$, and the Stokes flow solution as :

$$\psi = \frac{U \sin^2 \theta}{4} \left(\frac{a^3}{r} - 3ar + 2r^2 \right)$$

$$u_r = \frac{U \cos \theta}{2r^2} \left(\frac{a^3}{r} - 3ar + 2r^2 \right) \quad u_\theta = -\frac{U \sin \theta}{2r^2} \left(-\frac{a^3}{2r} - \frac{3ar}{2} + 2r^2 \right)$$

F.3 Stokes formula

We can then use this to obtain the constraints on the sphere. We must first determine pressure p from Navier Stokes equations :

$$\begin{cases} \partial_r p &= \eta U \cos \theta \, 3a/r^3 \\ \partial_\theta p &= \eta U \sin \theta \, 3a/2r^2 \end{cases}$$

with the solution

$$p = p_\infty - \frac{3a\eta U \cos \theta}{2r^2}$$

This flow is not subject to the d'Alembert paradox, since we see that pressure at the rear ($\theta < \pi/2$) is less than in front of the sphere. Lastly we compute the total constraint (viscous + pressure) on the sphere :

$$\vec{T} = -p\vec{e}_r + 2\eta\varepsilon_{rr}\vec{e}_r + 2\eta\varepsilon_{r\theta}\vec{e}_\theta|_{r=a}$$

$\varepsilon_{rr}(r=a) = 0$, $2\varepsilon_{r\theta}(r=a) = r\partial_r u_\theta(r=a) = -3U/2a \sin \theta$ and finally the constraint takes the very simple form

$$\vec{T} = -p_\infty\vec{e}_r + 3\eta/2a \vec{U}$$

We see that the viscous stress is everywhere constant on the sphere, a non obvious result which immediately leads by trivial integration to the expression of the Stokes force on the sphere

$$\vec{F} = 4\pi a^2 3\eta/2a \vec{U} = 6\pi a\eta \vec{U}$$

as stated at the beginning of this chapter. Note that this very simple result is not easy to guess nor to compute as you may have painfully experienced following this appendix !

Appendix G

The kinematic boundary condition at a free surface

The surface equation is written in a very general way as $F(x, y, z, t) = 0$ and we want a condition stating that 'particles at the surface have the same velocity as the surface'. As usual there are two ways to proceed, using the Lagrange or the Euler description. Let us start with the simpler Lagrange approach

G.1 Lagrangian derivation of the BC

Let us denote by $\vec{r}_s = (x_s(t), y_s(t), z_s(t))$ the trajectory of a given particle at the surface. We then have $F(x_s(t), y_s(t), z_s(t), t) = 0$. Taking the derivative with respect to the time t gives

$$\frac{\partial}{\partial t} F(\vec{r}_s, t) + \vec{v}(\vec{r}_s, t) \cdot \vec{\nabla} F(\vec{r}_s, t) = 0$$

as stated.

G.2 Eulerian derivation of the BC

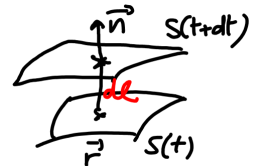
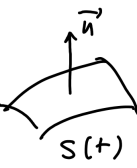
Since the trick seems too easy, let us stay in the Euler description. The surface $S(t)$ is described at time t by the condition $F(x, y, z, t) = 0$. The normal to the surface is given by $\vec{n} = \vec{\nabla} F / \|\vec{\nabla} F\|$. To compute the normal velocity of the surface, we have to find the point $\vec{r} + d\vec{r}$ on the surface $S(t + dt)$, in the direction of $\vec{n}$: $\vec{r} + d\vec{r} = \vec{r} + dl \vec{n}$. We then have

$$\begin{aligned} F(\vec{r}, t) &= 0 \\ F(\vec{r} + dl \vec{n}, t + dt) &= 0 \end{aligned}$$

Taking the difference of these two equation gives $dl \vec{n} \cdot \vec{\nabla} F + dt \partial_t F = 0$ and

$$v_{\perp} = \frac{dl}{dt} = - \frac{\partial_t F}{\|\vec{\nabla} F\|}$$

This velocity has to be equal to the normal velocity at the surface expressed with the Euler description $v_{\perp}(\vec{r}_s, t) = \vec{n} \cdot \vec{v}(\vec{r}_s, t)$. Using the expression for $v_{\perp}$ we end up once again with the



condition

$$\partial_t F + \vec{v} \cdot \vec{\nabla} F \Big|_{\text{surface}} = 0$$

Appendix H

Remark on the continuity of pressure at a wave interface.

We consider here the situation described in paragraph 4.5 of chapter 4. In order to obtain the dispersion relation for gravity wave between the two fluids, one writes the dynamical condition, or continuity of the stress tensor at the interface as :

$$P_0 - \rho_1 gh + p_1(0^-) = P_0 - \rho_2 gh + p_2(0^-)$$

However this classic procedure leads to a discontinuity of the dynamical part of the pressure

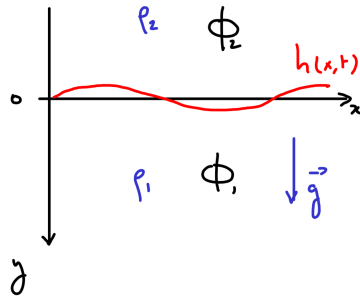


Figure H.1: Superposed fluids with different densities.

p which is due to a discontinuity of the hydrostatic pressure $P_s \equiv P_0 - \rho g z$ there. Although this leads to the correct dispersion relation, this is a bit annoying and may lead to some difficulties as a discontinuity in the energy current. The blunder comes from an incorrect definition of the hydrostatic pressure, which does not satisfy $\vec{\nabla} P_s = \rho \vec{g}$ at the interface. The correct procedure would be :

- write a correct continuous expression for the hydrostatic pressure :

$$\begin{aligned} P_s &= P_0 - \rho_1 g z & z < h \\ P_s &= P_0 - \rho_2 g z + (\rho_2 - \rho_1) g h & z > h \end{aligned}$$

- write the wave equations in perturbation for $z < h$ as

$$\begin{cases} -i\omega u &= -\frac{ik}{\rho_1} p(z) \\ -i\omega w &= -\frac{1}{\rho_1} p'(z) \\ iku + w' &= 0 \end{cases} \quad (\text{H.1})$$

This results in the Rayleigh equation $w'' - k^2 w = 0$ which is solved up to an arbitrary constant as

$$\begin{cases} w &= \exp(k(z - h)) \\ u &= i \exp(k(z - h)) \\ p &= i\rho_1 c \exp(k(z - h)) \end{cases} \quad (\text{H.2})$$

- write the wave equations in perturbation for $z > h$ as

$$\begin{cases} -i\omega u &= -\frac{ik}{\rho_2} (p(z) + (\rho_2 - \rho_1)gh) \\ -i\omega w &= -\frac{1}{\rho_2} p'(z) \\ iku + w' &= 0 \end{cases} \quad (\text{H.3})$$

the crucial point here being the added term $ik(\rho_2 - \rho_1)gh$ in the first equation, which comes from the $\exp(ikz)$ factor hidden in h . This system results in the same Rayleigh equation $w'' - k^2 w = 0$ and is solved as

$$\begin{cases} w &= \exp(-k(z - h)) \\ u &= -i \exp(-k(z - h)) \end{cases} \quad (\text{H.4})$$

Using then the kinematic condition at the interface $-i\omega h = w = 1$ and the continuity of the pressure $p = i\rho_1 c$ at the interface yields

$$\omega^2 = \frac{\rho_1 - \rho_2}{\rho_1 + \rho_2} g k \quad (\text{H.5})$$

as expected. As a conclusion this procedure is obviously more involved, but may help to convince oneself that it is possible to obtain a correct dispersion relation with a continuous dynamical pressure. This will become handy when we will consider energy transfer, as for example in the case of the Kelvin-Helmholtz or Rayleigh Miles instability.

Appendix I

The Miles mechanism

Soon to appear !

Bibliography

- [1] D.J.Aitchison, Elementary Fluid Dynamics, Clarendon Press Oxford (1990). Not so elementary as the title suggests, but an excellent introductory course, very physical and much used in these lectures.
- [2] D.J. Tritton : Physical Fluid Dynamics, 2nd Ed., Oxford University Press, (1988). Complementary to the previous reference, a bit in the same spirit but with more experimental details. A very good introductory part : if you want the basic ideas quickly, read this.
- [3] E. Guyon, J.P. Hulin, L. Petit, Hydrodynamique Physique, CNRS Ed. 2ème édition (2001), In French. Definitely not my favorite, but a lot of information and some subjects not covered in the previous books (as complex fluids).
- [4] L.Landau, E.Lifshitz, Fluid Mechanics, Course of Theoretical Physics vol. 6, Pergamon Press 1st ed. 1959, reprinted 1982. This is a classic, with an original presentation of Hydrodynamics by Lev Landau, who learned the subject by himself in order to write his course, and in the process derived some original results, as e.g. the immersed jet solution. The level is high as usual with Landau, but the course is self contained, in contradistinction with other volumes.
- [5] A. Sommerfeld, Mechanics of Deformables Bodies, Lectures on Theoretical Physics vol. 2, Academic Press, New York (1950). This course is older than the previous one. Actually the reference course in theoretical physics before Landau and Lifschitz classic series was Sommerfeld's course. A lot of original results and derivations given in details for all the vector algebra given in the appendix. Also a very useful historical account about Hydrodynamics by one of its major contributors.
- [6] G.K. Batchelor : An Introduction to Fluid Dynamics, Cambridge University Press, 2000. This is 'the' classic book, going in much more details than Landau. No account on turbulence, which is treated in the following reference. I never could spot any error in this book !
- [7] G.K. Batchelor : The theory of homogeneous turbulence, Cambridge Science Classics, Cambridge University Press, 1953, reprinted 1993. This is complementary to the previous one, a classic introduction to hydrodynamics turbulence despite his age. Batchelor introduced Kolmogorov's ideas on turbulence to western science.
- [8] G.B. Whitham : Linear and Non Linear Waves, Wiley Sons, 1999. A great classic on waves, with a detailed account on water waves spread in many chapters. A quite high level but very complete and useful.

- [9] P. Janssen : The Interaction of Ocean Waves and Wind, Cambridge University Press, 2004. Very detailed account on ocean waves and their formation, with an account of the Miles mechanism, and more recent results on this important subject (formation of rogue waves, tsunami...).
- [10] / P. Bergé, Y. Pomeau, C. Vidal, L'Ordre dans le chaos : vers une approche déterministe de la turbulence, Hermann, Paris 1984, reprinted 1997. Introduces the ideas of instability and chaos at an elementary level, in French.
- [11] Steven H. Strogatz, Nonlinear Dynamics and Chaos With Applications to Physics, Biology, Chemistry, and Engineering, Westview Press, 2015. 2nd ed. More mathematically oriented and going deeper than the preceding one, a classic reference to the field.
- [12] NSF Films Fluid Mechanics Series, Youtube, (Barry Belmont Youtube channel). A series of 21 short movies presenting different aspects of Fluid mechanics. Highly recommended. "Low Reynolds Number" by G.I. Taylor (see sequence at 13' 15") is legendary. Taylor, together with Prandtl (and Sommerfeld), should have been awarded the Nobel price, but Hydrodynamics was considered an engineering subject in the XX-th century.